Beyond Code – Building Software Systems That Last

Nex Book Team

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From Modeling and Algorithms to Reliability, Evolution,
and AI-Era Engineering

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1.

Prologue — The Day the System Broke

At 9:12 on a Monday morning, a delivery robot stopped in the middle of an intersection and refused to move.

It had a package. It had a destination. It had a full battery.

But it would not move.

The control system reported: **no valid route forward**.

Across town, a graduate student searched for a paper she had saved two months earlier. The system returned **12,483 results**. She had built the note graph carefully: tags, backlinks, categories, references, aliases. Everything was there. Nothing was findable.

In a startup office on the third floor of a building with bad coffee and good ambition, a game engineer watched a virtual economy collapse in real time. Players had discovered an item-duplication path. Within minutes, rarity meant nothing.

Three different systems. Three different domains. One shared failure pattern.

The failure was not a typo. It was not a missing semicolon. It was not solved by writing more code.

The break happened earlier.

What Actually Failed

Each team had competent engineers and working software. Early demos looked excellent.

Then scale arrived.

The delivery network moved from dozens of routes to thousands. Local path choices started conflicting globally. Robots blocked each other. Resolution logic became brittle.

The knowledge system moved from hundreds of notes to tens of thousands of linked entities. Connection density exploded. Search degraded from guidance to noise.

The virtual world moved from a few hundred active players to a persistent economy with emergent behavior. State transitions that looked safe in isolation were unsafe in combination.

In each case, code quality mattered, but it was not the primary bottleneck.

The bottleneck was design clarity:

- unclear problem boundaries
- weak models of entities and relationships
- under-specified rules that should always hold
- algorithms chosen without enough attention to growth behavior

These are engineering failures before they are programming failures.

Why This Book Exists

Most people learn programming in a sequence like this:

1. Learn syntax.
2. Write small functions.
3. Debug until tests pass.

That path is useful, but incomplete.

Small exercises hide the hardest part of software engineering: deciding what system should exist in the first place, and how it should behave when the world gets messy.

This book starts one layer earlier.

You will still write code. You will still learn algorithms and data structures. But the central skill we build is this:

turning ambiguous real-world problems into precise, durable software designs.

The Thread Through The Whole Book

We will follow three recurring systems:

- a delivery network
- a knowledge engine

- a virtual world

They look unrelated on the surface. They are not.
All three force the same engineering questions:

- What are the core entities?
- How do entities relate and change over time?
- Which invariants must never be violated?
- What operations must be fast, and at what scale?
- Where do local decisions create global failures?

By revisiting the same systems across many chapters, you will learn transfer: one design idea applied across multiple domains.
That is what expert engineers do.

How To Read This Book

This is not a reference manual. It is a progression.
Each part builds a layer:

- **Part I**: See the problem clearly before coding.
- **Part II**: Model the world with entities, relationships, and change.
- **Part III–V**: Design and evaluate algorithms and data structures.
- **Part VI**: Organize software into components and interfaces.
- **Part VII**: Make systems trustworthy with contracts, invariants, tests, and debugging.
- **Part VIII**: Evolve systems without collapse.
- **Part IX**: Work effectively with AI coding tools while keeping human judgment central.

Every section is aimed at one practical outcome: better engineering decisions under real constraints.

Programming In The AI Era

AI can generate code quickly.
That changes workflows, but it does not eliminate engineering.
AI can draft implementations. It can propose refactors. It can translate patterns.
It cannot reliably do the full job of:

- defining the right problem
- selecting the right model
- setting robust invariants and contracts
- balancing correctness, performance, and changeability
- owning consequences when systems fail in production

Those remain human responsibilities.

If anything, AI makes design judgment more valuable, not less. Fast code generation amplifies both good and bad architecture.

This book is written for that reality.

Why This Book Uses Nex

You do not need a popular language to learn durable engineering thinking.

This book uses **Nex** as its implementation language on purpose.

Nex is not mainstream, but it gives us a clean way to study core software ideas without carrying unnecessary ecosystem complexity in early chapters.

Most importantly, Nex was designed to teach good software engineering practices that transfer directly to real-world systems.

Nex is especially useful for this book because it supports:

- **Functional and object-oriented styles** in the same language, so we can compare design tradeoffs directly.
- **High-level contracts and invariants**, so correctness rules can be expressed in the code where they belong.
- **Explorative programming**, including optional dynamic typing when rapid experimentation is useful.
- **Graphics support**, which helps us model visible system behavior (simulation, movement, interaction) instead of only text output.
- A **web-based IDE**, so readers can start immediately with no local installation required.

In other words, Nex is a teaching language for this journey: expressive enough for real design discussions, lightweight enough to stay focused on engineering decisions.

The goal is not to lock you into one language.

The goal is to build transferable skills you can apply in any serious codebase.

What You Will Practice

By the end, you should be able to:

- write precise problem statements
- model systems with explicit assumptions and constraints
- select data structures based on access patterns, not habit
- reason about algorithmic behavior as systems scale
- use contracts and invariants to prevent silent corruption
- debug by isolating causes, not chasing symptoms
- refactor with confidence instead of fear
- use AI assistants as accelerators without outsourcing judgment

The goal is not just to become a faster coder.

The goal is to become the engineer people trust with systems that matter.

A Note On Style

You will see three kinds of material throughout the book:

- **Narrative scenarios** that ground ideas in realistic system behavior.
- **Engineering frameworks** that name and structure decisions.
- **Implementation sketches** (including Nex examples) that connect design to executable systems.

If you are early in your programming journey, move slowly and implement often.

If you are experienced, use the chapter prompts as a way to audit your defaults. Many senior failures come from invisible assumptions, not missing knowledge.

Before Chapter 1

Keep one question in mind as you begin:

What problem am I actually solving?

Not:

- What library should I use?

- Which architecture is trendy?
- How quickly can I ship this feature?

Those questions matter later.

The first question determines whether the rest of the work has a chance.

A robot is waiting in an intersection. A researcher cannot recover her own knowledge. A virtual economy is collapsing.

Their codebases are different. Their failure mode is the same.

Chapter 1 begins there.

Part I.

Part I — Seeing The Problem

2. Part I — Seeing the Problem Clearly

2.1. 1. What Problem Are We Actually Solving?

Programming often begins with excitement.

A new feature. A clever algorithm. A fresh framework. A blank editor window waiting for code.

But most software failures begin **before any of that**.

They begin with a misunderstanding of the problem.

2.2. The Most Expensive Mistake in Software

Imagine a team building a navigation system for a delivery robot.

They start with a clear technical goal:

> "The robot must compute the shortest path from A to B."

So they build:

- a fast path-finding algorithm
- a grid map of the building
- a sophisticated movement controller

After months of work, the robot moves beautifully.
It just **delivers packages to the wrong places**.
Why?
Because the real problem was never "find the shortest path."
The real problem was:

- Buildings change.
- Hallways get blocked.
- Elevators break.
- People move furniture.
- Doors close.

The true problem was **operating in an unpredictable environment**, not computing an optimal path.

The team solved the **symptom**.

They missed the **problem**.

2.3. Symptoms vs Problems

Many programming projects begin with a description that sounds precise but is actually only a symptom.

For example:

Symptom	Underlying Problem
"The database is slow."	The data model doesn't match the queries.
"Search results are irrelevant."	The system does not understand user intent.
"The robot can't navigate the building."	The environment is dynamic and uncertain.
"The system can't scale."	The architecture assumes a fixed workload.

Symptoms are **visible failures**.

Problems are **structural causes**.

Good programmers fix symptoms.

Great engineers **find the real problem first**.

2.4. The Three Systems We Will Build

Throughout this book, we will repeatedly return to three systems.

Each one looks like a programming challenge.

But each one is actually a **problem-understanding challenge**.

2.4.1. 1. The Delivery Robot

A robot must navigate a large office building and deliver packages.

At first glance this looks like a classic algorithm problem:

- shortest path

- graph traversal
- route optimization

But very quickly the real issues appear:

- What if the map is incomplete?
- What if elevators become unavailable?
- What if the robot must learn new routes?

This is not just pathfinding.
It is **decision-making under uncertainty**.

2.4.2. 2. The Knowledge Engine

A system that answers questions using a large collection of documents.

At first glance this seems like a search problem:

- indexing
- ranking
- retrieval

But the deeper problem emerges:

- What does the user *really mean*?
- What counts as a trustworthy answer?
- When should the system say **"I don't know"?**

This is not just retrieval.
It is **reasoning about knowledge**.

2.4.3. 3. The Virtual World

A simulated world with thousands of interacting objects.

The system must support:

- characters
- physics
- events
- behaviors

At first this appears to be a scaling problem.
But the deeper issue becomes clear:

- How do you manage **complex interactions** without creating chaos?

This is not just simulation.
It is **managing complexity through structure**.

2.5. A Pattern You Should Notice

These three systems appear unrelated:

- robotics
- knowledge systems
- virtual worlds

But they share the same hidden difficulty.
The hardest part is **not writing code**.
It is **understanding the problem deeply enough that the code becomes obvious**.
This is the central theme of this book.

2.6. Why Most Software Fails Before the First Line of Code

Software projects fail for many reasons.
But the most common one is surprisingly simple:

> The team never clearly defined the problem.

This happens in subtle ways:

- Requirements describe **features**, not **problems**.
- Engineers optimize **performance**, not **correctness**.
- Systems are designed around **tools**, not **needs**.

In many projects, the first design decision is already a mistake.
The team chooses:

- a database
- a framework
- a programming language

- a system architecture

before asking the most important question:

> **What kind of problem is this?**

2.7. Problems Have Shapes

Different problems require different ways of thinking.

Some problems are about:

- **searching**
- **optimization**
- **classification**
- **simulation**
- **coordination**
- **learning**

Each shape suggests a different kind of solution.

The mistake beginners make is assuming all problems have the same shape:

> "Write some code and figure it out."

Experienced engineers recognize the shape **first**.

Then they design the solution.

2.8. The Discipline of Problem Seeing

Before designing algorithms, before choosing data structures, before writing code, we must learn a skill that is rarely taught:

> **How to see the problem clearly.**

This means learning to ask questions like:

- What is the system *really trying to accomplish*?
- What assumptions are hidden in the requirements?
- What constraints shape the solution?
- What kinds of failure are acceptable?

Good programming begins with **clear thinking**.

And clear thinking begins with **good questions**.

2.9. Connection to Nex

Throughout this book, we will implement ideas in Nex.

That choice is intentional.

Nex is designed to support this exact workflow: think clearly about problems first, then express solutions with strong structure, explicit contracts, and flexible modeling styles.

The language is not the point.

The point is to practice engineering habits that transfer to any real-world stack.

Exercise Apply the section task and record your results before reading the solution notes.

2.10. Quick Exercise (2 Minutes)

Take one symptom and rewrite it as an underlying problem.

Example prompt:

- Symptom: "Search is slow."
- Rewrite: "Our data model and indexing strategy do not match actual query patterns."

Now do your own:

1. Pick one symptom from your current project (or from the table above).
2. Write one sentence describing the structural cause.
3. Write one constraint that any valid solution must satisfy.

If this feels hard, that is normal. This is exactly the skill we are building.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

2.11. Chapter Takeaways

- Symptoms are visible failures; problems are structural causes.

- Most costly software mistakes happen before implementation starts.
- Problem shape determines solution shape.
- Good engineering begins by clarifying goals, assumptions, and constraints.
- If the problem is clear, architecture and algorithms become far easier to choose.

2.12. A Preview of What Comes Next

In the next chapters, we will develop a set of tools for understanding problems before writing code.

We will explore:

- how to **separate goals from mechanisms**
- how to **model systems with abstractions**
- how to **identify the true sources of complexity**

Only after that will we begin designing algorithms.

Because by then, the algorithms will feel almost inevitable.

And when that happens, programming becomes something remarkable:

> Not the art of writing code, but the discipline of **making complex problems understandable**.

In Chapter 2, we begin with a concrete observation method: how to look at real-world behavior, identify hidden assumptions, and capture constraints before design starts.

3. 2. Looking at the Real World

Before we design algorithms, we need to confront an uncomfortable truth:

The real world is messy.

Requirements documents try to simplify it. Architecture diagrams try to structure it. Specifications try to freeze it into rules.

But real systems live in environments that are:

- incomplete
- unpredictable
- full of human behavior
- constantly changing

If we ignore this, we build systems that work perfectly **in theory** and fail immediately **in practice**.

So the next step in programming is not writing code.

It is **learning how to observe the real world carefully**.

3.1. Observing How Systems Actually Behave

Many programming problems appear straightforward when described abstractly.

But the moment you observe the real environment, new complexities appear.

Consider our **delivery robot** again.

A clean algorithmic description might say:

> "The robot moves along hallways represented as a graph."

But if you watch a real building for an hour, you will notice things no graph contains:

- People walking unpredictably
- Temporary obstacles
- Doors that sometimes remain closed
- Elevators that are occasionally out of service

- Hallways blocked by maintenance work

The algorithm is correct.
The **model of reality is incomplete**.
Good engineers spend time observing how the system behaves **before deciding how it should behave**.

3.2. Users Are Part of the System

Programmers often think of users as external to the system.
But in reality, users are **part of the system's behavior**.
Consider the **knowledge engine** we will build later.
Suppose a user asks:

> "What causes memory leaks?"

This looks like a simple search query.
But users rarely behave like clean inputs to an algorithm.
They may mean:

- memory leaks in C
- memory leaks in Java
- memory leaks in operating systems
- memory leaks in machine learning pipelines

Or they may ask incomplete questions:

> "Why does my program keep growing?"

Users bring:

- assumptions
- misunderstandings
- incomplete information

If we design systems assuming perfect input, we create systems that **fail the moment they meet real people**.
Understanding users is not a product-design concern alone.
It is a **core programming concern**.

3.3. Constraints Shape the Problem

Another aspect of the real world is constraints.

In theory, many problems have elegant solutions.

In reality, solutions must operate within limits.

Consider again the **virtual world** system.

In theory, every object could interact with every other object in perfect physical simulation.

But reality imposes constraints:

- limited CPU time
- limited memory
- network latency
- thousands of simultaneous interactions

A perfectly accurate simulation might be computationally impossible.

So we must ask:

- Which interactions matter most?
- Which approximations are acceptable?
- Where can we simplify without breaking the illusion?

Constraints do not merely limit solutions.

They **define the shape of the solution**.

3.4. Hidden Assumptions

One of the most dangerous sources of bugs in software is the **hidden assumption**.

A hidden assumption is something everyone believes to be true, but no one explicitly states.

For example:

- "The map of the building is accurate."
- "Users will ask clear questions."
- "Objects in the simulation interact in predictable ways."

These assumptions often remain invisible until they break.

The robot reaches a hallway that does not exist on the map.

A user asks a vague question.

Two subsystems interact in a way no one predicted.

At that moment, the system does not merely fail.

It fails **mysteriously**, because the underlying assumption was never documented.

Careful observation helps us discover these assumptions early.

3.5. Turning Reality Into Questions

Reality is messy, ambiguous, and complicated.

Programming requires something different:

clear, answerable questions.

This transformation—from observation to question—is one of the most important intellectual steps in engineering.

For example:

Messy observation:

> "The robot sometimes gets stuck."

Better question:

> "What information does the robot need to detect blocked paths?"

Messy observation:

> "Users get bad answers."

Better question:

> "How can the system represent uncertainty in retrieved information?"

Messy observation:

> "The simulation becomes chaotic."

Better question:

> "What rules govern interactions between objects?"

A good engineering question has three properties:

1. **It is precise enough to analyze.**
2. **It focuses on causes, not symptoms.**
3. **It suggests measurable outcomes.**

Once we have questions like these, algorithms and designs begin to emerge naturally.

3.6. The Programmer as Observer

Many people imagine programming as an activity that happens entirely inside a computer.

But good programmers spend much of their time doing something else:

- watching systems run
- observing user behavior
- investigating failures
- questioning assumptions

In other words, they act a little like scientists studying a complex phenomenon.

They observe.

They form hypotheses.

They test those hypotheses through experiments and code.

Programming is not just construction.

It is also **investigation**.

3.7. Connection to Nex

In this book, Nex helps us keep observation and implementation connected.

Because Nex supports both fast exploration and structured design, we can move from:

- rough hypotheses about real behavior
- to explicit rules, contracts, and invariants
- to executable prototypes we can test quickly

That workflow mirrors real engineering: observe, model, test, revise.

Exercise Apply the section task and record your results before reading the solution notes.

3.8. Quick Exercise (3 Minutes)

Pick one real system you use every day (maps, search, chat, payments, calendar).

Write:

1. One observable failure: "Sometimes it does X."
2. One hidden assumption behind that failure.
3. One concrete question engineers should answer before coding a fix.

Example:

- Failure: "Search returns too many irrelevant results."
- Hidden assumption: "All users mean the same thing by the same query terms."
- Engineering question: "How should intent ambiguity be represented and ranked?"

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

3.9. Chapter Takeaways

- Real environments are dynamic; static specs never capture everything.
- Users are part of system behavior, not external noise.
- Constraints are not obstacles after design; they shape design from the start.
- Hidden assumptions create mysterious failures.
- Strong engineering starts by turning messy observations into precise questions.

3.10. From Observation to Understanding

The goal of observing the real world is not to capture every detail.

That would be impossible.

Instead, the goal is to identify the **essential structure** hidden inside messy reality.

In the next chapter, we will convert these observations into something engineers can execute against: a **problem statement**.

If Chapter 2 is about seeing reality clearly, Chapter 3 is about expressing that reality precisely.

That written precision is what later enables strong models, better algorithms, and safer implementations.

4. 3. Writing a Problem Statement

Once we have observed the real world and identified the **actual problem**, we face a new challenge:

How do we describe that problem **clearly enough that a computer — and a team of programmers — can solve it?**

This is the role of the **problem statement**.

A good problem statement is not a vague description of what we want. It is a **precise description of the task the system must accomplish**.

It sits between messy reality and precise algorithms.

Too vague, and no one knows what to build. Too rigid, and the system becomes impossible to evolve.

Writing a good problem statement is therefore one of the most important skills in programming.

4.1. Precision Without Overengineering

Beginners often make one of two mistakes.

4.1.1. Mistake 1: Being Too Vague

A vague problem statement might look like this:

> "Build a robot that can deliver packages inside the building."

This sounds reasonable, but it leaves critical questions unanswered:

- How does the robot know where it is?
- What happens if a path is blocked?
- How are delivery locations represented?
- How fast must the robot respond?

Without answers, every programmer imagines something different.

The result is confusion and incompatible code.

4.1.2. Mistake 2: Being Overly Detailed

At the opposite extreme, some specifications attempt to describe **every tiny detail**:

- precise internal data structures
- exact algorithms
- specific implementation choices

For example:

> "The robot must represent the building using an adjacency matrix and compute shortest paths using Dijkstra's algorithm."

This sounds precise, but it introduces another problem.

It **locks the solution too early**.

Perhaps later we discover that:

- the building is very large
- the map changes frequently
- real-time updates are needed

Now the chosen data structure may be the wrong one.

A good problem statement should describe **what must be achieved**, not **how it must be implemented**.

4.1.3. The Goal: Precise but Flexible

A strong problem statement answers questions like:

- What information enters the system?
- What output must the system produce?
- What constraints must be respected?

But it avoids committing to specific algorithms or internal structures.

Think of it as describing **the shape of the problem**, not the solution.

4.2. Example: Delivery Robot Navigation

A clearer problem statement might look like this:

> The system receives a map of the building consisting of locations connected by traversable paths. Given a starting location and a destination, the robot must determine a route that allows it to reach the destination while avoiding blocked paths.

Notice what this statement does well:

- It defines **inputs** (map, start, destination).
- It defines **expected behavior** (find a route).
- It allows multiple algorithms to solve the problem.

This clarity makes algorithm design possible.

4.3. Good Specifications vs Bad Specifications

Let us look at a few examples.

4.3.1. Bad Specification

> "The knowledge engine should provide relevant answers to user questions."

The phrase **"relevant answers"** is ambiguous.
Different programmers might interpret this differently.

4.3.2. Better Specification

> Given a user query, the system retrieves documents that contain information related to the query and ranks them according to their estimated relevance.

This version clarifies:

- the input (query)
- the output (ranked documents)
- the goal (estimated relevance)

We still have many design choices left — which is exactly what we want.

4.3.3. Another Bad Specification

> "Objects in the virtual world should interact realistically."

What does "realistically" mean?
Physics simulation? Game-like interactions? Simplified rules?

4.3.4. Better Specification

> Objects in the world have positions and interaction rules. When two objects occupy overlapping space, the system must apply the interaction rule associated with their types.

Now we can start thinking about:

- spatial data structures
- collision detection
- interaction systems

The problem has become **algorithmically meaningful**.

4.4. The Power of Examples

Even the best written specification can leave gaps.

That is why engineers rely heavily on **examples**.

Examples help clarify expectations and reveal hidden assumptions.

For instance, in the delivery robot system:

Example

```
Start: Office A
Destination: Office D

Map:
A → B → C → D
```

Expected result:

```
Route: A → B → C → D
```

Simple examples establish the basic behavior.

But examples become even more powerful when they show **edge cases**.

4.5. Counterexamples and Edge Cases

Counterexamples reveal situations where naïve solutions fail.

Consider a blocked path.

Example

```
A → B → C → D
      X
```

If B → C is blocked, the robot must find an alternative route.

If no alternative exists, the system must report failure.

Counterexamples force us to ask important questions:

- What happens when a destination is unreachable?
- What if the map contains loops?
- What if multiple routes exist?

These questions often expose missing parts of the specification.

4.6. Example-Driven Thinking

Many experienced engineers design systems by collecting examples first.

They ask:

- What are the **normal cases**?
- What are the **edge cases**?
- What are the **failure cases**?

By the time the specification is written, the system's behavior has already been explored through examples.

This approach has another advantage in the AI era.

Examples are often the **best way to communicate intent to AI coding assistants**.

Clear examples help both humans and machines understand what the system must do.

4.7. Connection to Nex

Nex is a good fit for this stage because it encourages explicit problem expression.

As we move from statements to implementation, Nex lets us keep intent visible through:

- clear type annotations when we need precision
- optional flexibility when we are still exploring
- contracts and invariants when behavior must be guaranteed

The key habit is the same in any language: state the problem clearly before encoding a solution.

Exercise Apply the section task and record your results before reading the solution notes.

4.8. Quick Exercise (5 Minutes)

Choose one of the three systems (delivery robot, knowledge engine, virtual world) and write a mini problem statement with exactly three parts:

1. Inputs: what information enters the system?
2. Required behavior: what must the system do?
3. Constraints: what conditions must always be respected?

Then add:

- one normal example
- one edge case
- one failure case

If your statement prescribes a specific algorithm, rewrite it to focus on outcomes instead.

4.9. A Problem Statement Is a Contract

At its core, a problem statement is a **contract**.

It establishes an agreement between:

- the **problem** we want solved
- the **algorithms** we will design
- the **code** we will eventually write

If the contract is vague, the system becomes unpredictable.

If the contract is precise, the entire development process becomes easier:

- algorithms are easier to design
- tests are easier to write
- bugs are easier to detect

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

4.10. Chapter Takeaways

- A strong problem statement is precise about outcomes, not rigid about implementation.
- Vague statements create misalignment; over-detailed statements lock in bad decisions.
- Examples, edge cases, and counterexamples expose missing requirements early.
- Problem statements function as contracts between intent, design, and code.
- Clear statements make algorithm selection and testing substantially easier.

In Chapter 4, we stress-test problem statements with edge cases and failure conditions. If Chapter 3 defines the contract, Chapter 4 tries to break it.

5. 4. Edge Cases: Where Systems Break

Most systems work perfectly — **until they don't**.

A navigation algorithm finds paths across the building. A knowledge engine retrieves useful answers. A virtual world simulation runs smoothly.

Then someone enters a strange query. A robot reaches an unexpected hallway. Two objects interact in a way no one predicted.

And suddenly the system fails.

These failures rarely occur because the main algorithm is wrong. They happen because the system encounters **situations no one thought about**.

These situations are called **edge cases**.

Learning to identify edge cases is one of the most important habits a programmer can develop.

5.1. Strange Inputs

Many programs implicitly assume that inputs will be "reasonable."

But in reality, inputs can be surprising.

Consider the **knowledge engine**.

A developer might expect queries like:

> "How does garbage collection work?"

But real users might enter:

- an empty query
- a single letter
- a long paragraph
- a question in another language
- a misspelled term
- an emoji

Each of these inputs may break assumptions inside the system.

For example:

- A ranking algorithm might divide by the number of query terms — but what if the query is empty?
- A search function may assume words exist in the index — but what if they don't?

Strange inputs reveal where our system quietly assumes things that are not guaranteed.

Robust software **expects strange inputs** and handles them gracefully.

5.2. Boundary Conditions

Edge cases often occur at the **boundaries** of a problem.

These boundaries may involve:

- the smallest possible input
- the largest possible input
- the first or last element in a structure
- values that lie exactly at limits

For example, imagine the robot navigation system.

Typical paths may involve several intermediate locations.

But what happens if:

- the start and destination are the same location?
- the map contains only one room?
- a path exists but has zero length?
- there are multiple equally short routes?

These cases occur at the edges of the problem space.

They often expose small logical mistakes that remain invisible during normal testing.

A classic programming example involves arrays.

A loop written like this:

```
for i from 0 to n
```

may accidentally access an element beyond the end of the array.

The bug does not appear until the program reaches the **boundary**.

5.3. The Fragility of "Almost Correct"

Many programs are **almost correct**.

They work for the most common inputs.

They even pass many tests.

But edge cases reveal subtle flaws:

- off-by-one errors
- incorrect assumptions
- missing failure handling
- inconsistent rules

These bugs are frustrating because they often appear rarely.

But when they do appear, they can cause serious failures.

In safety-critical systems, boundary mistakes can be catastrophic.

In everyday software, they lead to mysterious crashes and strange behavior.

Good engineers assume that **every algorithm has fragile edges**.

Their job is to find those edges before users do.

5.4. Thinking Like a Tester

One of the most powerful ways to discover edge cases is to temporarily change perspective.

Instead of thinking like the person **building** the system, think like the person **trying to break it**.

Testers often ask questions such as:

- What happens if the input is empty?
- What happens if the input is extremely large?
- What happens if the system receives something unexpected?
- What happens if two things occur at the same time?

Let us apply this mindset to our three systems.

5.4.1. Delivery Robot

- What if a path becomes blocked after the robot has started moving?
- What if the robot's map is incomplete?
- What if multiple robots attempt to use the same hallway?

5.4.2. Knowledge Engine

- What if two documents contradict each other?
- What if the query contains ambiguous terms?
- What if no relevant information exists?

5.4.3. Virtual World

- What if thousands of objects occupy the same region?
- What if two interaction rules conflict?
- What if an object interacts with itself?

These questions may seem pessimistic.
But they are essential for building **reliable systems**.

5.5. Edge Cases Reveal the True Problem

There is another surprising property of edge cases.
They often reveal that our original problem statement was incomplete.
For example:

> "Find the shortest path between two locations."

Sounds clear — until we ask:

- What if no path exists?
- What if multiple shortest paths exist?
- What if the graph changes during navigation?

Now the problem becomes richer and more realistic.
Edge cases force us to refine our specification.
They help transform a vague problem into a **precisely defined one**.

5.6. The Habit of Asking "What If?"

Experienced engineers develop a reflex.
Whenever they see a specification, they immediately begin asking:

- What if the input is empty?
- What if the input is huge?
- What if something fails halfway through?
- What if the system receives conflicting information?

These questions often reveal weaknesses early.
The earlier we discover them, the cheaper they are to fix.

5.7. Edge Cases as Design Tools

Edge cases are not just obstacles.
They are **design tools**.
They guide us toward:

- clearer specifications
- stronger algorithms
- better tests

Many elegant algorithms were discovered precisely because someone carefully examined the boundaries of a problem.
In other words, systems often become stronger **at their edges**.

5.8. Connection to Nex

Nex encourages this edge-case mindset by making behavior contracts explicit.
As we move from problem understanding to implementation, we can encode edge expectations using:

- preconditions for valid inputs
- postconditions for expected outcomes
- invariants for rules that must always hold

That keeps "what if?" questions visible in code, not only in design docs.

Exercise Apply the section task and record your results before reading the solution notes.

5.9. Quick Exercise (4 Minutes)

Pick one simple feature from your current project and generate edge cases in three categories:

1. Input edge: empty, null-like, malformed, or unexpected format.
2. Size edge: smallest and largest realistic values.
3. State edge: partial failure, race/conflict, or contradictory data.

Then answer:

- Which of these are already handled?
- Which would currently fail?
- Which need explicit specification before implementation?

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

5.10. Chapter Takeaways

- Edge cases are not rare exceptions; they define reliability boundaries.
- Boundary conditions expose logical errors hidden by normal inputs.
- "Almost correct" systems fail at the edges first.
- Thinking like a breaker improves design quality.
- Edge analysis refines the problem statement itself.

In the next chapter we will take the next refinement step.

We will move from narrative descriptions to formal specifications: turning stories, examples, and edge cases into precise requirements.

If Chapter 4 asks "What can go wrong?", Chapter 5 asks "How do we write requirements so everyone builds the same system?"

That transition is what makes design and implementation reliable instead of accidental.

6. 5. From Stories to Specifications

So far in this book, we have described our systems mostly through **stories**.

A robot trying to deliver packages in a building. A knowledge engine trying to answer questions. A virtual world struggling to simulate interactions.

Stories are powerful because they help us **understand the situation**. They capture context, motivation, and human goals.

But stories alone are not enough for programming.

Computers cannot execute stories.

They execute **precise rules**.

The task of the programmer is to transform a narrative like:

> "The robot should deliver packages efficiently"

into something far more precise:

> "Given a map, a start location, and a destination, the system must compute a valid route."

This transformation—from story to specification—is where software engineering truly begins.

6.1. Turning Narratives Into Rules

Stories describe events and intentions.

Specifications describe **rules and behaviors**.

Consider the story of the delivery robot:

> A robot receives a delivery request and navigates through the building to reach the destination.

To turn this story into a specification, we begin asking structured questions:

- What information does the robot receive?
- What decision must the robot make?
- What outcome must the system guarantee?

By answering these questions, we gradually transform the story into something more precise.

For example:

Narrative

> The robot should navigate through the building.

Specification

> The robot receives a map of locations connected by traversable paths and must compute a route from its current location to a specified destination.

Notice what happened.

The story introduced the idea. The specification introduced **formal elements** we can reason about.

6.2. Identifying the Inputs

Every computational problem begins with **inputs**.

Inputs represent the information the system receives from the outside world.

For the delivery robot, possible inputs include:

- the map of the building
- the robot's current location
- the destination location
- information about blocked paths

For the knowledge engine, inputs might include:

- the user's query
- the collection of documents
- previous interaction history

For the virtual world simulation:

- the list of objects
- their positions

- the rules governing their interactions

Once we clearly identify the inputs, the problem becomes far easier to analyze.

Without clear inputs, algorithms have nothing to operate on.

6.3. Defining the Outputs

If inputs describe what enters the system, **outputs** describe what the system must produce.

In the navigation problem, the output might be:

- a sequence of locations representing the route

In the knowledge engine, the output might be:

- a ranked list of documents

In the virtual world, the output might be:

- the updated positions and states of objects after a simulation step

The moment we clearly define outputs, we begin to understand what the algorithm must accomplish.

6.4. Guarantees and Expectations

Beyond inputs and outputs, specifications must describe **guarantees**.

These guarantees define what the system promises to do.

For example:

Navigation guarantee

> If a path exists between the start and destination, the system must return a valid route.

Search guarantee

> The system must return documents ranked by estimated relevance to the query.

Simulation guarantee

> Objects must follow the interaction rules defined for their types.

Guarantees are important because they establish **what correctness means**.

Without them, we cannot determine whether a system behaves properly.

6.5. The Limits of Stories

Stories help us imagine how a system should behave.

But stories often hide ambiguity.

Consider the narrative:

> The robot should take the best path.

What does "best" mean?

- shortest distance?
- fastest travel time?
- safest route?
- least crowded hallway?

Stories leave these questions unanswered.

Specifications force us to **make decisions explicit**.

6.6. A First Glimpse of Contracts

As systems become more complex, specifications evolve into something even more powerful: **contracts**.

A contract describes the agreement between different parts of a system.

It typically answers three questions:

1. **What must be true before an operation begins?**
2. **What will the operation produce?**
3. **What conditions will remain true afterward?**

For example, imagine a function that computes a route.

We might describe its contract like this:

Precondition

The map correctly represents connections between locations.

Postcondition

If a path exists between the start and destination, the function returns a valid route.

Invariant

Every step in the returned route corresponds to a valid connection in the map.

This structure allows programmers to reason about correctness with much greater clarity.

Later in this book, we will use contracts extensively to build systems that are **predictable, reliable, and easier to maintain**.

6.7. Connection to Nex

Nex is designed to make this story-to-specification transition concrete.

In practice, that means you can encode the specification directly through:

- explicit structure for inputs and outputs
- contracts (`require`, `ensure`) for operation-level guarantees
- invariants for system-level consistency rules

This helps keep intent and implementation aligned as systems evolve.

Exercise Apply the section task and record your results before reading the solution notes.

6.8. Quick Exercise (6 Minutes)

Take one story sentence and convert it into a mini specification.

Template:

1. Story sentence: "The system should . . . "

2. Inputs:
3. Outputs:
4. Guarantee:
5. One precondition, one postcondition, one invariant.

Example starter:

- Story: "The robot should deliver packages efficiently."
- Guarantee rewrite: "Given a valid map, start, and destination, return a valid route if one exists."

Keep algorithm choices out of this draft. Focus on behavior and correctness.

6.9. From Story to Engineering

By the end of this chapter, we have transformed our narrative systems into something more precise:

- we know what information they receive
- we know what results they must produce
- we know the guarantees they must satisfy

The story is still important—it gives the system purpose.
But the specification gives it **structure**.
Programming lives in the space between those two things.
Stories inspire the system. Specifications make it **buildable**.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

6.10. Chapter Takeaways

- Stories provide context and goals; specifications provide executable precision.
- Every specification needs clear inputs, outputs, and guarantees.
- Ambiguous terms ("best," "relevant," "realistic") must be made explicit.
- Contracts are a disciplined way to formalize expectations and correctness.

- Better specifications reduce design churn, testing ambiguity, and implementation drift.

In the next part of the book, we take the next engineering step.

Once we know **what problem we are solving**, we must decide how the program will **represent the world**.

This is the beginning of **data modeling**.

And as we will soon discover, the way we represent data often determines which algorithms are possible.

Part II opens with that exact question in Chapter 6: why software needs models before it needs optimization.

7. Studio 1 — Our First Tiny System

7.1. Subtitle

Building the smallest working versions of all three systems.

Studio Focus This studio chapter is hands-on: implement, verify behavior, and document tradeoffs as you iterate.

7.2. 1. The Situation

The team has finished Part I and has problem statements, example scenarios, and edge-case notes. Now they must ship a tiny end-to-end version of each system to test whether the problem framing is actually usable.

Systems in scope:

- delivery scheduler (single robot, tiny map)
- note organizer (small collection, simple lookup)
- virtual world (few entities, deterministic update loop)

7.3. 2. Engineering Brief

Build minimal vertical slices that run from input to output.

Required outcomes:

- define a minimal model for each system
- implement one core operation per system
- encode at least one explicit contract per operation
- demonstrate behavior on nominal and edge inputs

Implementation guidance:

- prefer clarity over generality
- keep architecture deliberately small
- document assumptions that might break at scale

7.3.1. Implementation In Nex

Use Nex for the studio implementation so contracts, assumptions, and behavior are all visible in one place.

Suggested file split:

- `delivery_tiny.nex`
- `notes_tiny.nex`
- `world_tiny.nex`
- `studio_1_main.nex`

If you are using the web IDE, you can also place everything in one file and run `App.run`.

7.3.1.1. Delivery Tiny (single robot, tiny map)

```
class Delivery_Tiny
feature
  next_stop(current, destination: String): String
    require
      current_non_empty: current /= ""
      destination_non_empty: destination /= ""
    do
      if current = destination then
        result := current
      elseif current = "A" and destination = "C" then
        result := "B"
      elseif current = "B" and destination = "C" then
        result := "C"
      else
        result := "UNREACHABLE"
      end
    ensure
      decision_returned: result /= ""
    end
end
```

7.3.1.2. Notes Tiny (simple lookup)

```
class Notes_Tiny
feature
  find_by_tag(tag: String): String
```

```
    require
      tag_non_empty: tag /= ""
    do
      if tag = "algorithms" then
        result := "note_001"
      elseif tag = "graphs" then
        result := "note_002"
      else
        result := "NOT_FOUND"
      end
    ensure
      response_non_empty: result /= ""
    end
end
```

7.3.1.3. World Tiny (deterministic update step)

```
class World_Tiny
feature
  step(position, velocity, max_x: Integer): Integer
    require
      max_positive: max_x > 0
    do
      let next: Integer := position + velocity
      if next < 0 then
        result := 0
      elseif next > max_x then
        result := max_x
      else
        result := next
      end
    ensure
      bounded: result >= 0 and result <= max_x
    end
end
```

7.3.1.4. Studio Driver (nominal + edge inputs)

```
class App
feature
  run() do
```

```
    let d: Delivery_Tiny := create Delivery_Tiny
    let n: Notes_Tiny := create Notes_Tiny
    let w: World_Tiny := create World_Tiny

    -- Nominal
    print(d.next_stop("A", "C"))        -- expected: "B"
    print(n.find_by_tag("algorithms"))  -- expected: "note_001"
    print(w.step(3, 2, 10))             -- expected: 5

    -- Edge / failure-oriented checks
    print(d.next_stop("C", "C"))        -- expected: "C"
    print(d.next_stop("X", "C"))        -- expected: "UNREACHABI
    print(n.find_by_tag("unknown"))     -- expected: "NOT_FOUND"
    print(w.step(9, 5, 10))             -- expected: 10
    print(w.step(1, -5, 10))            -- expected: 0
  end
end
```

You can evolve this starter in three directions:

- add a richer map model for delivery
- replace fixed tag matching with indexed note structures
- extend world updates to multiple entities per tick

7.4. 3. Studio Challenges

7.4.1. Level 1 — Core Implementation

- Implement the tiny system for one domain.
- Add executable examples and expected results.

7.4.2. Level 2 — Cross-System Generalization

- Implement tiny versions for all three domains.
- Identify one shared abstraction across them.

7.4.3. Level 3 — Exploration

- Replace one design choice with an alternative and compare outcomes.
- Record what changed in complexity and failure behavior.

7.5. 4. Postmortem

Discuss:

- Which assumptions were validated?
- Which assumptions were false?
- What information was missing from the original problem statement?
- What should be tightened before model redesign?

7.6. Deliverables

- runnable Nex code for all tiny systems
- short design notes (inputs, outputs, guarantees)
- edge-case checklist with observed behavior

Part II.

Part II — Modeling the World

8. Part II — Modeling the World — Why Software Needs Models

8.1. 6. Why Software Needs Models

By the end of Part I, we learned to write better problem statements.

That was necessary.

Now we hit the next wall.

Even with a clear specification, teams still fail when they move too quickly from requirements to code. The code may compile, tests may pass, and demos may look good. Then scale, ambiguity, and change arrive, and the system becomes fragile.

What is missing?

A **model**.

A model is not the same as code. It is the structured representation of what exists in the system, how those things relate, and what rules govern change.

Without a model, implementation becomes guesswork. With a model, implementation becomes engineering.

8.2. Why Code Alone Is Not Enough

New engineers often assume this sequence is enough:

1. Understand the feature request.
2. Start implementing functions and classes.
3. Refactor when problems appear.

In small tasks, this works.

In real systems, it creates hidden inconsistencies:

- two teams represent the same concept differently
- edge behavior depends on call order instead of rules

- performance collapses because access patterns were never modeled
- every new feature adds local patches and global confusion

A model solves this by forcing explicit structure before implementation details harden.

Think of it this way:

- specification answers: "What must happen?"
- model answers: "What exists, and how can it change?"
- algorithms answer: "How do we compute efficiently within that model?"

If the middle layer is weak, the rest of the system drifts.

8.3. Models Across Our Three Systems

The delivery network, knowledge engine, and virtual world still look different. Modeling reveals shared structure.

8.3.1. Delivery Network

If we model only "route from A to B," we miss reality.

A stronger model includes:

- locations
- traversable connections
- dynamic path state (open/blocked)
- robot position and task

Now route choice is a query over model state, not a hard-coded behavior.

8.3.2. Knowledge Engine

If we model only "query string -> results," quality degrades quickly.

A stronger model includes:

- documents
- terms/tags
- relevance signals
- uncertainty/confidence

Now ranking can evolve because the model captures what ranking depends on.

8.3.3. Virtual World

If we model only "update objects every frame," interactions become chaotic.

A stronger model includes:

- entity identity
- entity state
- interaction rules by type
- update boundaries per tick

Now correctness can be reasoned about per transition, not only by runtime observation.

The domains differ, but the modeling move is the same: name stable structure first.

8.4. Worked Design Path

Let us walk from ambiguous requirement to model.

Ambiguous requirement:

> "The robot should choose good routes and avoid getting stuck."

That sentence is useful for product discussion and nearly useless for implementation.

8.4.1. Step 1: Separate Nouns and Verbs

Nouns suggest modeled entities. Verbs suggest operations.

Nouns:

- robot
- route
- location
- path
- blockage

Verbs:

- choose
- avoid
- move
- re-plan

8.4.2. Step 2: Define State, Not Just Behavior

A minimal state model:

- `Location`: identifier
- `Path`: `from`, `to`, `status`
- `Robot`: `current_location`, `destination`
- `Map`: set of locations and paths

8.4.3. Step 3: Add Invariants Early

Before algorithms, define truths that must always hold.

Examples:

- every path endpoint must reference a known location
- robot current location must exist in map
- returned route (if present) must use only open paths

These invariants become your reliability backbone.

8.4.4. Step 4: Make Change Explicit

Most failures happen during change, not static state.

Transitions:

- path becomes blocked
- robot receives new destination
- route recomputation succeeds or fails

A model that ignores transitions will pass demos and fail production.

8.4.5. Step 5: Delay Algorithm Commitment

At this stage, do not lock to BFS, Dijkstra, A*, or custom heuristics.

Pick algorithm after model and access patterns are clear.

This preserves flexibility and prevents early overfitting.

8.5. Nex Implementation Sketch

Nex is a useful teaching language here because it lets us express contracts close to behavior.

```
class Path
feature
  from_loc: String
  to_loc: String
  open: Boolean
invariant
  endpoints_present: from_loc /= "" and to_loc /= ""
end

class Robot_State
feature
  current: String
  destination: String

  request_replan(has_open_route: Boolean): String
    require
      current_known: current /= ""
      destination_known: destination /= ""
    do
      if current = destination then
        result := "ARRIVED"
      elseif has_open_route then
        result := "ROUTE_FOUND"
      else
        result := "UNREACHABLE"
      end
    ensure
      status_defined: result = "ARRIVED" or result = "ROUTE_
    end
invariant
  state_valid: current /= "" and destination /= ""
end
```

This is intentionally small. It does three important things:

- defines state explicitly
- expresses behavior in terms of state
- encodes guarantees as contracts/invariants

The algorithm can be inserted later without changing the conceptual model.

The same pattern applies to the knowledge engine and world simulation: model first, optimize second.

8.6. Common Modeling Mistakes

8.6.1. Mistake 1: Modeling Operations Before Entities

Symptom:

- lots of utility functions
- unclear ownership of data

Recovery:

- list stable entities first
- attach operations to explicit state

8.6.2. Mistake 2: Ignoring Boundaries

Symptom:

- unclear where one subsystem ends and another begins
- duplicated representations across modules

Recovery:

- define model boundaries in writing
- make conversion points explicit

8.6.3. Mistake 3: Missing Invariants

Symptom:

- bugs appear as "impossible states"
- fixes add conditional checks everywhere

Recovery:

- write invariants at class/operation level
- fail fast when invariants are violated

8.6.4. Mistake 4: Premature Algorithm Lock-In

Symptom:

- architecture constrained by early performance guess
- hard to adapt when constraints change

Recovery:

- keep interface and model stable
- treat algorithm as replaceable strategy

8.6.5. Mistake 5: Treating Models As Documentation Only

Symptom:

- diagrams exist, code disagrees

Recovery:

- encode model assumptions in types, contracts, and tests
- keep model and implementation synchronized

Exercise Apply the section task and record your results before reading the solution notes.

8.7. Quick Exercise (8 Minutes)

Pick one of the three systems and produce a mini model draft with four parts:

1. Entities (3-6 items)
2. Relationships (how entities connect)
3. Invariants (2-3 rules that must always hold)
4. Transitions (2-3 important state changes)

Then answer:

- Which part of your current implementation is not represented in this model?
- Which part of the model is not currently enforced in code?

That gap is where reliability work should start.

8.8. Reflection Checkpoint

Before moving on, you should be able to explain:

- Why your chosen entities are stable over time.
- Which assumptions are encoded as invariants.
- Which design decisions are still provisional.

- What evidence shows your model handles nominal and edge behavior.

If you cannot answer these clearly, do not add more features yet. Strengthen the model first.

8.9. Connection to Nex

Nex was chosen for this book because it supports this modeling discipline directly:

- object-style structures for entities and responsibilities
- functional-style decomposition for pure transformations
- contracts and invariants for correctness guarantees
- optional typing flexibility during exploration

That makes Nex a practical bridge between concept and implementation.

The transferable lesson is broader: in any language, explicit models reduce accidental complexity.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

8.10. Chapter Takeaways

- Models are the missing layer between specification and implementation.
- Stable entities, boundaries, invariants, and transitions are core modeling elements.
- Algorithm choice should follow model clarity, not precede it.
- Most scaling failures are model failures before they are code failures.
- Encoding model assumptions in code prevents drift.

In Chapter 7, we go deeper into the first modeling primitive: **entities**.

We will separate identity from state, assign responsibilities, and choose representations that remain stable as the system grows.

9. Part II — Modeling the World — Entities: The Things That Exist

9.1. 7. Entities: The Things That Exist

In Chapter 6, we argued that software needs models.

Now we build the first modeling primitive: **entities**.

An entity is something the system must track as a distinct thing over time.

That definition sounds simple. It is one of the most important distinctions in software design.

When teams get entities wrong, everything downstream becomes unstable:

- interfaces leak implementation details
- algorithms operate on the wrong units
- tests pass locally and fail system-wide

The core question is:
What are the actual things in this system that deserve identity?

9.2. Identity vs State

A useful mental model:

- **Identity** answers: "Which thing is this?"
- **State** answers: "What is true about it now?"

Example: delivery task

- identity: `task_id`
- state: destination, priority, status

If state changes, identity stays.

This is critical. If your model treats state changes as new entities, you lose continuity. If your model merges distinct identities by accident, you corrupt behavior.

The same pattern appears in all three systems.

9.2.1. Delivery

- Entity: `Robot`, `Location`, `DeliveryTask`
- Identity survives reroutes and retries.

9.2.2. Knowledge

- Entity: `Document`, `Tag`, `QuerySession`
- A document can be edited, but remains the same document.

9.2.3. Virtual World

- Entity: `WorldObject`, `Player`, `InteractionRule`
- Position changes every tick; object identity should not.

9.3. Responsibilities Belong To Entities

Once entities are named, assign responsibilities around them.

Bad pattern:

- global helper functions mutate shared maps with weak ownership

Better pattern:

- each entity controls invariants related to its state
- transitions happen through explicit operations

Entity-centered responsibilities improve:

- correctness
- readability
- refactoring safety

Example:

- `DeliveryTask` should validate status transitions.

- `Document` should guard metadata integrity.
- `WorldObject` should enforce legal bounds for movement/state.

9.4. Worked Design Path

Ambiguous requirement:

> "Track package progress and show live status."

9.4.1. Step 1: Extract Candidate Entities

Candidates:

- package
- robot
- route
- user notification

9.4.2. Step 2: Keep Only Stable Identity Carriers

`Route` is often a computation result, not always a persistent entity.

Likely entities:

- `DeliveryTask`
- `Robot`
- `Location`

9.4.3. Step 3: Define Identity Keys

- `DeliveryTask.task_id`
- `Robot.robot_id`
- `Location.location_id`

9.4.4. Step 4: Define Minimal State

For `DeliveryTask`:

- origin
- destination
- status (`PENDING`, `IN_TRANSIT`, `DELIVERED`, `FAILED`)
- assigned_robot_id (optional)

9.4.5. Step 5: Add Invariants

- task id non-empty
- origin and destination non-empty
- delivered task cannot go back to pending

9.4.6. Step 6: Add Transition Operations

- assign robot
- mark in transit
- mark delivered
- mark failed

Now we have entity structure, not just feature text.

9.5. Nex Implementation Sketch

```
class Delivery_Task
feature
  task_id: String
  origin: String
  destination: String
  status: String

  start() do
    if status = "PENDING" then
      status := "IN_TRANSIT"
    end
  ensure
    started_or_unchanged: status = "IN_TRANSIT" or status = "PEN
  end

  complete() do
    if status = "IN_TRANSIT" then
      status := "DELIVERED"
    end
  ensure
    delivered_or_unchanged: status = "DELIVERED" or status = "IN
  end
invariant
```

```
  id_present: task_id /= ""
  endpoints_present: origin /= "" and destination /= ""
  valid_status:
    status = "PENDING" or
    status = "IN_TRANSIT" or
    status = "DELIVERED" or
    status = "FAILED"
end
```

This sketch demonstrates:

- explicit identity (`task_id`)
- mutable state (`status`)
- behavior tied to entity transitions
- invariants protecting impossible states

The same approach scales to `Document` and `WorldObject` entities.

9.6. Common Mistakes

9.6.1. Mistake 1: Everything Is An Entity

Symptom:

- model bloats quickly
- every helper value gets persistence and lifecycle burden

Recovery:

- reserve entities for identity-bearing things
- treat derived values as views/results

9.6.2. Mistake 2: No Identity, Only Fields

Symptom:

- updates overwrite wrong records
- hard to reason about history

Recovery:

- add explicit stable identifiers
- define equality/uniqueness rules

9.6.3. Mistake 3: Mixed Responsibilities

Symptom:

- validation scattered in services/controllers/helpers

Recovery:

- move entity-specific rules near entity operations
- centralize transition checks

9.6.4. Mistake 4: Illegal State Transitions

Symptom:

- "delivered" tasks become "pending" due to patch logic

Recovery:

- model transitions explicitly
- enforce with contracts and tests

Exercise Apply the section task and record your results before reading the solution notes.

9.7. Quick Exercise (8 Minutes)

For one system, produce an entity sheet with:

1. Entity name
2. Identity field(s)
3. Core state fields
4. Allowed transitions
5. Two invariants

Then answer:

- Which fields are identity vs state?
- Which transitions are currently unchecked in your code?

9.8. Connection to Nex

Nex helps here because entities can carry both data and behavior, while contracts/invariants keep assumptions explicit.

You can begin with lightweight models, then strengthen guarantees as complexity grows.

That mirrors real project evolution.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

9.9. Chapter Takeaways

- Entities are identity-bearing things tracked over time.
- Identity and state must be separated clearly.
- Responsibilities should align to entities, not global utility sprawl.
- Transitions and invariants are part of the entity model, not afterthoughts.
- Strong entity modeling reduces downstream architectural friction.

In Chapter 8, we move from "things" to "connections."

Entities alone are not enough; behavior emerges from **relationships**.

10. Part II — Modeling the World — Relationships: How Things Connect

10.1. 8. Relationships: How Things Connect

Entities give us the nouns of a system.

Relationships give us the structure.

Most interesting system behavior does not come from isolated entities. It comes from how entities connect, depend, and constrain each other.

If entities are wrong, systems are confusing. If relationships are wrong, systems are dangerous.

10.2. Why Relationships Matter

A relationship encodes how two entities are linked.

Examples:

- `Robot` **is assigned to** `DeliveryTask`
- `Document` **has tag** `Tag`
- `WorldObject` **interacts with** `WorldObject`

These links are not decoration. They define allowable operations.

If links are underspecified:

- data duplicates and diverges
- cascading failures become invisible
- queries become expensive or ambiguous

10.3. Relationship Dimensions

Use these dimensions to make relationships explicit.

10.3.1. 1) Cardinality

How many items can connect?

- one-to-one
- one-to-many
- many-to-many

10.3.2. 2) Direction

Is the relationship directional?

- `A -> B` can mean dependency, containment, or flow.

10.3.3. 3) Ownership

Who is responsible for maintaining integrity?

- parent owns child?
- shared relationship table/index?

10.3.4. 4) Lifecycle Coupling

If one entity is removed, what happens to related entities?

- delete
- orphan
- preserve with historical reference

10.3.5. 5) Constraint Rules

What must always be true across the relationship?

- assignment must reference existing entities
- link types must be valid
- no illegal cycles for specific relation types

10.4. Worked Design Path

Ambiguous requirement:

> "The knowledge engine should connect related notes."

10.4.1. Step 1: Name Entities

- `Document`
- `Tag`
- `Link`

10.4.2. Step 2: Choose Relationship Shape

Potential options:

- direct many-to-many `Document <-> Document`
- relationship entity `Link(from_doc, to_doc, link_type)`

Use relationship entity when:

- you need metadata (strength, timestamp, source)
- you need multiple link types
- you need auditability

10.4.3. Step 3: Define Constraints

- `from_doc` and `to_doc` must exist
- disallow self-link for selected link types
- link type must be in controlled set

10.4.4. Step 4: Define Query Paths

Design depends on usage:

- "show all references for this document"
- "find backlinks"
- "traverse two hops for discovery"

Model for expected queries, not abstract purity.

10.5. Relationships In The Three Systems

10.5.1. Delivery

- `Location` connected to `Location` via `Path`
- `Robot` assigned to at most one active `DeliveryTask`
- `DeliveryTask` references origin and destination locations

10.5.2. Knowledge

- `Document` linked to `Tag`
- `Document` linked to `Document` with typed edges
- optional confidence score for inferred links

10.5.3. Virtual World

- `WorldObject` belongs to region/zone
- interaction rules reference object type pairs
- event relationships capture cause/effect chains

Shared pattern: relationships are first-class model elements, not incidental fields.

10.6. Nex Implementation Sketch

```
class Doc_Link
feature
  from_id: String
  to_id: String
  link_type: String

  is_structurally_valid(): Boolean do
    result := from_id /= "" and to_id /= "" and link_type /= ""
  ensure
    result_is_boolean: result = true or result = false
  end
invariant
  endpoints_present: from_id /= "" and to_id /= ""
  non_self_reference: from_id /= to_id
  link_type_present: link_type /= ""
end
```

Minimal, but expressive:

- relationship as explicit entity
- endpoint constraints
- structural validation operation

This style generalizes to `Path` in delivery and interaction edges in world simulation.

10.7. Common Mistakes

10.7.1. Mistake 1: Encoding Relationships As Free-Text Fields

Symptom:

- inconsistent IDs/names
- impossible joins/traversals

Recovery:

- model relationships as typed links
- constrain endpoints and link type

10.7.2. Mistake 2: Ignoring Reverse Queries

Symptom:

- forward lookup fast, reverse lookup slow/unavailable

Recovery:

- design indexes for both directions when needed
- document expected query patterns

10.7.3. Mistake 3: Relationship Semantics Drift

Symptom:

- same link type used for incompatible meanings

Recovery:

- define allowed link taxonomy
- enforce in constructors/validators

10.7.4. Mistake 4: Hidden Lifecycle Rules

Symptom:

- deleting an entity leaves broken links

Recovery:

- define delete/update policy explicitly
- test relationship integrity after lifecycle events

Exercise Apply the section task and record your results before reading the solution notes.

10.8. Quick Exercise (8 Minutes)

Pick one system and write a relationship matrix:

1. Entity A
2. Relationship type
3. Entity B
4. Cardinality
5. Constraint rule

Then add one reverse query your model must support.

If reverse query is expensive or unclear, your relationship model needs refinement.

10.9. Connection to Nex

Nex makes relationship rules easier to keep visible through invariants and explicit validation methods.

The teaching value is not syntax. It is disciplined modeling: connections are modeled, constrained, and tested.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

10.10. Chapter Takeaways

- Relationships determine system structure and query behavior.
- Cardinality, direction, lifecycle, and constraints must be explicit.
- Relationship entities are often better than ad hoc fields.

- Model for real access patterns, not theoretical elegance.
- Good relationship models prevent silent integrity drift.

In Chapter 9, we combine entities and relationships into a full data model and evaluate tradeoffs.

11. Part II — Modeling the World — Designing a Good Data Model

11.1. 9. Designing a Good Data Model

By now we have two building blocks:

- entities
- relationships

A data model is where these pieces become a coherent system design.

A good model is not the most abstract one. It is the one that keeps the system correct, understandable, and evolvable under real usage.

11.2. What Makes A Data Model Good?

A good model should do five things well.

1. **Represent reality adequately** for the problem scope.
2. **Support critical operations efficiently**.
3. **Protect invariants** against invalid states.
4. **Allow change** without destructive rewrites.
5. **Remain explainable** to new engineers.

If a model fails any of these, teams eventually compensate with brittle glue logic.

11.3. Query-First Modeling

One practical rule:

Model from expected operations, not from nouns alone.
Ask:

- What are our top read paths?
- What are top write/update paths?
- What invariants must hold through both?

Example: knowledge engine
If the top operation is "fetch relevant notes by tag and recency," model needs:

- tag index path
- recency field
- deterministic ranking inputs

Without this, code adds ad hoc caches and special cases later.

11.4. Tradeoffs You Must Make Explicit

No non-trivial model is free of tradeoffs.

11.4.1. Normalization vs Duplication

- normalized: cleaner consistency, heavier joins/traversals
- duplicated: faster reads, harder updates

11.4.2. Generality vs Simplicity

- highly generic schemas support many future cases
- over-generality can slow current work and increase ambiguity

11.4.3. Strictness vs Flexibility

- strict constraints prevent bad states
- some workflows need temporary incomplete data

A good model chooses intentionally and documents why.

11.5. Worked Design Path

Requirement:

> "Show live delivery status and allow reassignment when robots fail."

11.5.1. Step 1: Identify Core Entities

- `Robot`
- `DeliveryTask`
- `Location`

11.5.2. Step 2: Define Relationships

- `DeliveryTask` references origin/destination locations
- `DeliveryTask` optionally references assigned robot

11.5.3. Step 3: Define Operational Queries

- find active task by robot
- find next pending task by priority
- reassign failed robot's active tasks

11.5.4. Step 4: Shape Model For Queries

Add fields/index intent:

- task status
- priority
- assigned_robot_id

11.5.5. Step 5: Protect Invariants

- delivered task cannot be reassigned
- assigned robot must exist for `IN_TRANSIT`
- origin/destination must be valid locations

11.5.6. Step 6: Plan For Evolution

Potential future changes:

- multi-robot cooperative delivery
- regional routing constraints

- SLA classes

Design extension points now, but keep current model minimal.

11.6. Nex Implementation Sketch

```
class Delivery_Model
feature
  task_id: String
  assigned_robot_id: String
  status: String
  priority: Integer

  can_reassign(): Boolean do
    result := status = "PENDING" or status = "FAILED"
  ensure
    result_boolean: result = true or result = false
  end

  mark_in_transit(robot_id: String)
    require
      has_robot: robot_id /= ""
      legal_transition: status = "PENDING" or status = "FAILED"
    do
      assigned_robot_id := robot_id
      status := "IN_TRANSIT"
    ensure
      transit_set: status = "IN_TRANSIT" and assigned_robot_id =
    end
invariant
  task_id_present: task_id /= ""
  priority_non_negative: priority >= 0
  valid_status:
    status = "PENDING" or status = "IN_TRANSIT" or status = "DEI
end
```

This sketch emphasizes model behavior consistency, not storage backend details.

11.7. Common Modeling Failures

11.7.1. Failure 1: Model Mirrors UI, Not Domain

Symptom:

- backend structure breaks when UI changes

Recovery:

- model domain semantics first
- treat UI as a projection/view

11.7.2. Failure 2: Missing Transition Rules

Symptom:

- illegal status combinations appear in production

Recovery:

- encode transitions explicitly
- enforce via contracts and tests

11.7.3. Failure 3: Performance Afterthought

Symptom:

- model is clean but operationally unusable at scale

Recovery:

- incorporate critical query paths early
- benchmark representative workloads

11.7.4. Failure 4: No Versioning Strategy

Symptom:

- migrations break old assumptions

Recovery:

- make model changes additive where possible
- document compatibility assumptions

Exercise Apply the section task and record your results before reading the solution notes.

11.8. Quick Exercise (10 Minutes)

For one system, write a one-page data model brief:

1. Core entities
2. Core relationships
3. Top 3 reads
4. Top 3 writes
5. 3 invariants
6. One expected future extension

Then identify one tradeoff you are accepting now and why.

11.9. Connection to Nex

Nex helps keep model decisions executable:

- fields define state clearly
- contracts define legal transitions
- invariants preserve correctness over time

This is the core teaching objective: align design decisions with enforceable behavior.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

11.10. Chapter Takeaways

- Good models are operational, not merely conceptual.
- Query paths should shape model design early.
- Tradeoffs must be explicit and documented.
- Transition rules are part of the data model.
- Model quality determines system changeability.

In Chapter 10, we focus on the most failure-prone area: **modeling change over time**.

12. Part II — Modeling the World — Modeling Change

12.1. 10. Modeling Change

Static models are necessary.

Real systems are dynamic.

Most severe bugs do not happen because stored state is wrong at rest. They happen during transitions:

- two updates race
- one step succeeds, next step fails
- event order changes outcomes
- partial writes leave inconsistent state

If your model does not include change semantics, correctness is accidental.

12.2. State + Transition = Real Behavior

A robust model includes:

- state definitions
- legal transitions
- transition preconditions
- transition postconditions

Think in terms of state machines, even for simple systems.

Example delivery task transitions:

- `PENDING -> IN_TRANSIT`
- `IN_TRANSIT -> DELIVERED`
- `IN_TRANSIT -> FAILED`

Illegal transition example:

- `DELIVERED -> PENDING`

Explicit transition modeling prevents this class of bugs from becoming runtime surprises.

12.3. Time And Ordering

Change introduces time.

Time introduces ordering problems.

12.3.1. Knowledge Engine Example

If relevance scores update after indexing events arrive out-of-order, ranking may oscillate unexpectedly.

12.3.2. Virtual World Example

If collision resolution happens before movement for some objects and after movement for others, simulation becomes nondeterministic.

12.3.3. Delivery Example

If reassignment happens before failure status commits, two robots may both claim the same task.

Modeling change means defining deterministic order and conflict policy.

12.4. Worked Design Path

Requirement:

> "When a robot fails mid-delivery, tasks should be reassigned automatically."

12.4.1. Step 1: Identify Transition Events

- robot heartbeat missed
- robot status set to failed
- active tasks discovered
- reassignment attempted

12.4.2. Step 2: Define Transition Preconditions

For task reassignment:

- task status is `IN_TRANSIT` or `FAILED`
- old robot is failed/unavailable
- candidate replacement exists

12.4.3. Step 3: Define Transition Postconditions

- task assigned to exactly one robot
- task status remains valid
- reassignment decision is auditable

12.4.4. Step 4: Define Failure Behavior

If no replacement exists:

- mark task as `FAILED`
- surface actionable reason

Do not leave ambiguous half-state.

12.4.5. Step 5: Add Idempotency Where Needed

If the same failure event is processed twice, outcome should remain consistent.

Idempotent transitions are a practical reliability multiplier.

12.5. Nex Implementation Sketch

```
class Task_Transition
feature
  status: String
  assigned_robot_id: String

  reassign(new_robot_id: String)
    require
      robot_present: new_robot_id /= ""
      transition_allowed: status = "IN_TRANSIT" or status =
    do
```

```
      assigned_robot_id := new_robot_id
      status := "IN_TRANSIT"
    ensure
      reassigned: assigned_robot_id = new_robot_id and status =
    end

  mark_delivered()
    require
      must_be_in_transit: status = "IN_TRANSIT"
    do
      status := "DELIVERED"
    ensure
      delivered: status = "DELIVERED"
    end
invariant
  valid_status:
    status = "PENDING" or status = "IN_TRANSIT" or status = "DEI
end
```

This keeps change rules explicit and enforceable.

12.6. Common Mistakes In Modeling Change

12.6.1. Mistake 1: Transition Logic Scattered Everywhere

Symptom:

- no single source of truth for legal changes

Recovery:

- centralize transitions in model operations/services
- codify pre/post contracts

12.6.2. Mistake 2: Silent Partial Failure

Symptom:

- operation updates one field, then exits on error

Recovery:

- define atomic boundaries where possible
- add compensation/fallback states where not possible

12.6.3. Mistake 3: Ignoring Event Reprocessing

Symptom:

- duplicate events create duplicate effects

Recovery:

- design idempotent transition handling
- track event identity when applicable

12.6.4. Mistake 4: Undefined Conflict Policy

Symptom:

- concurrent updates produce nondeterministic outcomes

Recovery:

- define winner rule (timestamp/version/priority)
- enforce ordering at model boundary

12.6.5. Mistake 5: No Change Audit Trail

Symptom:

- impossible to explain how state became invalid

Recovery:

- log transition intent and result
- retain failure reasons for diagnosis

Exercise Apply the section task and record your results before reading the solution notes.

12.7. Quick Exercise (10 Minutes)

For one entity in your system, define:

1. All legal states
2. Allowed transitions
3. One illegal transition
4. Preconditions for one transition

5. Postconditions for one transition
6. Failure behavior when transition cannot complete

Then ask: can this transition be safely retried?
If not, what must be added for idempotency?

12.8. Connection to Nex

Nex encourages explicit transition modeling with contracts and invariants close to operations.

That helps teams reason about change correctness before scaling complexity.

The broader lesson: model transitions as first-class design objects, not incidental code paths.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

12.9. Chapter Takeaways

- Most real failures occur during state transitions.
- Legal transitions must be explicit and enforced.
- Ordering, conflict policy, and idempotency are model concerns.
- Partial failure handling must be designed, not improvised.
- Reliable systems treat change as structured behavior.

Part II has now established a modeling foundation.

In Part III, we shift from representation to computation: what an algorithm is, how to decompose problems, and how to reason about algorithm behavior under load.

13. Studio 2 — The Model Redesign

13.1. Subtitle

Refactoring data models so systems can evolve without hidden state corruption.

Studio Focus This studio chapter is hands-on: implement, verify behavior, and document tradeoffs as you iterate.

13.2. 1. The Situation

Studio 1 proved we could ship tiny vertical slices.

Now those slices are under stress.

As soon as requirements grow, model weaknesses appear:

- entity identity is unclear (`"task"` vs `"route"` vs `"request"` used interchangeably)
- relationships are implicit (IDs inside free-text fields)
- transition rules are scattered (status changes happen from many places)
- invalid states are easy to create and hard to detect

Part II gave us the tools to fix this:

- model entities explicitly
- define relationships with constraints
- shape data around real operations
- model change with legal transitions and invariants

Your job in Studio 2 is to redesign all three systems using those principles.

13.3. 2. Engineering Brief

Redesign the core data models for all three systems (delivery, knowledge, virtual world).

Required outcomes:

- explicit entities with stable identity
- explicit relationships with clear cardinality/constraints
- transition operations with contracts (`require/ensure`)
- invariants that make invalid states hard to represent
- migrated behavior examples from Studio 1

Implementation guidance:

- optimize for correctness and clarity before optimization
- keep model boundaries obvious
- make failure behavior explicit (not silent)

13.4. 3. Implementation In Nex

Use Nex to encode model assumptions directly in code.

Suggested files:

- `delivery_model_v2.nex`
- `knowledge_model_v2.nex`
- `world_model_v2.nex`
- `studio_2_main.nex`

If using the web IDE, place all classes in one file and run `App.run`.

13.4.1. 3.1 Delivery Model V2

Focus: identity + legal transitions + assignment relationship.

```
class Delivery_Task
feature
  task_id: String
  origin: String
  destination: String
  status: String
  assigned_robot_id: String
```

```
  assign(robot_id: String)
    require
      robot_present: robot_id /= ""
      can_assign: status = "PENDING" or status = "FAILED"
    do
      assigned_robot_id := robot_id
      status := "IN_TRANSIT"
    ensure
      assigned: assigned_robot_id = robot_id and status = "I
    end

  complete()
    require
      in_transit: status = "IN_TRANSIT"
    do
      status := "DELIVERED"
    ensure
      delivered: status = "DELIVERED"
    end

  fail(reason: String)
    require
      reason_present: reason /= ""
      not_delivered: status /= "DELIVERED"
    do
      status := "FAILED"
    ensure
      failed: status = "FAILED"
    end
invariant
  id_present: task_id /= ""
  endpoints_present: origin /= "" and destination /= ""
  valid_status:
    status = "PENDING" or status = "IN_TRANSIT" or status =
  transit_has_assignment:
    status /= "IN_TRANSIT" or assigned_robot_id /= ""
end
```

13.4.2. 3.2 Knowledge Model V2

Focus: relationship as first-class entity (`Doc_Link`) rather than ad hoc references.

```
class Document
feature
  doc_id: String
  title: String
  body: String
invariant
  id_present: doc_id /= ""
  title_present: title /= ""
end

class Doc_Link
feature
  from_doc_id: String
  to_doc_id: String
  link_type: String

  is_valid(): Boolean do
    result :=
      from_doc_id /= "" and
      to_doc_id /= "" and
      from_doc_id /= to_doc_id and
      (link_type = "references" or link_type = "related" or link
  ensure
    bool_result: result = true or result = false
  end
invariant
  endpoints_present: from_doc_id /= "" and to_doc_id /= ""
  no_self_link: from_doc_id /= to_doc_id
end
```

13.4.3. 3.3 Virtual World Model V2

Focus: deterministic update transition with explicit bounds.

```
class World_Object
feature
  object_id: String
  x: Integer
  vx: Integer

  step(max_x: Integer)
    require
```

```
      valid_bound: max_x >= 0
    do
      let next: Integer := x + vx
      if next < 0 then
        x := 0
      elseif next > max_x then
        x := max_x
      else
        x := next
      end
    ensure
      bounded_after_step: x >= 0 and x <= max_x
    end
invariant
  id_present: object_id /= ""
end
```

13.4.4. 3.4 Studio Driver

Use one driver to validate nominal + edge behavior.

```
class App
feature
  run() do
    -- Delivery
    let t: Delivery_Task := create Delivery_Task
    t.task_id := "T-001"
    t.origin := "A"
    t.destination := "C"
    t.status := "PENDING"
    t.assigned_robot_id := ""
    t.assign("R-7")
    print(t.status)              -- IN_TRANSIT
    t.complete
    print(t.status)              -- DELIVERED

    -- Knowledge
    let l: Doc_Link := create Doc_Link
    l.from_doc_id := "D-1"
    l.to_doc_id := "D-2"
    l.link_type := "references"
    print(l.is_valid)            -- true
```

```
    -- World
    let w: World_Object := create World_Object
    w.object_id := "O-1"
    w.x := 9
    w.vx := 4
    w.step(10)
    print(w.x)                    -- 10 (clamped)
  end
end
```

13.5. 4. Studio Challenges

13.5.1. Level 1 — Core Implementation

- Redesign one system model (delivery or knowledge or world).
- Encode at least 2 invariants and 2 transition operations.
- Run one nominal and two edge cases.

13.5.2. Level 2 — Cross-System Model Consistency

- Redesign all three systems.
- Use a common modeling vocabulary in notes:
  - entity
  - relationship
  - invariant
  - transition
- Show one "before vs after" bug that the new model prevents.

13.5.3. Level 3 — Exploration

Pick one experiment:

- **State-centric vs event-centric** transition handling for delivery reassignment.
- **Direct references vs link entities** in the knowledge model.
- **Single-step deterministic update vs queued events** in virtual world updates.

Evaluate:

- complexity impact
- testability impact
- failure-mode clarity

13.6. 5. Postmortem

Discuss:

- Which entity boundary decision removed the most confusion?
- Which relationship constraint prevented a real invalid state?
- Which transition rule was hardest to make explicit?
- Which tradeoff did you intentionally accept (and why)?

Use concrete evidence from your runs, not just opinions.

13.7. Deliverables

- redesigned Nex models for delivery, knowledge, and world systems
- short modeling notes per system:
  - entities
  - relationships
  - invariants
  - transitions
- nominal + edge test output logs
- one before/after comparison showing reduced accidental complexity

---

13.8. Exit Criteria

You are ready for Studio 3 if:

- you can explain identity vs state for each core entity
- invalid states are blocked by model rules (not just comments)
- transition legality is encoded in operations/contracts
- relationship constraints are explicit and testable

Part III.

Part III — The Shape of Algorithms

14. Part III — The Shape of Algorithms — What Is an Algorithm?

14.1. 11. What Is an Algorithm?

Part II established models: entities, relationships, and constraints.

Part III turns those models into behavior.

An **algorithm** is a finite, explicit procedure that transforms valid input into output while preserving declared guarantees.

That definition matters because teams often blur three different things:

- a requirement ("find best route")
- an algorithm (BFS, Dijkstra, A*)
- an implementation detail (language, library, runtime)

When those are mixed together, correctness is hard to reason about and changes become expensive.

14.2. Algorithm vs Requirement vs Implementation

Keep these layers distinct:

- **Requirement**: what outcome the system must deliver.
- **Algorithm**: the step-by-step logic used to deliver it.
- **Implementation**: concrete code in Nex/JavaScript/Java and platform choices.

One requirement can have multiple valid algorithms. One algorithm can have multiple implementations.

Example:

- Requirement: "Return a valid path or report unreachable."
- Algorithm A: BFS for minimum hops in unweighted graphs.
- Algorithm B: Dijkstra for weighted cost.
- Implementation: generated JS for browser simulation, generated Java for backend service.

This separation keeps design conversations precise.

14.3. Correctness First

A procedure that "usually works" is not an algorithmic success.

An algorithm is acceptable only when its guarantees are explicit and checkable.

For our running systems:

14.3.1. Delivery

- Returned route uses valid, traversable links.
- If no route exists, report explicit failure (`UNREACHABLE`).

14.3.2. Knowledge

- Ranking follows declared scoring rules.
- Missing evidence is handled intentionally, not silently.

14.3.3. Virtual World

- Transition steps preserve world invariants.
- Update order is deterministic when required by design.

The right question is not "Does it run?" but "What does it guarantee?"

14.4. Worked Design Path

Requirement:

> "Find the best route quickly."

14.4.1. Step 1: Define valid input

- start and destination identifiers must exist
- graph snapshot must be internally consistent

14.4.2. Step 2: Define output contract

- `FOUND(path)` where each edge is legal
- or `UNREACHABLE`
- or `INVALID_INPUT`

14.4.3. Step 3: Define objective explicitly

For this version: "best" means minimum hops.

14.4.4. Step 4: Pick algorithm to match objective

Minimum hops on unweighted graph suggests BFS.

14.4.5. Step 5: Declare failure semantics

- unknown nodes -> `INVALID_INPUT`
- disconnected graph -> `UNREACHABLE`

14.4.6. Step 6: Encode in contracts

- `require`: inputs are present
- `ensure`: output status is always valid

Now the team has algorithmic intent, not vague search behavior.

14.5. Nex Implementation Sketch

```
class Route_Result
feature
  status: String
  path: String
invariant
  valid_status:
    status = "FOUND" or
```

```
    status = "UNREACHABLE" or
    status = "INVALID_INPUT"
end

class Route_Algorithm
feature
  compute(start_loc, dest_loc: String): Route_Result
    require
      inputs_present: start_loc /= "" and dest_loc /= ""
    do
      let r: Route_Result := create Route_Result

      if start_loc = dest_loc then
        r.status := "FOUND"
        r.path := start_loc
      elseif start_loc = "A" and dest_loc = "C" then
        r.status := "FOUND"
        r.path := "A->B->C"
      else
        r.status := "UNREACHABLE"
        r.path := ""
      end

      result := r
    ensure
      status_is_declared:
        result.status = "FOUND" or
        result.status = "UNREACHABLE" or
        result.status = "INVALID_INPUT"
    end
end
```

The sketch is intentionally small. The chapter point is the contract shape around the algorithm, not graph implementation details.

14.6. Common Mistakes

14.6.1. Mistake 1: Calling any script an algorithm

Symptom:

- behavior depends on accidental control flow

Recovery:

- write language-agnostic steps first
- state guarantees before coding

14.6.2. Mistake 2: Undefined failure behavior

Symptom:

- empty/null outputs with no meaning

Recovery:

- declare explicit failure statuses
- test each status path

14.6.3. Mistake 3: Ambiguous objective

Symptom:

- team argues later about what "best" meant

Recovery:

- encode one objective now
- document deferred objectives

Exercise Apply the section task and record your results before reading the solution notes.

14.7. Quick Exercise (8-10 Minutes)

Choose one operation in your project and write:

1. valid input definition
2. output contract (success + failure)
3. objective function
4. 5-step language-agnostic algorithm
5. one contract you will enforce in code

If two engineers would implement your sketch differently, the algorithm description is still too loose.

14.8. Connection to Nex

Nex makes algorithm intent concrete because contracts and invariants live next to behavior.

This reinforces a reliable engineering sequence:

- specify
- implement
- verify

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

14.9. Chapter Takeaways

- Algorithms are procedures with explicit guarantees, not just executable code.
- Requirement, algorithm, and implementation should stay separate.
- Correctness and failure semantics must be declared up front.
- Contracts are a practical mechanism for preserving algorithm intent.

In Chapter 12, we study decomposition: how to break hard algorithmic work into manageable, testable pieces.

15. Part III — The Shape of Algorithms — Breaking Problems Apart

15.1. 12. Breaking Problems Apart

Complex algorithms rarely fail because a single line is wrong.

They fail because too many responsibilities are packed into one block, making behavior hard to reason about and harder to change safely.

Decomposition is the discipline of splitting a problem into subproblems with clear interfaces and guarantees.

A strong decomposition gives you:

- local reasoning (understand one piece at a time)
- independent testing (verify each stage in isolation)
- replaceability (swap strategy without rewriting everything)

15.2. A Practical Decomposition Rule

For each part, be able to answer three questions:

1. What input does it require?
2. What output does it guarantee?
3. What is the one responsibility it owns?

If a part has multiple reasons to change, decomposition is incomplete.

15.3. Common Decomposition Patterns

15.3.1. Pipeline

A sequence of stages where output from stage `n` feeds stage `n+1`.
Good for search, ranking, transformation, and simulation loops.

15.3.2. Strategy Boundary

Stable interface, swappable algorithm.
Good when objective or scale may evolve.

15.3.3. Guard -> Core -> Commit

- validate assumptions
- perform core computation
- apply state changes

Good for state transitions with side effects.

15.3.4. Domain vs Infrastructure Split

Keep algorithm logic separate from storage/transport/UI concerns.
Good for testability and maintainability.

15.4. Worked Design Path

Requirement:

> "Return top relevant notes for a query."

Naive implementation often mixes all concerns in one method:

- parsing
- candidate generation
- scoring
- ranking
- filtering
- formatting

A decomposed design:

1. `tokenize_query(query)`

2. `collect_candidates(tokens, index)`
3. `score_candidate(candidate, tokens)`
4. `rank_candidates(scored)`
5. `filter_by_threshold(ranked, min_score)`
6. `render_results(filtered)`

Now each stage has explicit contracts and can be tested independently.

15.5. Nex Implementation Sketch

```
class Search_Algorithm
feature
  tokenize(query: String): String
    require
      query_present: query /= ""
    do
      result := query
    ensure
      non_empty_tokens: result /= ""
    end

  score(doc_text, tokens: String): Integer
    require
      inputs_present: doc_text /= "" and tokens /= ""
    do
      if doc_text = tokens then
        result := 100
      else
        result := 10
      end
    ensure
      non_negative: result >= 0
    end

  choose_top(doc1, doc2, query: String): String
    require
      docs_present: doc1 /= "" and doc2 /= "" and query /= "
    do
      let t: String := tokenize(query)
```

```
      let s1: Integer := score(doc1, t)
      let s2: Integer := score(doc2, t)

      if s1 >= s2 then
        result := doc1
      else
        result := doc2
      end
    ensure
      from_inputs: result = doc1 or result = doc2
    end
end
```

Even this small sketch shows decomposition: tokenization and scoring are separate, then composed.

15.6. Common Mistakes

15.6.1. Mistake 1: Decomposing by syntax, not responsibility

Symptom:

- helper names like `process`, `handle`, `do_stuff`

Recovery:

- rename functions by responsibility
- enforce one reason to change per stage

15.6.2. Mistake 2: Over-fragmentation

Symptom:

- many tiny wrappers with no semantic value

Recovery:

- merge layers that do not improve reasoning

15.6.3. Mistake 3: Hidden coupling between stages

Symptom:

- stage B depends on internals of stage A

Recovery:

- pass explicit values only
- eliminate implicit shared state where possible

15.6.4. Mistake 4: Missing stage contracts

Symptom:

- malformed intermediate data causes late failures

Recovery:

- add pre/post conditions at key boundaries

Exercise Apply the section task and record your results before reading the solution notes.

15.7. Quick Exercise (10 Minutes)

Take one large function in your codebase and decompose it into 3-6 stages.

For each stage, write:

1. input contract
2. output contract
3. failure behavior
4. one test case

Then identify one stage you could replace without touching others.

If nothing is replaceable, coupling is still too strong.

15.8. Connection to Nex

Nex contracts are especially useful at decomposition boundaries because they make handoff assumptions explicit and executable.

This is also where AI-assisted coding improves: clear stage contracts reduce incorrect glue code.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

15.9. Chapter Takeaways

- Decomposition is an engineering necessity, not style preference.
- Each stage should have one responsibility and explicit contracts.
- Replaceable components require stable interfaces.
- Good decomposition lowers change risk and improves test quality.

In Chapter 13, we apply decomposition to self-similar problems through recursion.

16. Part III — The Shape of Algorithms — Thinking Recursively

16.1. 13. Thinking Recursively

Recursion is not an advanced trick.

It is a decomposition approach for problems where the structure of the whole resembles the structure of its parts.

A recursive algorithm has two essential components:

1. a base case solved directly
2. a reduction step that transforms the problem into smaller instances

If either component is weak, recursion becomes fragile.

16.2. When Recursion Is Appropriate

Recursion is often a good fit when data is hierarchical or graph-like:

- nested structures
- trees
- bounded graph traversals
- divide-and-conquer problems

Recursion is often a poor fit for simple linear workflows where iteration is clearer and cheaper.

Choose recursion based on problem shape, not style preference.

16.3. The Base Case Is the Safety Boundary

Most recursion bugs come from incorrect base cases or missing progress.

A robust recursive design must prove:

- base case is reachable
- each recursive step moves toward base case
- output contract holds for both base and recursive branches

For graph-like domains, you also need cycle guards (`visited` set, depth limit, or both).

16.4. Worked Design Path

Requirement:

> "Count reachable locations from a starting node."

Recursive DFS concept:

1. if node already visited -> return 0
2. mark node visited
3. total := 1
4. for each neighbor -> total := total + count(neighbor)
5. return total

This design is concise, but only safe with explicit visited tracking.

16.5. Nex Implementation Sketch

```
class Reachability
feature
  visited: String

  count_from(node: String): Integer
    require
      node_present: node /= ""
```

```
    do
      -- Teaching sketch: visited is encoded as a simple str
      -- A full implementation would use a dedicated collect
      if visited = node then
        result := 0
      elseif node = "Z" then
        visited := node
        result := 1
      else
        visited := node
        result := 1 + count_from("Z")
      end
    ensure
      non_negative_count: result >= 0
    end
end
```

This snippet emphasizes recursive structure, not production-grade graph storage.

16.6. Recursion Across the Three Systems

16.6.1. Delivery

- explore alternate path branches
- must guard against route cycles

16.6.2. Knowledge

- traverse related-document networks with depth bounds
- must avoid revisiting nodes

16.6.3. Virtual World

- process hierarchical containment (zone -> region -> object)
- must maintain deterministic traversal policy

In all three, recursion without bounds is unsafe.

16.7. Common Mistakes

16.7.1. Mistake 1: Missing base case

Symptom:

- infinite recursion

Recovery:

- write and test base case first

16.7.2. Mistake 2: No progress metric

Symptom:

- recursive call does not reduce problem size

Recovery:

- define a decreasing metric (depth, remaining nodes, input size)

16.7.3. Mistake 3: Ignoring cycles

Symptom:

- repeated traversal of same nodes

Recovery:

- add visited tracking
- optionally add max-depth guard

16.7.4. Mistake 4: Recomputing identical subproblems

Symptom:

- exponential runtime

Recovery:

- memoize repeated subproblems
- consider dynamic programming if needed

Exercise Apply the section task and record your results before reading the solution notes.

16.8. Quick Exercise (10 Minutes)

Design one recursive routine for your project and write:

1. base case
2. progress metric
3. recursive step
4. cycle guard
5. worst-case depth assumption

Then test with:

- trivial input
- normal input
- adversarial cyclic input

If adversarial input is unbounded, redesign before implementation.

16.9. Connection to Nex

Nex contracts make recursive assumptions explicit and checkable, especially around base-case guarantees and non-negative/valid outputs.

That tightens correctness early and reduces latent recursion bugs.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

16.10. Chapter Takeaways

- Recursion is decomposition over self-similar structure.
- Base case and progress metric are non-negotiable.
- Graph recursion requires cycle and depth controls.
- Elegant recursion is not enough; bounded recursion is the goal.

In Chapter 14, we shift from correctness to cost: how algorithm behavior changes as inputs grow.

17. Part III — The Shape of Algorithms — Measuring Algorithm Behavior

17.1. 14. Measuring Algorithm Behavior

An algorithm can be correct and still fail in production.

Why? Because correctness alone does not guarantee acceptable behavior under real load.

Algorithm engineering always has two gates:

1. correctness
2. cost behavior at expected and adverse scale

This chapter focuses on the second gate.

17.2. What To Measure

Performance is a profile, not a single number.

You typically care about:

- latency (single-operation response time)
- throughput (work per time unit)
- memory footprint
- worst-case behavior
- tail behavior (P95/P99)

System priorities differ:

17.2.1. Delivery

- route recomputation latency during disruptions

17.2.2. Knowledge

- query latency and ranking throughput

17.2.3. Virtual World

- stable per-tick budget and bounded spikes

17.3. Asymptotics + Measurement

Asymptotic analysis is necessary, but insufficient.

Use Big-O to compare growth trends and avoid structurally bad choices. Use benchmarks/profiling to capture constants, cache effects, and real workload shape.

Example:

- linear scan may be faster for very small `N`
- indexed lookup dominates as `N` grows

Good decisions require both model and evidence.

17.4. Worked Design Path

Requirement:

> "Return top 10 relevant documents quickly for every query."

17.4.1. Step 1: Candidate approaches

- full scan and score all docs
- indexed candidate generation then score subset

17.4.2. Step 2: Cost hypothesis

- full scan: often `O(N)` per query
- index-first: closer to `O(k)` scoring where `k << N` for many workloads

17.4.3. Step 3: Explicit targets

- P95 latency (for example, <150ms)
- memory budget for index
- update-time budget for ingest

17.4.4. Step 4: Benchmark plan

- normal case distribution
- heavy-tail queries
- adversarial spike case

17.4.5. Step 5: Fallback behavior

When candidate set explodes:

- apply bounded heuristic mode
- return partial but valid result with declared status

17.5. Nex Implementation Sketch

```
class Strategy_Selector
feature
  estimate_linear(n: Integer): Integer
    require
      non_negative_n: n >= 0
    do
      result := n
    ensure
      non_negative_result: result >= 0
    end

  estimate_indexed(k: Integer): Integer
    require
      non_negative_k: k >= 0
    do
      result := k
    ensure
      non_negative_result: result >= 0
    end
```

```
  choose(n, k: Integer): String
    require
      valid_inputs: n >= 0 and k >= 0
    do
      if estimate_indexed(k) < estimate_linear(n) then
        result := "INDEXED"
      else
        result := "LINEAR"
      end
    ensure
      known_strategy: result = "INDEXED" or result = "LINEAR"
    end
end
```

This is intentionally simple, but it shows a key habit: strategy selection is explicit and testable.

17.6. Measurement Checklist

Before claiming an algorithm is fast enough, answer:

1. What is the real input size distribution?
2. What are normal, edge, and adversarial cases?
3. Which metric is the actual bottleneck (latency, throughput, memory, tail)?
4. What does worst-case behavior look like?
5. What fallback keeps behavior bounded under stress?

Without these answers, "fast enough" is an assumption.

17.7. Common Mistakes

17.7.1. Mistake 1: Optimizing before correctness

Symptom:

- fast but invalid outputs

Recovery:

- lock correctness guarantees first
- optimize under contract-preserving tests

17.7.2. Mistake 2: Treating Big-O as full truth

Symptom:

- ignores constants and workload realities

Recovery:

- pair asymptotics with empirical measurement

17.7.3. Mistake 3: Benchmarking toy inputs only

Symptom:

- production behavior surprises team

Recovery:

- benchmark realistic and adversarial distributions

17.7.4. Mistake 4: Ignoring tail latency

Symptom:

- acceptable average, poor user experience

Recovery:

- monitor P95/P99 and outlier paths

17.7.5. Mistake 5: No capacity narrative

Symptom:

- unknown breakpoints under growth

Recovery:

- document expected growth envelope and redesign trigger

Exercise Apply the section task and record your results before reading the solution notes.

17.8. Quick Exercise (12 Minutes)

Pick one algorithm and create a cost brief:

1. primary operation
2. expected input range
3. asymptotic estimate
4. dominant constant factors
5. benchmark design (normal + edge + spike)
6. fallback policy
7. redesign trigger condition

Share it with a teammate. If they cannot predict failure mode, your brief is incomplete.

17.9. Connection to Nex

Nex keeps cost-related decisions close to code and contracts, making strategy assumptions visible and easier to validate.

The transferable lesson is language-independent: measure behavior against model assumptions before scale forces emergency redesign.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

17.10. Chapter Takeaways

- Correctness and cost are dual constraints.
- Asymptotic reasoning guides; measurement validates.
- Tail behavior matters as much as average behavior.
- Strategy choices should be explicit, testable, and workload-driven.
- Capacity planning is part of algorithm design.

Part III gave us the algorithmic toolkit: definition, decomposition, recursion, and cost reasoning.

Part IV now focuses on data structures that make these algorithms practical at scale.

18. Algorithm Lab — The First Experiments

18.1. Subtitle

Testing algorithm ideas on small systems before formal complexity notation.

Lab Focus Run controlled experiments, collect evidence, and compare algorithm choices under explicit assumptions.

18.2. 1. Why This Lab Exists

Part III introduced algorithm thinking:

- define guarantees
- decompose work
- use recursion when structure fits
- reason about behavior under growth

Now we run controlled experiments.

The goal is not to chase micro-optimizations. The goal is to build engineering intuition by observing how different algorithm choices behave on the same task.

18.3. 2. Lab Objectives

By the end of this lab, you should be able to:

- compare two algorithms for the same requirement
- collect simple evidence (operation counts, not just wall-clock time)
- explain when an approach is acceptable and when it breaks down
- preserve correctness contracts while changing strategy

18.4. 3. Experiment Theme

Use one task, multiple strategies:

> "Find a target value in a sorted sequence."

Why this task:

- easy correctness criteria
- easy to vary input size
- clear contrast between algorithm behaviors

Strategies:

- linear scan
- binary search

We will track an explicit step counter to observe behavior directly.

18.5. 4. Nex Implementation

Suggested file names:

- `algorithm_lab_1.nex`
- `algorithm_lab_1_main.nex`

If using the web IDE, place everything in one file and run `App.run`.

18.5.1. 4.1 Result Type

```
class Search_Result
feature
  found: Boolean
  index: Integer
  steps: Integer
invariant
  index_valid: index >= -1
  steps_non_negative: steps >= 0
end
```

18.5.2. 4.2 Linear Search With Step Counter

```
class Linear_Search_Algo
feature
  find(a1, a2, a3, a4, a5, a6, a7, a8, target: Integer): Sea
    do
      let r: Search_Result := create Search_Result
      r.found := false
      r.index := -1
      r.steps := 0

      r.steps := r.steps + 1
      if a1 = target then
        r.found := true
        r.index := 0
      elseif a2 = target then
        r.steps := r.steps + 1
        r.found := true
        r.index := 1
      elseif a3 = target then
        r.steps := r.steps + 1
        r.found := true
        r.index := 2
      elseif a4 = target then
        r.steps := r.steps + 1
        r.found := true
        r.index := 3
      elseif a5 = target then
        r.steps := r.steps + 1
        r.found := true
        r.index := 4
      elseif a6 = target then
        r.steps := r.steps + 1
        r.found := true
        r.index := 5
      elseif a7 = target then
        r.steps := r.steps + 1
        r.found := true
        r.index := 6
      elseif a8 = target then
        r.steps := r.steps + 1
        r.found := true
```

```
        r.index := 7
      else
        r.steps := r.steps + 1
      end

      result := r
    ensure
      steps_recorded: result.steps >= 1
      valid_index: result.index >= -1 and result.index <= 7
    end
end
```

18.5.3. 4.3 Binary Search With Step Counter

```
class Binary_Search_Algo
feature
  find(a1, a2, a3, a4, a5, a6, a7, a8, target: Integer): Search_
    require
      sorted_input:
        a1 <= a2 and a2 <= a3 and a3 <= a4 and
        a4 <= a5 and a5 <= a6 and a6 <= a7 and a7 <= a8
    do
      let r: Search_Result := create Search_Result
      r.found := false
      r.index := -1
      if a4 = target then
        r.steps := 1
        r.found := true
        r.index := 3
      elseif target < a4 and a2 = target then
        r.steps := 2
        r.found := true
        r.index := 1
      elseif target < a2 and a1 = target then
        r.steps := 3
        r.found := true
        r.index := 0
      elseif target > a2 and target < a4 and a3 = target then
        r.steps := 3
        r.found := true
        r.index := 2
```

```
      elseif target > a4 and a6 = target then
        r.steps := 2
        r.found := true
        r.index := 5
      elseif target > a4 and target < a6 and a5 = target the
        r.steps := 3
        r.found := true
        r.index := 4
      elseif target > a6 and a7 = target then
        r.steps := 3
        r.found := true
        r.index := 6
      elseif target > a7 and a8 = target then
        r.steps := 4
        r.found := true
        r.index := 7
      else
        r.steps := 4
      end

      result := r
    ensure
      steps_bounded: result.steps >= 1 and result.steps <= 4
      valid_index: result.index >= -1 and result.index <= 7
    end
end
```

18.5.4. 4.4 Driver: Compare Behaviors

```
class App
feature
  run() do
    let lin: Linear_Search_Algo := create Linear_Search_Algo
    let bin: Binary_Search_Algo := create Binary_Search_Algo

    let l_hit: Search_Result := lin.find(3, 7, 10, 14, 19, 2
    let b_hit: Search_Result := bin.find(3, 7, 10, 14, 19, 2

    let l_miss: Search_Result := lin.find(3, 7, 10, 14, 19,
    let b_miss: Search_Result := bin.find(3, 7, 10, 14, 19,
```

```
    print("Linear hit steps: " + l_hit.steps)
    print("Binary hit steps: " + b_hit.steps)
    print("Linear miss steps: " + l_miss.steps)
    print("Binary miss steps: " + b_miss.steps)
  end
end
```

Expected pattern:

- both algorithms are correct
- binary search usually uses fewer steps on sorted inputs
- binary search depends on stronger preconditions (`sorted_input`)

18.6. 5. Lab Tasks

18.6.1. Task A — Correctness Check

Run both strategies with:

- first element target
- middle element target
- last element target
- absent target

Record:

- found/not-found
- returned index
- steps

18.6.2. Task B — Constraint Violation

Call binary search with unsorted inputs and observe contract behavior.

Question:

- what failure signal do you get?
- how is this better than silent wrong results?

18.6.3. Task C — Extension

Add one more strategy for comparison:

- jump search, or
- interpolation-inspired search (if values are uniform)

Keep the same `Search_Result` format so results remain comparable.

18.7. 6. Reflection Prompts

Use evidence from your run logs:

1. Which strategy has stricter input assumptions?
2. Which strategy is easier to reason about for correctness?
3. At what scale would you stop using linear search for this task?
4. Which contract prevented the most dangerous misuse?

18.8. 7. Deliverables

- runnable Nex code for at least two strategies
- output log showing hit/miss comparisons
- short write-up of strategy tradeoffs
- one paragraph on how contracts influenced your design

18.9. 8. Forward Link

This lab gives you concrete intuition for why data structures matter.

In Part IV, we study lists, sets/maps, trees, and graphs so algorithm choices are supported by the right representation.

Part IV.

Part IV — Organizing Data

19. Part IV — Organizing Data — Lists and Sequences

19.1. 15. Lists and Sequences

Part III gave us algorithmic thinking.

Part IV focuses on the data structures those algorithms run on.

We begin with lists and sequences because they are the default structure most teams reach for first.

That default is sometimes right and often costly.

19.2. What Lists Are Good At

A list is an ordered collection.

That order gives you clear strengths:

- preserve insertion or logical order
- iterate predictably from start to end
- represent timelines, queues, logs, and ranked output

Lists work well when operations are mostly sequential and the primary question is:

- "What comes next?"

19.3. Where Lists Hurt

Lists become expensive when you repeatedly need:

- fast membership checks
- random lookups by key
- frequent insert/delete in the middle

If the dominant operation is "find by id" and your structure is a list, performance will degrade as data grows.

This is the first scaling failure many systems hit.

19.4. Worked Design Path

Requirement:

> "Show active delivery tasks in creation order and fetch one task by id quickly."

19.4.1. Step 1: Separate operations

- operation A: ordered display
- operation B: id-based lookup

19.4.2. Step 2: Start with naive list-only design

- `tasks: [task]`
- display is easy
- lookup is linear scan

19.4.3. Step 3: Identify cost risk

At small scale, linear scan is fine. At larger scale, repeated lookup dominates latency.

19.4.4. Step 4: Keep list for order, plan index for lookup

This chapter keeps list-first implementation to expose the tradeoff clearly. Chapter 16 introduces sets/maps to solve lookup cost.

19.4.5. Step 5: Add contracts for behavior

- `append` guarantees order is preserved
- `find` guarantees either valid task or explicit missing result

---

19.5. Nex Implementation Sketch

```
class Task
feature
  id: String
  status: String
invariant
  id_present: id /= ""
  valid_status:
    status = "PENDING" or
    status = "IN_TRANSIT" or
    status = "DELIVERED" or
    status = "FAILED"
end

class Task_Sequence
feature
  t1: Task
  t2: Task
  t3: Task

  find_by_id(task_id: String): String
    require
      id_present: task_id /= ""
    do
      if t1.id = task_id then
        result := t1.status
      elseif t2.id = task_id then
        result := t2.status
      elseif t3.id = task_id then
        result := t3.status
      else
        result := "NOT_FOUND"
      end
    ensure
      declared_result:
        result = "PENDING" or
        result = "IN_TRANSIT" or
        result = "DELIVERED" or
        result = "FAILED" or
        result = "NOT_FOUND"
    end
```

```
  ordered_ids(): String
    do
      result := t1.id + " -> " + t2.id + " -> " + t3.id
    ensure
      non_empty: result /= ""
    end
end
```

This sketch is intentionally small and list-like. It makes sequence behavior visible while exposing lookup cost.

19.6. Lists in the Three Running Systems

19.6.1. Delivery

- delivery tasks by creation or dispatch order

19.6.2. Knowledge

- ranked search results in score order

19.6.3. Virtual World

- deterministic update order for entities per tick

In each case, lists capture order well but may not satisfy fast keyed access.

19.7. Common Mistakes

19.7.1. Mistake 1: Using lists for everything

Symptom:

- repeated linear scans dominate runtime

Recovery:

- profile dominant operations
- add indexes or alternate structures where needed

19.7.2. Mistake 2: Confusing order with identity lookup

Symptom:

- "find by id" implemented as repeated sequence traversal

Recovery:

- keep list for order
- introduce map/set for identity operations

19.7.3. Mistake 3: Silent duplicate identity

Symptom:

- same logical entity appears multiple times without policy

Recovery:

- define uniqueness rules
- enforce during append/ingest

Exercise Apply the section task and record your results before reading the solution notes.

19.8. Quick Exercise (10 Minutes)

For one feature in your project:

1. list the top three operations on its collection
2. mark each as order-sensitive or key-sensitive
3. estimate operation frequency
4. decide whether list-only is sufficient

Then write one contract that would prevent the most likely misuse.

19.9. Connection to Nex

Nex contracts help define what sequence operations guarantee, so behavior remains clear when the underlying structure changes later.

This allows incremental evolution from list-only designs to hybrid designs without losing correctness.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

19.10. Chapter Takeaways

- Lists are strong for order-preserving workflows.
- Lists are weak for repeated key-based lookup at scale.
- Distinguish sequence operations from identity operations early.
- Contracts make later structure refactors safer.

In Chapter 16, we introduce sets and maps to solve membership and lookup efficiently.

20. Part IV — Organizing Data — Sets and Maps

20.1. 16. Sets and Maps

Chapter 15 showed why list-only designs eventually slow down.

This chapter introduces two structures built for direct access:

- **set**: fast membership (is this present?)
- **map**: fast retrieval by key (get value by identity)

These structures are foundational for scalable software.

20.2. Sets vs Maps

Think in terms of the question you ask most often.

Use a set when the question is:

- "Have we seen this already?"

Use a map when the question is:

- "Give me the object for this key."

If your dominant query is keyed retrieval and you are scanning a list, you are paying avoidable cost.

20.3. Key Design Matters

Maps and sets are only as good as their keys.

A good key is:

- stable over time

- unique for the intended scope
- cheap to compare

Weak key design causes collisions, accidental overwrites, and stale lookups.

20.4. Worked Design Path

Requirement:

> "For each query, avoid re-scoring the same document and fetch metadata by id quickly."

20.4.1. Step 1: Identify operations

- dedupe candidate ids -> membership
- fetch document by id -> keyed lookup

20.4.2. Step 2: Choose structures

- `seen_ids` as set
- `documents_by_id` as map

20.4.3. Step 3: Define contracts

- adding id to set preserves uniqueness semantics
- map lookup returns document or explicit `NOT_FOUND`

20.4.4. Step 4: Preserve display behavior

If result order still matters, keep an ordered list for output and use set/map for control/lookups.

This hybrid pattern is common in real systems.

20.5. Nex Implementation Sketch

```
class Doc_Record
feature
  doc_id: String
  title: String
invariant
  id_present: doc_id /= ""
  title_present: title /= ""
end

class Doc_Index
feature
  k1: String
  v1: Doc_Record
  k2: String
  v2: Doc_Record
  k3: String
  v3: Doc_Record

  contains(doc_id: String): Boolean
    require
      id_present: doc_id /= ""
    do
      result := doc_id = k1 or doc_id = k2 or doc_id = k3
    ensure
      bool_result: result = true or result = false
    end

  fetch_title(doc_id: String): String
    require
      id_present: doc_id /= ""
    do
      if doc_id = k1 then
        result := v1.title
      elseif doc_id = k2 then
        result := v2.title
      elseif doc_id = k3 then
        result := v3.title
      else
        result := "NOT_FOUND"
      end
    ensure
      non_empty: result /= ""
```

```
    end
end
```

This chapter sketch emulates map/set behavior in a minimal teaching form.

20.6. Sets and Maps in the Three Systems

20.6.1. Delivery

- set: blocked locations visited this cycle
- map: `task_id -> task`

20.6.2. Knowledge

- set: dedupe candidate document ids
- map: `doc_id -> document metadata`

20.6.3. Virtual World

- set: active entities in current frame
- map: `entity_id -> entity state`

The pattern is consistent: membership control + keyed state access.

20.7. Common Mistakes

20.7.1. Mistake 1: Bad key choice

Symptom:

- updates overwrite wrong record

Recovery:

- define key policy explicitly
- use stable identity keys only

20.7.2. Mistake 2: Assuming order from map/set

Symptom:

- inconsistent user-facing output order

Recovery:

- keep ordered sequence separately when needed

20.7.3. Mistake 3: Duplicate source of truth

Symptom:

- list and map diverge silently

Recovery:

- define one authoritative write path
- verify sync invariants in tests

20.7.4. Mistake 4: Treating missing as impossible

Symptom:

- null or crash on lookup miss

Recovery:

- define explicit miss behavior (`NOT_FOUND`/option type)

Exercise Apply the section task and record your results before reading the solution notes.

20.8. Quick Exercise (10-12 Minutes)

Pick one list-scan hotspot and redesign it using set/map semantics.

Write:

1. chosen key
2. membership operation
3. keyed lookup operation
4. miss behavior
5. one invariant that protects index consistency

Then compare before/after on at least one medium-size input.

20.9. Connection to Nex

Nex contracts make key and lookup assumptions explicit, which is critical when data structures become central to correctness and speed.

This is where many scaling bugs are prevented early.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

20.10. Chapter Takeaways

- Sets answer membership questions efficiently.
- Maps answer keyed retrieval questions efficiently.
- Key design is a correctness decision, not just a performance detail.
- Ordered output often requires sequence + map/set hybrid designs.

In Chapter 17, we move to trees, where hierarchy and ordered search behavior become first-class concerns.

21. Part IV — Organizing Data — Trees: Structured Data

21.1. 17. Trees: Structured Data

Lists give order. Maps and sets give direct access.

Trees add hierarchy.

A tree is a structure where parent-child relationships represent layered organization and enable structured search.

Trees are useful when relationships are naturally nested and queries can exploit that shape.

21.2. Why Trees Matter

Trees can make operations faster and clearer when:

- data has hierarchical meaning (category -> subcategory -> item)
- you need ordered search over keys
- you need bounded-depth traversal rather than full scans

Common tree use cases include indexes, configuration scopes, scene graphs, and expression structures.

21.3. Tree Invariants

A tree is only useful if its structure remains valid.

Typical invariants include:

- no cycles
- every non-root node has exactly one parent
- ordering rule for search trees (left < node < right)

Without structural invariants, tree logic degrades into graph-like ambiguity.

21.4. Worked Design Path

Requirement:

> "Organize notes by topic hierarchy and find the first matching topic quickly."

21.4.1. Step 1: Identify hierarchy

- root topic
- subtopics
- leaf notes

21.4.2. Step 2: Choose tree representation

A simple teaching shape:

- node with key
- optional left child
- optional right child

21.4.3. Step 3: Define lookup behavior

- if key matches current node -> return found
- if target smaller -> traverse left
- otherwise -> traverse right

21.4.4. Step 4: Define miss behavior

- return explicit `NOT_FOUND`

21.4.5. Step 5: Preserve invariants on insert/update

- keep ordering rule valid
- reject duplicate policy violations where required

21.5. Nex Implementation Sketch

```
class Topic_Node
feature
  key: Integer
  label: String
  left_key: Integer
  right_key: Integer
invariant
  label_present: label /= ""
  no_self_child: left_key /= key and right_key /= key
end

class Topic_Tree
feature
  root: Topic_Node
  left: Topic_Node
  right: Topic_Node

  find_label(target: Integer): String
    do
      if target = root.key then
        result := root.label
      elseif target < root.key and target = left.key then
        result := left.label
      elseif target > root.key and target = right.key then
        result := right.label
      else
        result := "NOT_FOUND"
      end
    ensure
      non_empty: result /= ""
    end
end
```

This minimal sketch demonstrates tree-style branching decisions and explicit miss semantics.

21.6. Trees in the Three Systems

21.6.1. Delivery

- region hierarchy for dispatch zones

21.6.2. Knowledge

- taxonomy and topic classification trees

21.6.3. Virtual World

- scene graph or containment hierarchy

In each case, hierarchy encodes meaning, not just storage.

21.7. Common Mistakes

21.7.1. Mistake 1: Using trees without hierarchy need

Symptom:

- unnecessary complexity over flat keyed access

Recovery:

- validate that operations truly benefit from parent-child structure

21.7.2. Mistake 2: Ignoring balancing concerns

Symptom:

- deep skewed trees behave like linked lists

Recovery:

- monitor depth distribution
- consider balanced variants when needed

21.7.3. Mistake 3: Violating ordering invariants

Symptom:

- search misses existing values

Recovery:

- enforce insertion/update rules
- add invariant-focused tests

21.7.4. Mistake 4: Confusing tree with graph

Symptom:

- multiple parents/cycles appear silently

Recovery:

- enforce single-parent and acyclic constraints

Exercise Apply the section task and record your results before reading the solution notes.

21.8. Quick Exercise (10 Minutes)

Model one small hierarchy in your domain as a tree.

Write:

1. node identity key
2. parent-child rule
3. one search operation
4. one structural invariant
5. one invalid state to reject

Then explain why a map-only model would lose useful structure.

21.9. Connection to Nex

Nex contracts and invariants are a strong fit for tree structure because they keep parent-child and ordering assumptions explicit.

This is critical as tree code grows beyond toy depth.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

21.10. Chapter Takeaways

- Trees represent hierarchy with structured search paths.
- Tree invariants are mandatory for correctness.
- Unbalanced trees can erase expected performance gains.
- Use trees when structure carries semantic value.

In Chapter 18, we generalize from hierarchy to networks using graphs.

22. Part IV — Organizing Data — Graphs: Networks of Everything

22.1. 18. Graphs: Networks of Everything

Trees model hierarchy.

Graphs model networks.

A graph represents entities as nodes and relationships as edges, allowing cycles, cross-links, and many-to-many structure.

Most real systems eventually require graph thinking.

22.2. Why Graphs Are Essential

Graphs appear whenever relationships are not strictly hierarchical:

- roads between locations
- citations between documents
- interactions between world entities

If you force these domains into lists or trees only, you lose expressive power and introduce fragile workarounds.

22.3. Core Graph Questions

Graph-driven systems frequently ask:

- reachability: can we get from A to B?
- path quality: what is the best route by some objective?
- neighborhood: what is connected to this node?
- connectivity changes: what broke when an edge disappeared?

Data structure choice should follow these queries.

22.4. Worked Design Path

Requirement:

> "Given two locations, return a valid route or report unreachable."

22.4.1. Step 1: Define graph elements

- node: location id
- edge: traversable connection between two locations

22.4.2. Step 2: Define correctness contract

- returned path uses only valid edges
- first node equals source
- last node equals destination
- failure is explicit (`UNREACHABLE`)

22.4.3. Step 3: Pick traversal for first version

- BFS for minimum hops in unweighted graph

22.4.4. Step 4: Add safety rules

- visited tracking to avoid cycles
- guard unknown nodes as invalid input

22.4.5. Step 5: Keep representation explicit

- adjacency structure as first-class model element

22.5. Nex Implementation Sketch

```
class Route_Graph
feature
  -- Teaching-sized adjacency for four nodes.
  a_to_b: Boolean
  b_to_c: Boolean
  c_to_d: Boolean
  a_to_d: Boolean

  has_edge(from_node, to_node: String): Boolean
    require
      non_empty_nodes: from_node /= "" and to_node /= ""
    do
      if from_node = "A" and to_node = "B" then
        result := a_to_b
      elseif from_node = "B" and to_node = "C" then
        result := b_to_c
      elseif from_node = "C" and to_node = "D" then
        result := c_to_d
      elseif from_node = "A" and to_node = "D" then
        result := a_to_d
      else
        result := false
      end
    ensure
      bool_result: result = true or result = false
    end

  route_a_to_d(): String
    do
      if has_edge("A", "D") then
        result := "A->D"
      elseif has_edge("A", "B") and has_edge("B", "C") and h
        result := "A->B->C->D"
      else
        result := "UNREACHABLE"
      end
    ensure
      declared_result:
        result = "A->D" or
        result = "A->B->C->D" or
        result = "UNREACHABLE"
```

```
    end
end
```

This sketch keeps graph intent explicit: edges define legal movement; route logic must respect them.

22.6. Graphs in the Three Systems

22.6.1. Delivery

- location network and route computation

22.6.2. Knowledge

- citation/relevance links among documents

22.6.3. Virtual World

- interaction network between entities and zones

Graphs are often the hidden structure behind system complexity.

22.7. Common Mistakes

22.7.1. Mistake 1: Implicit graph representation

Symptom:

- connectivity logic scattered across code paths

Recovery:

- centralize adjacency representation
- expose graph operations explicitly

22.7.2. Mistake 2: Missing cycle protection in traversal

Symptom:

- infinite loops or repeated work

Recovery:

- add visited tracking
- bound traversal depth when appropriate

22.7.3. Mistake 3: Conflating reachability with best path

Symptom:

- any found route treated as optimal

Recovery:

- define objective separately (hops, cost, risk)
- choose traversal/search accordingly

22.7.4. Mistake 4: Weak failure semantics

Symptom:

- partial/empty path with no meaning

Recovery:

- return explicit unreachable/invalid status

Exercise Apply the section task and record your results before reading the solution notes.

22.8. Quick Exercise (12 Minutes)

For one networked feature in your project:

1. define nodes and edges
2. define one reachability query
3. define one path-quality objective
4. define failure status
5. list one cycle-related risk and mitigation

Then decide whether current code treats the graph explicitly or accidentally.

22.9. Connection to Nex

Nex contracts make graph assumptions explicit at operation boundaries, especially around valid inputs, declared failure modes, and route correctness guarantees.

That is essential for systems where graph mistakes become production incidents.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

22.10. Chapter Takeaways

- Graphs model real-world networks more faithfully than tree-only designs.
- Reachability and optimality are different questions.
- Traversal safety requires cycle-aware logic.
- Explicit graph models improve correctness and maintainability.

Part IV established the core data-structure families that underpin scalable software.

In Part V, we apply them to core algorithm families: searching, sorting, traversals, and path finding.

23. Studio 3 — The Scaling Crisis

23.1. Subtitle

Refactoring Studio 2 models for scale without losing correctness.

Studio Focus This studio chapter is hands-on: implement, verify behavior, and document tradeoffs as you iterate.

23.2. 1. The Situation

Studio 2 gave us explicit entities, relationships, and invariants.

Now volume has grown to thousands of tasks, documents, links, and world objects. The model is still correct, but some operations are too slow because access paths are still scan-heavy.

Symptoms observed:

- latency spikes on common lookups
- repeated linear scans in hot paths
- update loops that do unnecessary work

This studio is a direct continuation of Studio 2: same core model classes, new access structures and algorithm choices.

23.3. 2. Engineering Brief

Keep the Studio 2 model semantics. Improve runtime behavior.

Required outcomes:

- identify one bottleneck operation per domain
- show a before/after refactor using Studio 2 classes
- measure operation-count improvement
- preserve behavior contracts and invariants

Implementation guidance:

- optimize access paths, not domain meaning
- use explicit failure statuses (`NOT_FOUND`, `NO_MATCH`, etc.)
- keep model classes intact; add indexing/organization layers around them

---

23.4. 3. Implementation In Nex

Use operation counts to compare baseline and refactored paths.

Suggested files:

- `delivery_scale_from_studio2.nex`
- `knowledge_scale_from_studio2.nex`
- `world_scale_from_studio2.nex`
- `studio_3_main.nex`

If using the web IDE, place all classes in one file and run `App.run`.

23.4.1. 3.1 Reused Studio 2 Models (Unchanged)

```
class Delivery_Task
feature
  task_id: String
  origin: String
  destination: String
  status: String
  assigned_robot_id: String
invariant
  id_present: task_id /= ""
end

class Document
feature
  doc_id: String
  title: String
  body: String
invariant
  id_present: doc_id /= ""
end

class Doc_Link
```

```
feature
  from_doc_id: String
  to_doc_id: String
  link_type: String
invariant
  endpoints_present: from_doc_id /= "" and to_doc_id /= ""
end

class World_Object
feature
  object_id: String
  x: Integer
  vx: Integer
invariant
  id_present: object_id /= ""
end

class Measure_Result
feature
  value: String
  steps: Integer
invariant
  value_present: value /= ""
  non_negative_steps: steps >= 0
end
```

23.4.2. 3.2 Delivery Refactor: Task Lookup

Before (Studio 2-style collection usage): linear scan over `Delivery_Task` objects.

After: index layer for direct key access while keeping `Delivery_Task` as source object.

```
class Delivery_Task_Store_V2
feature
  t1: Delivery_Task
  t2: Delivery_Task
  t3: Delivery_Task
  t4: Delivery_Task
  t5: Delivery_Task

  find_status(task_id: String): Measure_Result
```

```
    require
      id_present: task_id /= ""
    do
      let r: Measure_Result := create Measure_Result
      r.value := "NOT_FOUND"
      r.steps := 0

      r.steps := r.steps + 1
      if t1.task_id = task_id then
        r.value := t1.status
      elseif t2.task_id = task_id then
        r.steps := r.steps + 1
        r.value := t2.status
      elseif t3.task_id = task_id then
        r.steps := r.steps + 1
        r.value := t3.status
      elseif t4.task_id = task_id then
        r.steps := r.steps + 1
        r.value := t4.status
      elseif t5.task_id = task_id then
        r.steps := r.steps + 1
        r.value := t5.status
      else
        r.steps := r.steps + 1
      end

      result := r
    ensure
      bounded_steps: result.steps >= 1 and result.steps <= 5
    end
end

class Delivery_Task_Store_V3
feature
  -- Index from task id to task object (teaching-sized map form)
  k1: String
  v1: Delivery_Task
  k2: String
  v2: Delivery_Task
  k3: String
  v3: Delivery_Task
  k4: String
```

```
  v4: Delivery_Task
  k5: String
  v5: Delivery_Task

  find_status(task_id: String): Measure_Result
    require
      id_present: task_id /= ""
    do
      let r: Measure_Result := create Measure_Result
      r.steps := 1

      if task_id = k1 then
        r.value := v1.status
      elseif task_id = k2 then
        r.value := v2.status
      elseif task_id = k3 then
        r.value := v3.status
      elseif task_id = k4 then
        r.value := v4.status
      elseif task_id = k5 then
        r.value := v5.status
      else
        r.value := "NOT_FOUND"
      end

      result := r
    ensure
      constant_step: result.steps = 1
    end
end
```

23.4.3. 3.3 Knowledge Refactor: Link Validation + Document Lookup

Before: for each `Doc_Link`, document existence check scans document list.

After: use document index by `doc_id` and validate links against indexed lookups.

```
class Knowledge_Model_V2
feature
  d1: Document
```

```
  d2: Document
  d3: Document
  l1: Doc_Link
  l2: Doc_Link

  validate_link(link: Doc_Link): Measure_Result
    do
      let r: Measure_Result := create Measure_Result
      r.steps := 2

      if (d1.doc_id = link.from_doc_id or d2.doc_id = link.from_
         (d1.doc_id = link.to_doc_id or d2.doc_id = link.to_doc_
        r.value := "VALID"
      elseif not (d1.doc_id = link.from_doc_id or d2.doc_id = li
        r.value := "MISSING_FROM"
      else
        r.value := "MISSING_TO"
      end

      result := r
    ensure
      status_known:
        result.value = "VALID" or
        result.value = "MISSING_FROM" or
        result.value = "MISSING_TO"
    end
end

class Knowledge_Model_V3
feature
  dk1: String
  dv1: Document
  dk2: String
  dv2: Document
  dk3: String
  dv3: Document

  has_doc(doc_id: String): Boolean
    do
      result := doc_id = dk1 or doc_id = dk2 or doc_id = dk3
    ensure
      bool_result: result = true or result = false
```

```
    end

  validate_link(link: Doc_Link): Measure_Result
    do
      let r: Measure_Result := create Measure_Result
      r.steps := 2

      if not has_doc(link.from_doc_id) then
        r.value := "MISSING_FROM"
      elseif not has_doc(link.to_doc_id) then
        r.value := "MISSING_TO"
      else
        r.value := "VALID"
      end

      result := r
    ensure
      status_known:
        result.value = "VALID" or
        result.value = "MISSING_FROM" or
        result.value = "MISSING_TO"
    end
end
```

23.4.4. 3.4 Virtual World Refactor: Targeted Update

Before: scan all `World_Object` entries to find target.

After: id index for direct target selection, then apply same position update semantics.

```
class World_Model_V2
feature
  w1: World_Object
  w2: World_Object
  w3: World_Object
  w4: World_Object

  move_by_id(object_id: String; delta: Integer): Measure_Res
    require
      id_present: object_id /= ""
    do
      let r: Measure_Result := create Measure_Result
```

```
      r.value := "NOT_FOUND"
      r.steps := 0

      r.steps := r.steps + 1
      if w1.object_id = object_id then
        w1.x := w1.x + delta
        r.value := "UPDATED"
      elseif w2.object_id = object_id then
        r.steps := r.steps + 1
        w2.x := w2.x + delta
        r.value := "UPDATED"
      elseif w3.object_id = object_id then
        r.steps := r.steps + 1
        w3.x := w3.x + delta
        r.value := "UPDATED"
      elseif w4.object_id = object_id then
        r.steps := r.steps + 1
        w4.x := w4.x + delta
        r.value := "UPDATED"
      else
        r.steps := r.steps + 1
      end

      result := r
    ensure
      bounded_steps: result.steps >= 1 and result.steps <= 4
    end
end

class World_Model_V3
feature
  wk1: String
  wv1: World_Object

  move_by_id(object_id: String; delta: Integer): Measure_Result
    require
      id_present: object_id /= ""
    do
      let r: Measure_Result := create Measure_Result
      r.steps := 1

      if object_id = wk1 then
```

```
        wv1.x := wv1.x + delta
        r.value := "UPDATED"
      else
        r.value := "NOT_FOUND"
      end

      result := r
    ensure
      constant_step: result.steps = 1
    end
end
```

23.4.5. 3.5 Studio Driver (Before/After on Studio 2 Models)

```
class App
feature
  run() do
    -- Delivery tasks (Studio 2 model objects)
    let t1: Delivery_Task := create Delivery_Task
    let t2: Delivery_Task := create Delivery_Task
    let t3: Delivery_Task := create Delivery_Task
    let t4: Delivery_Task := create Delivery_Task
    let t5: Delivery_Task := create Delivery_Task

    t1.task_id := "T-1"; t1.status := "PENDING"
    t2.task_id := "T-2"; t2.status := "IN_TRANSIT"
    t3.task_id := "T-3"; t3.status := "FAILED"
    t4.task_id := "T-4"; t4.status := "DELIVERED"
    t5.task_id := "T-5"; t5.status := "PENDING"

    let d_v2: Delivery_Task_Store_V2 := create Delivery_Task
    d_v2.t1 := t1; d_v2.t2 := t2; d_v2.t3 := t3; d_v2.t4 :=

    let d_v3: Delivery_Task_Store_V3 := create Delivery_Task
    d_v3.k1 := t1.task_id; d_v3.v1 := t1
    d_v3.k2 := t2.task_id; d_v3.v2 := t2
    d_v3.k3 := t3.task_id; d_v3.v3 := t3
    d_v3.k4 := t4.task_id; d_v3.v4 := t4
    d_v3.k5 := t5.task_id; d_v3.v5 := t5
```

```
    let d2r: Measure_Result := d_v2.find_status("T-5")
    let d3r: Measure_Result := d_v3.find_status("T-5")
    print("Delivery V2 steps: " + d2r.steps)
    print("Delivery V3 steps: " + d3r.steps)

    -- Knowledge documents + links (Studio 2 model objects)
    let doc1: Document := create Document
    let doc2: Document := create Document
    let doc3: Document := create Document
    doc1.doc_id := "D-1"; doc1.title := "Graphs"
    doc2.doc_id := "D-2"; doc2.title := "Trees"
    doc3.doc_id := "D-3"; doc3.title := "Maps"

    let link: Doc_Link := create Doc_Link
    link.from_doc_id := "D-1"
    link.to_doc_id := "D-3"
    link.link_type := "references"

    let k_v2: Knowledge_Model_V2 := create Knowledge_Model_V2
    k_v2.d1 := doc1; k_v2.d2 := doc2; k_v2.d3 := doc3; k_v2.l1 :

    let k_v3: Knowledge_Model_V3 := create Knowledge_Model_V3
    k_v3.dk1 := doc1.doc_id; k_v3.dv1 := doc1
    k_v3.dk2 := doc2.doc_id; k_v3.dv2 := doc2
    k_v3.dk3 := doc3.doc_id; k_v3.dv3 := doc3

    let k2r: Measure_Result := k_v2.validate_link(link)
    let k3r: Measure_Result := k_v3.validate_link(link)
    print("Knowledge V2 status/steps: " + k2r.value + "/" + k2r.
    print("Knowledge V3 status/steps: " + k3r.value + "/" + k3r.

    -- World objects (Studio 2 model objects)
    let o1: World_Object := create World_Object
    let o2: World_Object := create World_Object
    let o3: World_Object := create World_Object
    let o4: World_Object := create World_Object
    o1.object_id := "E-1"; o1.x := 1
    o2.object_id := "E-2"; o2.x := 2
    o3.object_id := "E-3"; o3.x := 3
    o4.object_id := "E-4"; o4.x := 4

    let w_v2: World_Model_V2 := create World_Model_V2
```

```
    w_v2.w1 := o1; w_v2.w2 := o2; w_v2.w3 := o3; w_v2.w4 :=

    let w_v3: World_Model_V3 := create World_Model_V3
    w_v3.wk1 := o4.object_id; w_v3.wv1 := o4

    let w2r: Measure_Result := w_v2.move_by_id("E-4", 3)
    let w3r: Measure_Result := w_v3.move_by_id("E-4", 3)
    print("World V2 steps: " + w2r.steps)
    print("World V3 steps: " + w3r.steps)
  end
end
```

Expected pattern:

- V2 keeps Studio 2 model semantics but uses scan-heavy access
- V3 keeps the same model objects and adds index layers for critical operations
- behavior stays consistent; operation counts improve

23.5. 4. Studio Challenges

23.5.1. Level 1 — One Refactor

- choose one Studio 2 operation with repeated scans
- implement V2 baseline and V3 indexed refactor
- compare step counts on hit and miss cases

23.5.2. Level 2 — Three-Domain Refactor

- implement before/after for delivery, knowledge, and world
- keep model classes unchanged
- document where indexes are derived from model identity

23.5.3. Level 3 — Competing Strategies

- for one domain, compare two refactors:
  - index-heavy approach
  - batch/partition approach
- state when each approach wins

23.6. 5. Postmortem

Discuss with evidence:

- Which Studio 2 operation became the worst bottleneck and why?
- Which refactor changed asymptotic behavior versus constants only?
- Which invariant or contract protected correctness during optimization?
- Which optimization should be postponed until higher load?

23.7. Deliverables

- Nex code showing V2 (baseline) and V3 (refactor) on Studio 2 model classes
- output logs with step-count comparisons
- one-page design note:
  - bottleneck operation
  - chosen refactor
  - correctness safeguards
  - limitations and next trigger point

23.8. Exit Criteria

You are ready for Part V if:

- you can optimize access paths without changing domain meaning
- you can preserve model contracts while changing data organization
- you can justify changes with measured evidence, not intuition
- you can explain when the next redesign will be necessary

Part V.

Part V — Algorithms That Power Systems

24. Part V — Algorithms That Power Systems — Searching for What Matters

24.1. 19. Searching for What Matters

Part IV gave us data-structure options.

Part V focuses on algorithm families that use those structures effectively.

We start with searching because most systems spend significant time answering one question:

- "Where is the thing we need?"

24.2. Search Is A Design Decision

Search strategy should match data organization and workload.

Common strategies:

- linear search for small or unsorted collections
- binary search for sorted sequences
- map/set lookup for keyed identity access
- graph/tree traversal for structural discovery

Choosing a strategy by habit instead of operation profile creates hidden latency costs.

24.3. Correctness Before Speed

A search algorithm is useful only if it clearly defines:

- what counts as a match

- what happens when no match exists
- what assumptions input must satisfy

Without explicit miss semantics, search paths tend to leak null/empty ambiguity.

24.4. Worked Design Path

Requirement:

> "Given a task id, return current status quickly and safely."

24.4.1. Step 1: Define match semantics

- exact id equality

24.4.2. Step 2: Define miss behavior

- return `NOT_FOUND`

24.4.3. Step 3: Check data properties

- if unsorted sequence: linear search baseline
- if sorted key list: binary search option
- if direct index exists: keyed lookup preferred

24.4.4. Step 4: Choose first implementation

Start with linear search if model is early and data is small.

24.4.5. Step 5: Define upgrade trigger

When scans dominate hot path latency, move to indexed/keyed strategy.

24.5. Nex Implementation Sketch

```
class Search_Result
feature
  status: String
  steps: Integer
invariant
  status_present: status /= ""
  steps_non_negative: steps >= 0
end

class Task_Search
feature
  id1: String
  st1: String
  id2: String
  st2: String
  id3: String
  st3: String
  id4: String
  st4: String

  linear_find(task_id: String): Search_Result
    require
      id_present: task_id /= ""
    do
      let r: Search_Result := create Search_Result
      r.status := "NOT_FOUND"
      r.steps := 0

      r.steps := r.steps + 1
      if task_id = id1 then
        r.status := st1
      elseif task_id = id2 then
        r.steps := r.steps + 1
        r.status := st2
      elseif task_id = id3 then
        r.steps := r.steps + 1
        r.status := st3
      elseif task_id = id4 then
        r.steps := r.steps + 1
        r.status := st4
      else
        r.steps := r.steps + 1
```

```
      end

      result := r
    ensure
      bounded_steps: result.steps >= 1 and result.steps <= 4
    end
end
```

This chapter example emphasizes explicit match/miss semantics and measurable behavior.

24.6. Search Across The Three Systems

24.6.1. Delivery

- locate task/robot/location by identity

24.6.2. Knowledge

- locate candidate docs by token/id/tag

24.6.3. Virtual World

- locate entity state by object id

In each system, search is often the first scaling bottleneck.

24.7. Common Mistakes

24.7.1. Mistake 1: One search strategy everywhere

Symptom:

- growing latency on high-frequency operations

Recovery:

- classify operation profile first
- use structure-appropriate search

24.7.2. Mistake 2: Undefined miss semantics

Symptom:

- null/empty meaning depends on caller

Recovery:

- return explicit status values

24.7.3. Mistake 3: Ignoring input assumptions

Symptom:

- binary search used on unsorted data

Recovery:

- enforce preconditions through contracts

Exercise Apply the section task and record your results before reading the solution notes.

24.8. Quick Exercise (10 Minutes)

Choose one search hotspot and document:

1. match rule
2. miss rule
3. current strategy
4. expected input size
5. redesign trigger

Then add one contract that prevents incorrect strategy use.

24.9. Connection to Nex

Nex contracts make search assumptions executable, reducing silent mismatches between caller expectations and implementation behavior.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

24.10. Chapter Takeaways

- Search strategy should follow operation profile and data organization.
- Correctness semantics (match/miss) are part of algorithm design.
- Linear search is often a baseline, not an end state.
- Contracts help prevent invalid search usage.

In Chapter 20, we examine sorting, where ordering becomes a leverage point for downstream speed.

25. Part V — Algorithms That Power Systems — Sorting the World

25.1. 20. Sorting the World

Searching finds items.

Sorting organizes them so later operations become simpler and faster.

In production systems, sorting is rarely cosmetic. It is usually a prerequisite for ranking, batching, deterministic processing, and efficient retrieval.

25.2. Why Sorting Matters

Sorting creates structure from raw collections.

Benefits include:

- deterministic output order
- easier duplicate detection
- compatibility with binary search and merge-like workflows
- clearer user-facing ranking behavior

The main tradeoff is upfront cost versus downstream savings.

25.3. Choosing A Sort Strategy

Strategy depends on constraints:

- collection size
- stability requirements

- memory budget
- partially ordered inputs

For teaching, a simple comparison sort is enough to reason about behavior.

In real systems, use library sorts unless domain constraints demand custom behavior.

25.4. Worked Design Path

Requirement:

> "Return delivery tasks ordered by priority, preserving insertion order for ties."

25.4.1. Step 1: Define comparison key

- primary: priority score

25.4.2. Step 2: Define tie behavior

- stable order by arrival sequence

25.4.3. Step 3: Choose baseline

- simple stable pass-based approach for clarity

25.4.4. Step 4: Define correctness checks

- output is non-decreasing by priority
- equal-priority items keep original relative order

25.4.5. Step 5: Define scale transition

- when list size grows, use optimized stable sort implementation

---

25.5. Nex Implementation Sketch

```
class Task_View
feature
  id: String
  priority: Integer
  arrival: Integer
invariant
  id_present: id /= ""
  arrival_non_negative: arrival >= 0
end

class Sort_Example
feature
  t1: Task_View
  t2: Task_View
  t3: Task_View

  sorted_ids_by_priority(): String
    do
      -- Teaching-sized fixed comparison network for three i
      let a: Task_View := t1
      let b: Task_View := t2
      let c: Task_View := t3

      if a.priority > b.priority then
        let tmp: Task_View := a
        a := b
        b := tmp
      end

      if b.priority > c.priority then
        let tmp2: Task_View := b
        b := c
        c := tmp2
      end

      if a.priority > b.priority then
        let tmp3: Task_View := a
        a := b
        b := tmp3
      end
```

```
      result := a.id + " -> " + b.id + " -> " + c.id
    ensure
      non_empty: result /= ""
    end
end
```

This sketch focuses on order semantics, not production-grade sorting APIs.

25.6. Sorting Across The Three Systems

25.6.1. Delivery

- prioritize task queues

25.6.2. Knowledge

- order ranked search results

25.6.3. Virtual World

- deterministic processing order for updates/events

Sorting often sits between raw data and high-value operations.

25.7. Common Mistakes

25.7.1. Mistake 1: Unclear comparison rule

Symptom:

- inconsistent order across runs

Recovery:

- define comparator explicitly
- document tie-break policy

25.7.2. Mistake 2: Ignoring stability needs

Symptom:

- equal-priority records reorder unexpectedly

Recovery:

- use stable strategy when required by semantics

25.7.3. Mistake 3: Re-sorting too often

Symptom:

- repeated full sort in hot loops

Recovery:

- batch updates or maintain partially ordered structures

Exercise Apply the section task and record your results before reading the solution notes.

25.8. Quick Exercise (10 Minutes)

Pick one ordered output in your system and define:

1. primary sort key
2. tie-break key
3. stability requirement
4. current sort frequency
5. optimization opportunity

Then write one test case with equal keys to validate tie behavior.

25.9. Connection to Nex

Nex contracts help encode ordering guarantees so changes to sort implementation do not silently change system behavior.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

25.10. Chapter Takeaways

- Sorting is a structural tool, not just presentation logic.
- Comparator and tie-break rules must be explicit.
- Stability can be a correctness requirement.
- Upfront sort cost can unlock faster downstream operations.

In Chapter 21, we explore traversal algorithms for trees and graphs.

26. Part V — Algorithms That Power Systems — Exploring Trees and Graphs

26.1. 21. Exploring Trees and Graphs

Once data is structured as trees or graphs, traversal becomes the core algorithmic operation.

Traversal answers questions like:

- what is reachable?
- what is connected?
- in what order should we process structure?

This chapter focuses on depth-first and breadth-first exploration patterns.

26.2. DFS vs BFS

Two foundational traversal styles:

- **Depth-first search (DFS)**: go deep along a branch before backtracking.
- **Breadth-first search (BFS)**: visit level by level from a frontier.

They solve different problems efficiently:

- DFS is natural for exhaustive structure exploration and recursive decomposition.
- BFS is natural for minimum-hop reachability in unweighted graphs.

26.3. Traversal Safety Rules

Traversal correctness requires explicit controls:

- visited tracking (avoid repeated loops)
- boundary conditions (depth/time/cost limits)
- deterministic policy when order matters

Without these, traversal may be correct on toy cases and unstable in real workloads.

26.4. Worked Design Path

Requirement:

> "From a starting document, discover connected documents up to depth 2."

26.4.1. Step 1: Define reachability scope

- max depth = 2

26.4.2. Step 2: Choose traversal

- BFS for depth-layer behavior

26.4.3. Step 3: Define duplicate policy

- each document visited once

26.4.4. Step 4: Define output semantics

- return discovered ids in visit order

26.4.5. Step 5: Define failure behavior

- unknown start id -> `INVALID_START`

26.5. Nex Implementation Sketch

```
class Explore_Result
feature
  status: String
  discovered: String
invariant
  status_present: status /= ""
end

class Graph_Explorer
feature
  -- Teaching-sized adjacency representation.
  a_to_b: Boolean
  a_to_c: Boolean
  b_to_d: Boolean
  c_to_d: Boolean

  bfs_from_a_depth2(): Explore_Result
    do
      let r: Explore_Result := create Explore_Result

      if not a_to_b and not a_to_c then
        r.status := "ISOLATED"
        r.discovered := "A"
      elseif a_to_b and b_to_d then
        r.status := "OK"
        r.discovered := "A,B,D"
      elseif a_to_c and c_to_d then
        r.status := "OK"
        r.discovered := "A,C,D"
      elseif a_to_b then
        r.status := "OK"
        r.discovered := "A,B"
      elseif a_to_c then
        r.status := "OK"
        r.discovered := "A,C"
      else
        r.status := "ISOLATED"
        r.discovered := "A"
      end
```

```
      result := r
    ensure
      declared_status: result.status = "OK" or result.status = "
    end
end
```

This simplified sketch emphasizes traversal outcomes and explicit status.

26.6. Traversal Across The Three Systems

26.6.1. Delivery

- explore route alternatives and reachable depots

26.6.2. Knowledge

- walk linked documents/concepts

26.6.3. Virtual World

- traverse scene/interactions for update and visibility

Traversal design determines both correctness and cost behavior.

26.7. Common Mistakes

26.7.1. Mistake 1: No visited policy

Symptom:

- repeated processing or infinite loops

Recovery:

- track visited nodes explicitly

26.7.2. Mistake 2: Wrong traversal for goal

Symptom:

- DFS used where minimum-hop answer required

Recovery:

- choose traversal to match objective

26.7.3. Mistake 3: Implicit depth/cost bounds

Symptom:

- traversal growth surprises under dense connectivity

Recovery:

- define max depth or expansion budget

Exercise Apply the section task and record your results before reading the solution notes.

26.8. Quick Exercise (12 Minutes)

Pick one graph/tree operation and specify:

1. traversal type (DFS/BFS)
2. visited policy
3. bound policy (depth/time)
4. output order rule
5. invalid-input behavior

Then justify the choice in one paragraph.

26.9. Connection to Nex

Nex contracts make traversal assumptions explicit, especially around entry validity and result guarantees under bounded exploration.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

26.10. Chapter Takeaways

- Traversal is the primary algorithm over structured relationships.
- DFS and BFS solve different classes of questions.
- Visited and boundary policies are correctness requirements.
- Traversal order should be explicit when behavior depends on it.

In Chapter 22, we build on traversal to solve best-path optimization problems.

27. Part V — Algorithms That Power Systems — Finding the Best Path

27.1. 22. Finding the Best Path

Reachability asks whether a path exists.

Best-path algorithms ask which path is optimal under an objective.

That objective must be explicit: minimum hops, minimum time, minimum risk, or weighted combination.

27.2. Path Optimality Requires A Cost Model

A path algorithm is only meaningful relative to a declared cost model.

Examples:

- unweighted hops -> BFS-style shortest path
- non-negative weighted edges -> Dijkstra-style approach
- heuristic-guided search -> A*-style tradeoff

If cost model is unclear, "best" becomes team interpretation, not system behavior.

27.3. Priority Matters

Best-path search usually depends on managing a frontier by priority.

Key design questions:

- what is frontier ordering key?
- how are ties handled?
- when is a node considered finalized?

These choices affect both correctness and performance.

27.4. Worked Design Path

Requirement:

> "Route from A to D minimizing travel cost."

27.4.1. Step 1: Define edge costs

- each edge has non-negative integer cost

27.4.2. Step 2: Define validity

- path cost = sum of edge costs
- result must be valid connected sequence

27.4.3. Step 3: Define failure behavior

- unreachable destination -> `UNREACHABLE`

27.4.4. Step 4: Choose algorithm

- Dijkstra-style priority expansion for non-negative costs

27.4.5. Step 5: Define determinism

- stable tie-break rule for equal-cost candidates

27.5. Nex Implementation Sketch

```
class Path_Result
feature
  status: String
  path: String
  total_cost: Integer
invariant
```

```
  status_present: status /= ""
  non_negative_cost: total_cost >= 0
end

class Best_Path_Example
feature
  a_b: Integer
  b_d: Integer
  a_c: Integer
  c_d: Integer
  a_d: Integer

  best_a_to_d(): Path_Result
    require
      non_negative_costs:
        a_b >= 0 and b_d >= 0 and a_c >= 0 and c_d >= 0 and
    do
      let r: Path_Result := create Path_Result
      let abd: Integer := a_b + b_d
      let acd: Integer := a_c + c_d

      if a_d <= abd and a_d <= acd then
        r.status := "FOUND"
        r.path := "A->D"
        r.total_cost := a_d
      elseif abd <= acd then
        r.status := "FOUND"
        r.path := "A->B->D"
        r.total_cost := abd
      else
        r.status := "FOUND"
        r.path := "A->C->D"
        r.total_cost := acd
      end

      result := r
    ensure
      known_status: result.status = "FOUND" or result.status
    end
end
```

This teaching sketch shows objective-driven path selection with explicit costs.

27.6. Best-Path In The Three Systems

27.6.1. Delivery

- route tasks by travel cost and constraints

27.6.2. Knowledge

- find lowest-cost explanation/link path between concepts

27.6.3. Virtual World

- navigate entities through weighted interaction spaces

Best-path design turns network models into operational decisions.

27.7. Common Mistakes

27.7.1. Mistake 1: Objective ambiguity

Symptom:

- different runs optimize different implicit goals

Recovery:

- document cost model explicitly

27.7.2. Mistake 2: Using greedy local choice as global optimum

Symptom:

- locally cheap edge leads to expensive total path

Recovery:

- use algorithm with global optimality guarantees for chosen assumptions

27.7.3. Mistake 3: Ignoring edge constraints

Symptom:

- algorithm treats blocked/illegal edges as available

Recovery:

- validate edge legality in expansion step

27.7.4. Mistake 4: Unclear unreachable semantics

Symptom:

- empty path interpreted inconsistently

Recovery:

- return explicit unreachable status

Exercise Apply the section task and record your results before reading the solution notes.

27.8. Quick Exercise (12 Minutes)

For one routing problem in your system, define:

1. objective function
2. edge validity rules
3. tie-break rule
4. unreachable behavior
5. algorithm choice and assumption set

Then add one adversarial case where greedy local choice fails.

27.9. Connection to Nex

Nex contracts keep path assumptions explicit and testable, especially around non-negative costs, legal edges, and declared failure semantics.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

27.10. Chapter Takeaways

- Best path is objective-dependent.
- Priority-driven frontier management is central to weighted search.
- Local greedy choice is not always globally optimal.
- Explicit cost and failure semantics are required for correctness.

Next: **Algorithm Lab — When Algorithms Compete**, where we benchmark competing strategies under different workload shapes before moving into Part VI.

28. Algorithm Lab — When Algorithms Compete

28.1. Subtitle

Benchmarking competing algorithm choices under realistic workload shapes.

Lab Focus Run controlled experiments, collect evidence, and compare algorithm choices under explicit assumptions.

28.2. 1. Why This Lab Exists

The first algorithm lab built intuition that different strategies behave differently.

This lab goes one step further: direct competition under controlled scenarios.

The goal is to answer engineering questions with evidence:

- Which algorithm wins for our workload profile?
- Where does each algorithm break down?
- Which assumptions must hold for the chosen strategy?

28.3. 2. Lab Objectives

By the end of this lab, you should be able to:

- compare competing algorithms for the same requirement
- evaluate normal-case vs adverse-case behavior
- separate asymptotic reasoning from measured behavior
- justify a strategy choice with explicit tradeoffs

28.4. 3. Competition Setup

Use one concrete problem:

> "Find the shortest path from A to G in an unweighted graph."

Competing strategies:

- DFS-style search (finds a path, not guaranteed shortest by hops)
- BFS-style search (guarantees minimum hops in unweighted graphs)

Why this setup works:

- same input graph
- same success/failure contract
- different quality/cost behavior

28.5. 4. Nex Implementation

Suggested files:

- `algorithm_lab_2.nex`
- `algorithm_lab_2_main.nex`

If using the web IDE, place everything in one file and run `App.run`.

28.5.1. 4.1 Shared Result Type

```
class Path_Run_Result
feature
  strategy: String
  status: String
  path: String
  hops: Integer
  steps: Integer
invariant
  strategy_present: strategy /= ""
  status_present: status /= ""
  hops_non_negative: hops >= 0
  steps_non_negative: steps >= 0
end
```

28.5.2. 4.2 Graph Fixture

```
class Graph_Fixture
feature
  -- Teaching-sized directed edges.
  a_b: Boolean
  a_c: Boolean
  b_d: Boolean
  d_e: Boolean
  e_g: Boolean
  c_f: Boolean
  f_g: Boolean

  setup_default() do
    a_b := true
    a_c := true
    b_d := true
    d_e := true
    e_g := true
    c_f := true
    f_g := true
  ensure
    initialized:
      a_b and a_c and b_d and d_e and e_g and c_f and f_g
  end
end
```

28.5.3. 4.3 DFS-Style Competitor

```
class DFS_Competitor
feature
  run(g: Graph_Fixture): Path_Run_Result
    do
      let r: Path_Run_Result := create Path_Run_Result
      r.strategy := "DFS"
      r.steps := 0

      -- Deterministic branch preference: A->B before A->C.
      r.steps := r.steps + 1
      if g.a_b and g.b_d and g.d_e and g.e_g then
        r.status := "FOUND"
```

```
        r.path := "A->B->D->E->G"
        r.hops := 4
      elseif g.a_c and g.c_f and g.f_g then
        r.status := "FOUND"
        r.path := "A->C->F->G"
        r.hops := 3
      else
        r.status := "UNREACHABLE"
        r.path := ""
        r.hops := 0
      end

      result := r
    ensure
      known_status: result.status = "FOUND" or result.status = "
    end
end
```

28.5.4. 4.4 BFS-Style Competitor

```
class BFS_Competitor
feature
  run(g: Graph_Fixture): Path_Run_Result
    do
      let r: Path_Run_Result := create Path_Run_Result
      r.strategy := "BFS"
      r.steps := 0

      -- Layer-aware logic returns minimum-hop route if availabl
      r.steps := r.steps + 1
      if g.a_c and g.c_f and g.f_g then
        r.status := "FOUND"
        r.path := "A->C->F->G"
        r.hops := 3
      elseif g.a_b and g.b_d and g.d_e and g.e_g then
        r.status := "FOUND"
        r.path := "A->B->D->E->G"
        r.hops := 4
      else
        r.status := "UNREACHABLE"
        r.path := ""
```

```
        r.hops := 0
      end

      result := r
    ensure
      known_status: result.status = "FOUND" or result.status
    end
end
```

28.5.5. 4.5 Driver: Run Competitions

```
class App
feature
  run() do
    let g: Graph_Fixture := create Graph_Fixture
    g.setup_default

    let dfs: DFS_Competitor := create DFS_Competitor
    let bfs: BFS_Competitor := create BFS_Competitor

    let r_dfs: Path_Run_Result := dfs.run(g)
    let r_bfs: Path_Run_Result := bfs.run(g)

    print("DFS: " + r_dfs.status + " " + r_dfs.path + " hops
    print("BFS: " + r_bfs.status + " " + r_bfs.path + " hops

    -- Adverse scenario: disable shorter branch.
    g.c_f := false
    let r_dfs_2: Path_Run_Result := dfs.run(g)
    let r_bfs_2: Path_Run_Result := bfs.run(g)

    print("DFS adverse: " + r_dfs_2.status + " " + r_dfs_2.p
    print("BFS adverse: " + r_bfs_2.status + " " + r_bfs_2.p
  end
end
```

Expected observations:

- both may find paths in nominal case
- BFS-style competitor returns shorter route in default fixture
- DFS-style competitor can return longer-but-valid route depending on branch order

---

28.6. 5. Lab Tasks

28.6.1. Task A — Nominal Competition

Run both competitors on the default graph and record:

- found/unreachable status
- returned path
- hop count
- step count

28.6.2. Task B — Workload Variation

Modify graph shape and rerun:

- remove one shortcut edge
- add one distracting dead-end edge

Document which strategy is more sensitive to each change.

28.6.3. Task C — Constraint Stress

Introduce an explicit depth cap and evaluate:

- when DFS fails due to depth bound
- when BFS frontier growth becomes expensive

28.7. 6. Decision Template

Use this template for final strategy choice:

1. problem objective (reachability or shortest path)
2. dominant workload shape
3. correctness requirement (any path vs best path)
4. preferred algorithm and why
5. fallback strategy under stress

28.8. 7. Deliverables

- runnable Nex code for both competitors
- experiment table across at least 3 graph variants
- one-page decision note with selected strategy
- explicit list of assumptions that must remain true

28.9. 8. Forward Link

This lab closes Part V by turning algorithm choice into measurable engineering practice.

In Part VI, we move from algorithm decisions to software architecture decisions: components, boundaries, and interfaces.

Part VI.

Part VI — Building Real Software

29. Part VI — Building Real Software — From Algorithms to Components

29.1. 23. From Algorithms to Components

Part V helped us choose good algorithms.

Part VI is about turning those algorithms into maintainable software systems.

An algorithm can be correct and fast, but still fail in production if component boundaries are unclear.

29.2. Why Components Matter

Components give structure to change.

A useful component boundary:

- groups related responsibilities
- hides internal data/steps
- exposes stable behavior contracts

Without boundaries, systems drift into tightly coupled code where every change risks unrelated behavior.

29.3. Boundary Design Rules

A practical rule:

- high cohesion inside a component
- low coupling across components

Questions to ask:

- Does this component have one clear reason to change?
- Are dependencies flowing in a consistent direction?
- Can we test this component in isolation?

29.4. Worked Design Path

Requirement:

> "Compute best route and notify clients when route changes."

Naive design mixes concerns in one service:

- route computation
- persistence updates
- notification delivery

Componentized design:

1. `Route_Component` (algorithm + route rules)
2. `Task_Component` (task state transitions)
3. `Notify_Component` (message delivery)
4. `Coordinator` (workflow orchestration)

Benefits:

- route algorithm can evolve without rewriting notification logic
- failures are localized by component contract

29.5. Nex Implementation Sketch

```
class Route_Component
feature
  compute(start_loc, end_loc: String): String
    require
      inputs_present: start_loc /= "" and end_loc /= ""
    do
      if start_loc = end_loc then
        result := start_loc
```

```
      elseif start_loc = "A" and end_loc = "C" then
        result := "A->B->C"
      else
        result := "UNREACHABLE"
      end
    ensure
      result_present: result /= ""
    end
end

class Notify_Component
feature
  send(task_id, message: String): String
    require
      inputs_present: task_id /= "" and message /= ""
    do
      result := "SENT"
    ensure
      status_known: result = "SENT" or result = "FAILED"
    end
end

class Delivery_Coordinator
feature
  route: Route_Component
  notify: Notify_Component

  reroute_and_notify(task_id, start_loc, end_loc: String): S
    require
      inputs_present: task_id /= "" and start_loc /= "" and
    do
      let p: String := route.compute(start_loc, end_loc)
      if p = "UNREACHABLE" then
        result := "NO_ROUTE"
      else
        result := notify.send(task_id, "route=" + p)
      end
    ensure
      known_result: result = "NO_ROUTE" or result = "SENT" o
    end
end
```

This sketch separates algorithm logic from side effects while preserving end-to-end behavior.

29.6. Components Across The Three Systems

29.6.1. Delivery

- routing, task state, notification as separate components

29.6.2. Knowledge

- retrieval, ranking, rendering as separate components

29.6.3. Virtual World

- simulation, collision, event output as separate components

Componentization is where algorithm quality becomes software quality.

29.7. Common Mistakes

29.7.1. Mistake 1: Utility blob component

Symptom:

- one module owns unrelated responsibilities

Recovery:

- split by reason-to-change

29.7.2. Mistake 2: Leaky boundaries

Symptom:

- callers depend on internal fields/steps

Recovery:

- expose narrow interface contracts only

29.7.3. Mistake 3: Wrong dependency direction

Symptom:

- core domain depends on transport/UI details

Recovery:

- invert dependencies through interfaces/adapters

Exercise Apply the section task and record your results before reading the solution notes.

29.8. Quick Exercise (12 Minutes)

Take one workflow and map:

1. core algorithm component
2. state component
3. side-effect component
4. coordinator/orchestrator
5. contract for each boundary

Then identify one dependency you should reverse.

29.9. Connection to Nex

Nex contracts make component boundaries explicit and executable, improving refactor safety as systems grow.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

29.10. Chapter Takeaways

- Good algorithms need good component boundaries.
- Cohesion and coupling are operational concerns, not style preferences.

- Coordinators should orchestrate, not absorb all logic.
- Contracts stabilize component interactions.

In Chapter 24, we examine functional thinking as a strategy for composable and testable components.

30. Part VI — Building Real Software — Functional Thinking

30.1. 24. Functional Thinking

Component boundaries define where code lives.

Functional thinking defines how behavior flows through those boundaries.

The core idea is simple: represent logic as explicit transformations with minimal hidden state.

30.2. What Functional Thinking Adds

Functional style emphasizes:

- pure functions (same input -> same output)
- explicit data flow
- composition of small transformations

Benefits:

- easier testability
- easier reasoning about correctness
- fewer side-effect bugs

This does not require eliminating all objects. It means using pure transformations where they give leverage.

30.3. Pure vs Effectful Layers

A practical pattern:

- pure core: compute decisions, scores, transitions
- effectful shell: persistence, network calls, logging, UI

This split reduces debugging complexity and keeps behavior reproducible.

30.4. Worked Design Path

Requirement:

> "Rank documents for a query and emit top 3 IDs."

Naive approach mixes ranking and I/O.
Functional approach:

1. `tokenize(query)`
2. `score(doc, tokens)`
3. `sort_by_score(scored)`
4. `take_top(sorted, 3)`
5. effect layer publishes result

Now ranking logic is deterministic and testable without infrastructure.

30.5. Nex Implementation Sketch

```
class Rank_Functions
feature
  score(doc, query: String): Integer
    require
      inputs_present: doc /= "" and query /= ""
    do
      if doc = query then
        result := 100
      else
        result := 20
      end
    ensure
```

```
      non_negative: result >= 0
    end

  pick_top(doc1, doc2, doc3, query: String): String
    require
      docs_present: doc1 /= "" and doc2 /= "" and doc3 /= ""
    do
      let s1: Integer := score(doc1, query)
      let s2: Integer := score(doc2, query)
      let s3: Integer := score(doc3, query)

      if s1 >= s2 and s1 >= s3 then
        result := doc1
      elseif s2 >= s1 and s2 >= s3 then
        result := doc2
      else
        result := doc3
      end
    ensure
      from_inputs: result = doc1 or result = doc2 or result
    end
end

class Rank_Publisher
feature
  publish(top_doc: String): String
    require
      doc_present: top_doc /= ""
    do
      result := "PUBLISHED"
    ensure
      known_status: result = "PUBLISHED" or result = "FAILED
    end
end
```

This keeps ranking logic pure while isolating effects in a separate component.

30.6. Functional Thinking Across The Three Systems

30.6.1. Delivery

- pure route-scoring function + effectful dispatch update

30.6.2. Knowledge

- pure ranking pipeline + effectful storage/telemetry

30.6.3. Virtual World

- pure next-state calculation + effectful render/output

Functional decomposition improves reliability regardless of domain.

30.7. Common Mistakes

30.7.1. Mistake 1: Hidden side effects in "pure" routines

Symptom:

- same input produces different output over time

Recovery:

- isolate mutable/environmental dependencies

30.7.2. Mistake 2: Over-fragmentation

Symptom:

- too many tiny functions with no semantic gain

Recovery:

- compose around meaningful transformations

30.7.3. Mistake 3: Avoiding all effects inside domain code

Symptom:

- awkward abstractions that reduce clarity

Recovery:

- keep pure/effectful separation pragmatic, not dogmatic

Exercise Apply the section task and record your results before reading the solution notes.

30.8. Quick Exercise (10 Minutes)

Pick one feature and split it into:

1. pure decision function
2. pure transformation function
3. effectful output function
4. one contract per function

Then run tests for pure functions without using any external systems.

30.9. Connection to Nex

Nex supports functional thinking naturally through contract-checked routines and clear data-flow style, even in mixed OOP systems.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

30.10. Chapter Takeaways

- Functional thinking improves predictability and testability.
- Pure core + effectful shell is a practical architecture pattern.

- Composition works best with explicit contracts and data flow.
- Pragmatic hybrid design beats ideological purity.

In Chapter 25, we turn to object-oriented thinking for behavior-centered collaboration.

31. Part VI — Building Real Software — Object-Oriented Thinking

31.1. 25. Object-Oriented Thinking

Functional design emphasizes transformations.

Object-oriented design emphasizes responsibility and collaboration.

When a system has rich state, lifecycle rules, and interacting actors, OOP can make behavior easier to organize.

31.2. OOP As Responsibility Design

Useful OOP is not about classes for everything.

It is about assigning behavior to the objects that own the relevant state and invariants.

Good OOP boundaries answer:

- who owns this data?
- who is allowed to change this state?
- who enforces transition legality?

31.3. Collaboration Protocols

Objects should collaborate through explicit protocols, not by reaching into each other's internal fields.

A robust collaboration model:

- object A asks object B for behavior via method contracts
- object B enforces its own invariants

- cross-object workflows happen in coordinators/services

This keeps state integrity local and interactions predictable.

31.4. Worked Design Path

Requirement:

> "Assign a task to a robot and start movement only when both are ready."

Design:

1. `Delivery_Task` owns assignment and status transitions
2. `Robot` owns readiness state
3. `Dispatch_Service` coordinates collaboration

Contracts:

- task can be assigned only from legal states
- robot must be ready before task starts

31.5. Nex Implementation Sketch

```
class Robot
feature
  robot_id: String
  ready: Boolean

  mark_ready() do
    ready := true
  ensure
    now_ready: ready = true
  end
invariant
  id_present: robot_id /= ""
end
```

```
class Delivery_Task
feature
  task_id: String
  status: String
  assigned_robot: String

  assign(robot_id: String)
    require
      robot_present: robot_id /= ""
      can_assign: status = "PENDING" or status = "FAILED"
    do
      assigned_robot := robot_id
      status := "IN_TRANSIT"
    ensure
      assigned: assigned_robot = robot_id and status = "IN_T
    end
invariant
  id_present: task_id /= ""
  valid_status:
    status = "PENDING" or
    status = "IN_TRANSIT" or
    status = "DELIVERED" or
    status = "FAILED"
end

class Dispatch_Service
feature
  dispatch(task: Delivery_Task; robot: Robot): String
    require
      robot_ready: robot.ready = true
      task_pending_or_failed: task.status = "PENDING" or tas
    do
      task.assign(robot.robot_id)
      result := "DISPATCHED"
    ensure
      known_result: result = "DISPATCHED"
    end
end
```

This pattern keeps domain invariants inside domain objects while coordinating workflows externally.

31.6. OOP Across The Three Systems

31.6.1. Delivery

- task and robot behavior with explicit transition methods

31.6.2. Knowledge

- document and link entities with relationship management methods

31.6.3. Virtual World

- world objects owning update/collision state transitions

Behavior ownership is the key strength of OOP in stateful domains.

31.7. Common Mistakes

31.7.1. Mistake 1: Anemic models

Symptom:

- objects store data while logic lives in services only

Recovery:

- move invariant-related behavior to owning object

31.7.2. Mistake 2: God objects

Symptom:

- one class controls too much workflow and state

Recovery:

- split by responsibility and collaboration protocol

31.7.3. Mistake 3: Cross-object field mutation

Symptom:

- callers directly mutate internal state of other objects

Recovery:

- expose methods with contracts instead of raw mutation

Exercise Apply the section task and record your results before reading the solution notes.

31.8. Quick Exercise (12 Minutes)

Choose one workflow and identify:

1. state-owning objects
2. transitions each object should enforce
3. coordinator/service role
4. one illegal transition to block
5. one collaboration contract to add

Then refactor one anemic object to own one key behavior.

31.9. Connection to Nex

Nex supports OOP well through class invariants and contract-checked feature routines, making behavior ownership explicit and enforceable.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

31.10. Chapter Takeaways

- OOP is responsibility design, not class count.
- Objects should own behavior tied to their invariants.

- Coordinators should orchestrate collaboration, not own all state logic.
- Contracts make collaboration protocols explicit.

In Chapter 26, we formalize interface design so component collaboration stays stable as systems evolve.

32. Part VI — Building Real Software — Designing Interfaces

32.1. 26. Designing Interfaces

By this point we have:

- algorithms
- data structures
- components
- functional and object-oriented design tools

Interfaces are what keep these pieces composable over time.

A weak interface leaks internals, couples modules, and blocks safe evolution.

32.2. Interface Design Goals

A good interface should be:

- minimal: only expose necessary operations
- explicit: make assumptions and failure modes clear
- stable: avoid forcing callers to track internals
- testable: behavior can be validated through contract

Interfaces are not just signatures; they are behavioral agreements.

32.3. Contract-First APIs

Interface contracts should define:

- input preconditions
- output guarantees
- error/failure semantics
- invariants that remain true across calls

When these are missing, integration bugs move from compile-time to production-time.

32.4. Worked Design Path

Requirement:

> "Expose route planning to multiple clients (web UI, scheduler, simulator)."

Bad approach:

- expose internal graph structures and mutable route buffers

Better approach:

- expose one operation: `plan_route(start, goal)`
- return explicit result status and path
- hide internal algorithm/data structure choices

Now algorithm upgrades do not break clients.

32.5. Nex Implementation Sketch

```
class Route_Response
feature
  status: String
  path: String
invariant
  valid_status:
    status = "FOUND" or
    status = "UNREACHABLE" or
    status = "INVALID_INPUT"
end
```

```
class Route_Interface
feature
  plan_route(start_loc, goal_loc: String): Route_Response
    require
      inputs_present: start_loc /= "" and goal_loc /= ""
    do
      let r: Route_Response := create Route_Response

      if start_loc = goal_loc then
        r.status := "FOUND"
        r.path := start_loc
      elseif start_loc = "A" and goal_loc = "C" then
        r.status := "FOUND"
        r.path := "A->B->C"
      else
        r.status := "UNREACHABLE"
        r.path := ""
      end

      result := r
    ensure
      declared_status:
        result.status = "FOUND" or
        result.status = "UNREACHABLE" or
        result.status = "INVALID_INPUT"
    end
end

class Route_Client
feature
  route_api: Route_Interface

  request(start_loc, goal_loc: String): String
    require
      inputs_present: start_loc /= "" and goal_loc /= ""
    do
      let resp: Route_Response := route_api.plan_route(start
      if resp.status = "FOUND" then
        result := "OK:" + resp.path
      elseif resp.status = "UNREACHABLE" then
        result := "NO_ROUTE"
      else
```

```
        result := "BAD_INPUT"
      end
    ensure
      known_result: result /= ""
    end
end
```

Clients depend on stable response contracts, not internal planner details.

32.6. Interface Design Across The Three Systems

32.6.1. Delivery

- route/task APIs with explicit statuses

32.6.2. Knowledge

- retrieval/ranking APIs with confidence and miss semantics

32.6.3. Virtual World

- update/query APIs with deterministic boundary guarantees

Strong interfaces preserve flexibility behind the boundary.

32.7. Common Mistakes

32.7.1. Mistake 1: Leaking internals

Symptom:

- callers manipulate internal structures directly

Recovery:

- return value objects and status codes, not internal state handles

32.7.2. Mistake 2: Ambiguous failure modes

Symptom:

- same return shape used for success and failure

Recovery:

- define explicit statuses and per-status expectations

32.7.3. Mistake 3: Over-wide interfaces

Symptom:

- many methods with overlapping behavior

Recovery:

- narrow interface to essential use cases

32.7.4. Mistake 4: Versioning by breakage

Symptom:

- small internal change breaks many consumers

Recovery:

- preserve stable contracts, introduce additive evolution

Exercise Apply the section task and record your results before reading the solution notes.

32.8. Quick Exercise (12 Minutes)

Pick one module interface and document:

1. operation signature
2. preconditions
3. success guarantees
4. failure statuses
5. one internal detail that must remain hidden

Then write one integration test that validates only contract behavior.

32.9. Connection to Nex

Nex contracts make interface agreements explicit and executable, reducing integration ambiguity across modules and teams.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

32.10. Chapter Takeaways

- Interfaces are behavioral contracts, not just method names.
- Contract-first design prevents integration ambiguity.
- Stable interfaces enable internal algorithm/data evolution.
- Narrow, explicit interfaces improve long-term system adaptability.

Part VI prepared the architecture layer: components, functional decomposition, object collaboration, and interfaces.

Next is **Studio 4 — The Architecture Refactor**, where these ideas are applied to a system-wide redesign.

33. Studio 4 — The Architecture Refactor

33.1. Subtitle

Evolving Studio 3 optimizations into stable component architecture.

Studio Focus This studio chapter is hands-on: implement, verify behavior, and document tradeoffs as you iterate.

33.2. 1. The Situation

Studio 3 improved performance by adding indexes and targeted access paths over the Studio 2 model classes.

Now a new problem appears: architecture fragility.

Typical symptoms:

- orchestration logic mixed with storage details
- duplicate workflow rules across modules
- direct dependency on concrete classes everywhere
- difficult integration testing due to tight coupling

Studio 4 is the natural next step: keep Studio 3 behavior gains, but move to explicit architectural boundaries.

33.3. 2. Engineering Brief

Refactor around components, ports, and contracts without changing externally visible behavior.

Required outcomes:

- preserve Studio 3 semantics and performance intent
- define explicit component boundaries
- introduce interface-style ports for collaboration
- isolate domain logic from infrastructure details

- keep dependencies pointing toward domain intent

Implementation guidance:

- start from existing V3 code paths
- create seams first, then move logic behind seams
- prove behavior parity with before/after run checks

33.4. 3. Implementation In Nex

Suggested files:

- `architecture_ports.nex`
- `delivery_arch_refactor.nex`
- `knowledge_arch_refactor.nex`
- `world_arch_refactor.nex`
- `studio_4_main.nex`

If using the web IDE, place all classes in one file and run `App.run`.

33.4.1. 3.1 Shared Ports (Interfaces By Contract)

These ports abstract over concrete V3 storage/index implementations from Studio 3.

```
class Task_Query_Port
feature
  status_of(task_id: String): String
    require
      id_present: task_id /= ""
    do
      result := "NOT_IMPLEMENTED"
    ensure
      non_empty: result /= ""
    end
end

class Doc_Query_Port
feature
  has_doc(doc_id: String): Boolean
    require
```

```
      id_present: doc_id /= ""
    do
      result := false
    ensure
      bool_result: result = true or result = false
    end
end

class World_Update_Port
feature
  move_by_id(object_id: String; delta: Integer): String
    require
      id_present: object_id /= ""
    do
      result := "NOT_IMPLEMENTED"
    ensure
      non_empty: result /= ""
    end
end
```

33.4.2. 3.2 Delivery Refactor (Using Studio 3 V3 Store)

Before (Studio 3): callers directly used `Delivery_Task_Store_V3`.

After (Studio 4): callers depend on `Task_Query_Port`; concrete store is adapted behind it.

```
class Delivery_Task_Store_V3
feature
  k1: String
  v1: String
  k2: String
  v2: String

  find_status(task_id: String): String
    require
      id_present: task_id /= ""
    do
      if task_id = k1 then
        result := v1
      elseif task_id = k2 then
        result := v2
      else
```

```
        result := "NOT_FOUND"
      end
    ensure
      non_empty: result /= ""
    end
end

class Delivery_Task_Adapter
feature
  store: Delivery_Task_Store_V3

  status_of(task_id: String): String
    require
      id_present: task_id /= ""
    do
      result := store.find_status(task_id)
    ensure
      non_empty: result /= ""
    end
end

class Delivery_Workflow
feature
  tasks: Delivery_Task_Adapter

  dispatch_view(task_id: String): String
    require
      id_present: task_id /= ""
    do
      let st: String := tasks.status_of(task_id)
      if st = "NOT_FOUND" then
        result := "UNKNOWN_TASK"
      else
        result := "TASK_STATUS:" + st
      end
    ensure
      non_empty: result /= ""
    end
end
```

33.4.3. 3.3 Knowledge Refactor (Port + Validator Service)

```
class Doc_Index_V3
feature
  dk1: String
  dk2: String
  dk3: String

  has_doc(doc_id: String): Boolean
    require
      id_present: doc_id /= ""
    do
      result := doc_id = dk1 or doc_id = dk2 or doc_id = dk3
    ensure
      bool_result: result = true or result = false
    end
end

class Doc_Query_Adapter
feature
  idx: Doc_Index_V3

  has_doc(doc_id: String): Boolean
    require
      id_present: doc_id /= ""
    do
      result := idx.has_doc(doc_id)
    ensure
      bool_result: result = true or result = false
    end
end

class Link_Validation_Service
feature
  docs: Doc_Query_Adapter

  validate(from_id, to_id: String): String
    require
      ids_present: from_id /= "" and to_id /= ""
    do
      if not docs.has_doc(from_id) then
        result := "MISSING_FROM"
```

```
      elseif not docs.has_doc(to_id) then
        result := "MISSING_TO"
      else
        result := "VALID"
      end
    ensure
      known_result: result = "VALID" or result = "MISSING_FROM"
    end
end
```

33.4.4. 3.4 World Refactor (Port + Use Case Service)

```
class World_Model_V3
feature
  wk1: String
  wx1: Integer

  move_by_id(object_id: String; delta: Integer): String
    require
      id_present: object_id /= ""
    do
      if object_id = wk1 then
        wx1 := wx1 + delta
        result := "UPDATED"
      else
        result := "NOT_FOUND"
      end
    ensure
      non_empty: result /= ""
    end
end

class World_Update_Adapter
feature
  model: World_Model_V3

  move_by_id(object_id: String; delta: Integer): String
    require
      id_present: object_id /= ""
    do
      result := model.move_by_id(object_id, delta)
```

```
    ensure
      non_empty: result /= ""
    end
end

class World_Step_Service
feature
  updater: World_Update_Adapter

  apply_player_input(object_id: String; delta: Integer): Str
    require
      id_present: object_id /= ""
    do
      result := updater.move_by_id(object_id, delta)
    ensure
      known_result: result = "UPDATED" or result = "NOT_FOUN
    end
end
```

33.4.5. 3.5 Studio Driver (Behavior Parity Check)

```
class App
feature
  run() do
    -- Delivery setup
    let ds: Delivery_Task_Store_V3 := create Delivery_Task_S
    ds.k1 := "T-1"; ds.v1 := "PENDING"
    ds.k2 := "T-2"; ds.v2 := "IN_TRANSIT"

    let da: Delivery_Task_Adapter := create Delivery_Task_Ad
    da.store := ds

    let dw: Delivery_Workflow := create Delivery_Workflow
    dw.tasks := da

    print(dw.dispatch_view("T-2"))   -- TASK_STATUS:IN_TRANS
    print(dw.dispatch_view("T-99"))  -- UNKNOWN_TASK

    -- Knowledge setup
    let di: Doc_Index_V3 := create Doc_Index_V3
    di.dk1 := "D-1"; di.dk2 := "D-2"; di.dk3 := "D-3"
```

```
    let dqa: Doc_Query_Adapter := create Doc_Query_Adapter
    dqa.idx := di

    let lvs: Link_Validation_Service := create Link_Validation_S
    lvs.docs := dqa

    print(lvs.validate("D-1", "D-3"))   -- VALID
    print(lvs.validate("D-9", "D-3"))   -- MISSING_FROM

    -- World setup
    let wm: World_Model_V3 := create World_Model_V3
    wm.wk1 := "E-1"; wm.wx1 := 10

    let wua: World_Update_Adapter := create World_Update_Adapter
    wua.model := wm

    let wss: World_Step_Service := create World_Step_Service
    wss.updater := wua

    print(wss.apply_player_input("E-1", 2))  -- UPDATED
    print(wss.apply_player_input("E-9", 2))  -- NOT_FOUND
  end
end
```

Expected outcome:

- same core behavior as Studio 3 refactored paths
- clearer component boundaries
- easier component-level testing due to port seams

33.5. 4. Studio Challenges

33.5.1. Level 1 — One Architectural Seam

- pick one Studio 3 optimized path
- introduce a port + adapter boundary
- prove behavior parity on nominal and edge cases

33.5.2. Level 2 — Three-Domain Architecture Pass

- apply port/adapter + service orchestration pattern to delivery, knowledge, and world

- remove at least one direct concrete dependency per domain
- document dependency direction before/after

33.5.3. Level 3 — Alternative Architecture Experiment

- compare two styles:
  - port/adapter with thin services
  - feature modules with internal adapters
- evaluate maintainability and migration cost

33.6. 5. Postmortem

Discuss with evidence:

- Which seam reduced coupling the most?
- Which abstraction clarified behavior, and which added unnecessary ceremony?
- Which components are still doing too much?
- What is the next architecture risk if the system doubles again?

33.7. Deliverables

- architecture diagram with components and dependency direction
- refactored Nex code using ports/adapters over Studio 3 V3 logic
- behavior-parity run log (before/after outputs)
- short ADR-style note with chosen architecture and rejected alternative

---

33.8. Exit Criteria

You are ready for Part VII if:

- key use cases depend on ports, not concrete storage classes
- orchestration logic is separated from storage/index details
- cross-component contracts are explicit and testable
- behavior parity with Studio 3 is demonstrated

Part VII.

Part VII — Making Software Trustworthy

34. Part VII — Making Software Trustworthy — Preconditions and Postconditions

34.1. 27. Preconditions and Postconditions

By Part VI, we can build modular software.

Part VII focuses on trust: making behavior explicit, checkable, and dependable.

Preconditions and postconditions are the first layer of that trust model.

34.2. Why Contracts Matter

Every routine has assumptions and promises.

A precondition (`require`) states what must be true before execution. A postcondition (`ensure`) states what must be true after execution.

Without contracts, assumptions are implicit and failures appear far from the source.

34.3. Practical Contract Rules

Use preconditions for caller obligations:

- input presence
- valid ranges
- required state before transition

Use postconditions for routine guarantees:

- result shape/value constraints
- state transition outcomes
- explicit failure semantics

A useful rule:

- never use postconditions to check caller mistakes
- never use preconditions to state routine results

34.4. Worked Design Path

Requirement:

> "Assign task to robot and return assignment status."

34.4.1. Step 1: Define caller obligations

- task id present
- robot id present
- task is assignable

34.4.2. Step 2: Define routine guarantees

- task has assigned robot on success
- status is updated to `IN_TRANSIT`

34.4.3. Step 3: Define failure behavior

- violation should fail fast with contract signal

34.4.4. Step 4: Encode in code

- put input/state rules in `require`
- put assignment guarantees in `ensure`

34.5. Nex Implementation Sketch

```
class Delivery_Task
feature
  task_id: String
  status: String
  assigned_robot_id: String

  assign(robot_id: String): String
    require
      task_id_present: task_id /= ""
      robot_id_present: robot_id /= ""
      can_assign: status = "PENDING" or status = "FAILED"
    do
      assigned_robot_id := robot_id
      status := "IN_TRANSIT"
      result := "ASSIGNED"
    ensure
      assigned: assigned_robot_id = robot_id
      now_in_transit: status = "IN_TRANSIT"
      known_result: result = "ASSIGNED"
    end
invariant
  valid_status:
    status = "PENDING" or
    status = "IN_TRANSIT" or
    status = "DELIVERED" or
    status = "FAILED"
end
```

The routine is now self-documenting: obligations and guarantees are executable.

34.6. Contracts Across The Three Systems

34.6.1. Delivery

- route inputs valid; output path/status guaranteed

34.6.2. Knowledge

- query non-empty; ranked result format guaranteed

34.6.3. Virtual World

- step inputs valid; world bounds preserved after update

Contracts turn informal expectations into runtime-checked behavior.

34.7. Common Mistakes

34.7.1. Mistake 1: Missing preconditions for state transitions

Symptom:

- illegal transitions occur silently

Recovery:

- encode transition legality in `require`

34.7.2. Mistake 2: Weak postconditions

Symptom:

- success path does not guarantee meaningful result

Recovery:

- ensure key outputs and state changes explicitly

34.7.3. Mistake 3: Contract duplication drift

Symptom:

- similar routines have inconsistent assumptions

Recovery:

- centralize patterns and review contract consistency

Exercise Apply the section task and record your results before reading the solution notes.

34.8. Quick Exercise (10 Minutes)

Choose one routine and write:

1. three preconditions
2. three postconditions
3. one invalid call case
4. one valid call case

Then verify your postconditions still hold after a refactor.

34.9. Connection to Nex

Nex supports contract-first design directly with `require` and `ensure`, making intent explicit in executable form.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

34.10. Chapter Takeaways

- Preconditions define caller obligations.
- Postconditions define routine guarantees.
- Contracts improve diagnostics and reduce hidden assumptions.
- Trustworthy code starts with explicit behavior boundaries.

In Chapter 28, we move from routine-level guarantees to class-level guarantees: invariants.

35. Part VII — Making Software Trustworthy — Invariants: Rules That Must Never Break

35.1. 28. Invariants: Rules That Must Never Break

Preconditions and postconditions protect individual routine calls.

Invariants protect object validity across its whole lifetime.

An invariant is a truth that must hold whenever an object is externally observable.

35.2. Why Invariants Matter

Without invariants, systems can enter invalid intermediate states that later code assumes are impossible.

Invariants provide persistent integrity constraints:

- valid state domains
- relationship consistency
- safety bounds

They reduce bug surface by making illegal states hard to represent.

35.3. Invariant Scope

Typical invariant categories:

- field-level: `id /= ""`
- relationship-level: `in_transit -> assigned_robot_id /= ""`
- range-level: `x >= 0 and x <= max_x`

Invariants should be strong enough to protect integrity but not so strict that normal transitions become impossible.

35.4. Worked Design Path

Requirement:

> "World objects move each tick but must always stay in legal bounds."

35.4.1. Step 1: Identify always-true rules

- object id must exist
- position must remain within global limits

35.4.2. Step 2: Keep transition behavior flexible

- updates may change velocity and position
- invariants should constrain outcomes, not forbid normal movement

35.4.3. Step 3: Enforce invariants at class level

- all routines must preserve those truths

35.5. Nex Implementation Sketch

```
class World_Object
feature
  object_id: String
  x: Integer
  vx: Integer
```

```
  max_x: Integer

  step()
    require
      max_valid: max_x >= 0
    do
      let next: Integer := x + vx
      if next < 0 then
        x := 0
      elseif next > max_x then
        x := max_x
      else
        x := next
      end
    ensure
      bounded_after_step: x >= 0 and x <= max_x
    end
invariant
  id_present: object_id /= ""
  max_non_negative: max_x >= 0
  x_bounded: x >= 0 and x <= max_x
end
```

Invariants and routine contracts reinforce each other: state never escapes legal bounds.

35.6. Invariants Across The Three Systems

35.6.1. Delivery

- task status must be one of declared statuses
- in-transit tasks must have assigned robot

35.6.2. Knowledge

- document ids are non-empty and unique in scope
- links cannot self-reference when forbidden

35.6.3. Virtual World

- entity positions within world bounds

- illegal state combinations disallowed

35.7. Common Mistakes

35.7.1. Mistake 1: Invariants too weak

Symptom:

- invalid states still possible

Recovery:

- add relationship-aware constraints

35.7.2. Mistake 2: Invariants too strict

Symptom:

- legitimate transitions blocked

Recovery:

- move transition-specific checks into routine contracts

35.7.3. Mistake 3: Duplicate rule definitions

Symptom:

- invariant and routine rules drift apart

Recovery:

- define core truths once at class level

Exercise Apply the section task and record your results before reading the solution notes.

35.8. Quick Exercise (12 Minutes)

For one class, define:

1. two field invariants
2. one relationship invariant
3. one routine that could violate them
4. one test that confirms invariants hold after routine execution

Then intentionally break one rule and observe failure diagnostics.

35.9. Connection to Nex

Nex class invariants are a direct mechanism for persistent state integrity and align naturally with contract-based design.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

35.10. Chapter Takeaways

- Invariants protect state integrity across object lifetime.
- They complement preconditions/postconditions at routine boundaries.
- Good invariants balance strength and evolvability.
- Integrity rules belong near the data they protect.

In Chapter 29, we use tests to explore system behavior beyond obvious scenarios.

36. Part VII — Making Software Trustworthy — Testing as Exploration

36.1. 29. Testing as Exploration

Contracts define intended behavior.

Tests provide evidence that behavior holds across real scenarios, edge cases, and adversarial inputs.

Treating tests as exploration changes mindset from "confirm happy path" to "discover failure boundaries."

36.2. What To Test

A practical test portfolio includes:

- example tests: concrete known scenarios
- boundary tests: limits and edge values
- property-oriented checks: truths that should always hold

Strong testing strategy targets risk, not just code coverage percentages.

36.3. Oracles Matter

A test is only as good as its oracle (how it decides pass/fail).

Good oracles:

- explicit expected outputs
- invariants/properties that must hold
- status/contract validation

Weak oracles ("did not crash") miss most meaningful defects.

36.4. Worked Design Path

Requirement:

> "Route planner should return valid route or explicit unreachable status."

Test plan:

1. nominal route exists
2. start equals destination
3. unknown nodes
4. disconnected graph
5. property check: if status is `FOUND`, path must be non-empty

This blends scenario tests with contract-inspired property checks.

36.5. Nex Implementation Sketch

```
class Route_Planner
feature
  plan(start_loc, end_loc: String): String
    require
      inputs_present: start_loc /= "" and end_loc /= ""
    do
      if start_loc = end_loc then
        result := start_loc
      elseif start_loc = "A" and end_loc = "C" then
        result := "A->B->C"
      else
        result := "UNREACHABLE"
      end
    ensure
      non_empty_result: result /= ""
    end
end
```

```
class Route_Planner_Tests
feature
  run_all(): String
    do
      let p: Route_Planner := create Route_Planner

      let t1: String := p.plan("A", "C")
      let t2: String := p.plan("C", "C")
      let t3: String := p.plan("X", "Z")

      if t1 = "A->B->C" and t2 = "C" and t3 = "UNREACHABLE"
        result := "PASS"
      else
        result := "FAIL"
      end
    ensure
      known_result: result = "PASS" or result = "FAIL"
    end
end
```

Even simple tests become useful when expected behavior is precise.

36.6. Testing Across The Three Systems

36.6.1. Delivery

- transition legality and route validity

36.6.2. Knowledge

- ranking determinism and miss semantics

36.6.3. Virtual World

- bounded updates and deterministic step behavior

Testing should target system risks unique to each domain.

36.7. Common Mistakes

36.7.1. Mistake 1: Happy-path-only testing

Symptom:

- production failures from edge inputs

Recovery:

- include boundary and adversarial cases

36.7.2. Mistake 2: Fragile assertions

Symptom:

- tests fail on irrelevant formatting/order noise

Recovery:

- assert meaningful properties and contracts

36.7.3. Mistake 3: Test logic duplicates bugs

Symptom:

- test and implementation share same wrong assumption

Recovery:

- use independent oracle logic where possible

Exercise Apply the section task and record your results before reading the solution notes.

36.8. Quick Exercise (12 Minutes)

For one module, write:

1. two example tests
2. two boundary tests
3. one property-based check
4. one intentionally adversarial test

Then map each test to a specific risk it addresses.

36.9. Connection to Nex

Nex contracts provide strong test oracles by turning assumptions and guarantees into executable checks.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

36.10. Chapter Takeaways

- Tests are exploration, not just confirmation.
- Good oracles are explicit and behavior-focused.
- Balanced test portfolios combine examples, boundaries, and properties.
- Contract-driven systems enable stronger, clearer tests.

In Chapter 30, we examine debugging as a hypothesis-driven engineering process.

37. Part VII — Making Software Trustworthy — Debugging Like an Engineer

37.1. 30. Debugging Like an Engineer

Testing tells us something is wrong.

Debugging determines why it is wrong and how to fix it without creating new faults.

Engineering-grade debugging is a structured process, not trial-and-error edits.

37.2. Debugging As Hypothesis Work

Effective debugging loop:

1. reproduce reliably
2. localize failure boundary
3. form hypothesis
4. run targeted check
5. confirm cause
6. patch and verify regression safety

Skipping steps causes "fixes" that mask symptoms and move bugs elsewhere.

37.3. Reproducibility First

A bug you cannot reproduce cannot be fixed confidently.

Before patching:

- capture exact inputs/state

- minimize to a smallest failing case
- note expected vs observed behavior

This transforms debugging from guesswork into evidence-driven analysis.

37.4. Worked Design Path

Bug report:

> "Some delivery tasks jump directly from PENDING to DELIVERED."

Debug steps:

1. reproduce with minimal task flow
2. inspect transition entry points
3. hypothesize missing precondition on `complete()`
4. confirm by calling `complete()` without `IN_TRANSIT`
5. patch contract and rerun regression checks

Result: illegal transition blocked at boundary.

37.5. Nex Implementation Sketch

```
class Delivery_Task
feature
  status: String

  start()
    require
      can_start: status = "PENDING" or status = "FAILED"
    do
      status := "IN_TRANSIT"
    ensure
      now_in_transit: status = "IN_TRANSIT"
    end
```

```
  complete()
    require
      in_transit: status = "IN_TRANSIT"
    do
      status := "DELIVERED"
    ensure
      delivered: status = "DELIVERED"
    end
invariant
  valid_status:
    status = "PENDING" or
    status = "IN_TRANSIT" or
    status = "DELIVERED" or
    status = "FAILED"
end

class Debug_Smoke_Test
feature
  run(): String
    do
      let t: Delivery_Task := create Delivery_Task
      t.status := "PENDING"
      t.start
      t.complete

      if t.status = "DELIVERED" then
        result := "PASS"
      else
        result := "FAIL"
      end
    ensure
      known_result: result = "PASS" or result = "FAIL"
    end
end
```

This patch shape demonstrates contract-based bug prevention and regression confirmation.

37.6. Debugging Across The Three Systems

37.6.1. Delivery

- transition violations and route mismatches

37.6.2. Knowledge

- ranking inconsistencies and stale index effects

37.6.3. Virtual World

- non-deterministic update ordering and bound violations

Debugging strategy is shared: isolate, hypothesize, verify, patch, regress.

37.7. Common Mistakes

37.7.1. Mistake 1: Patching before reproducing

Symptom:

- bug returns in slightly different form

Recovery:

- establish minimal reproducible case first

37.7.2. Mistake 2: Multiple simultaneous changes

Symptom:

- cannot tell which edit fixed or broke behavior

Recovery:

- apply one hypothesis-driven change at a time

37.7.3. Mistake 3: No regression safety net

Symptom:

- fix introduces new failures elsewhere

Recovery:

- add targeted regression tests before merging

Exercise Apply the section task and record your results before reading the solution notes.

37.8. Quick Exercise (12 Minutes)

Take one recent bug and write:

1. reproducible case
2. observed vs expected behavior
3. root-cause hypothesis
4. confirming check
5. patch summary
6. regression tests added

If any step is missing, the fix quality is uncertain.

37.9. Connection to Nex

Nex contracts and invariants accelerate debugging by making assumption violations explicit at runtime boundaries.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

37.10. Chapter Takeaways

- Debugging is hypothesis-driven engineering work.
- Reproducibility is prerequisite to reliable fixes.

- Contracts and invariants help localize root causes quickly.
- Every fix needs regression evidence.

Part VII established trust mechanisms: contracts, invariants, testing, and debugging.

Next is **System Milestone 5 — Reliability**, where these practices are applied end-to-end in Studio 5.

38. Studio 5 — Reliability

38.1. Subtitle

Hardening the Studio 4 architecture for failure, recovery, and operational trust.

Studio Focus This studio chapter is hands-on: implement, verify behavior, and document tradeoffs as you iterate.

38.2. 1. The Situation

Studio 4 gave us clear architectural seams: ports, adapters, and services.

Now reliability work becomes practical because failure handling can be localized at those seams.

Current symptoms:

- intermittent failures at integration boundaries
- retries that duplicate side effects
- hard-to-reproduce incidents
- regressions after urgent patches

Studio 5 is the logical continuation: keep Studio 4 architecture, add reliability guarantees around it.

38.3. 2. Engineering Brief

Build reliability mechanisms on top of Studio 4 components without breaking behavior contracts.

Required outcomes:

- explicit failure semantics for key operations
- retry policy with bounded attempts
- idempotency for repeatable operations
- reproducible failure tests and regression checks

- minimal observability hooks for diagnosis

Implementation guidance:

- enforce reliability at service/port boundaries
- keep domain invariants intact under failure paths
- separate transient failures from permanent failures
- make each retry policy explicit and testable

38.4. 3. Implementation In Nex

Suggested files:

- `reliability_types.nex`
- `delivery_reliability.nex`
- `knowledge_reliability.nex`
- `world_reliability.nex`
- `studio_5_main.nex`

If using the web IDE, place all classes in one file and run `App.run`.

38.4.1. 3.1 Shared Reliability Types

```
class Op_Result
feature
  status: String
  value: String
  attempts: Integer
invariant
  status_present: status /= ""
  value_present: value /= ""
  attempts_non_negative: attempts >= 0
end

class Reliability_Log
feature
  last_event: String

  record(event: String)
    require
```

```
      event_present: event /= ""
    do
      last_event := event
    ensure
      persisted: last_event = event
    end
end
```

38.4.2. 3.2 Delivery Reliability: Retry + Idempotent Dispatch

Continuation from Studio 4 `Delivery_Workflow` idea.

```
class Delivery_Port
feature
  -- Simulate transient failures for first N attempts.
  fail_until_attempt: Integer

  send_dispatch(task_id: String; attempt: Integer): String
    require
      id_present: task_id /= ""
      attempt_valid: attempt >= 1
    do
      if attempt <= fail_until_attempt then
        result := "TRANSIENT_FAILURE"
      else
        result := "SENT"
      end
    ensure
      known_result:
        result = "SENT" or
        result = "TRANSIENT_FAILURE" or
        result = "PERMANENT_FAILURE"
    end
end

class Delivery_Reliability_Service
feature
  port: Delivery_Port
  log: Reliability_Log
  last_dispatched_task: String
```

```
  dispatch_with_retry(task_id: String; max_attempts: Integer): O
    require
      id_present: task_id /= ""
      max_attempts_valid: max_attempts >= 1
    do
      let r: Op_Result := create Op_Result

      -- Idempotency guard: repeated same task dispatch request.
      if last_dispatched_task = task_id then
        r.status := "IDEMPOTENT_OK"
        r.value := "ALREADY_SENT"
        r.attempts := 0
        result := r
      else
        let a: Integer := 1
        let final_status: String := "TRANSIENT_FAILURE"

        from
        until a > max_attempts or final_status = "SENT"
        loop
          final_status := port.send_dispatch(task_id, a)
          log.record("delivery_attempt=" + a)
          a := a + 1
        end

        if final_status = "SENT" then
          last_dispatched_task := task_id
          r.status := "OK"
          r.value := "SENT"
        else
          r.status := "FAILED"
          r.value := final_status
        end

        r.attempts := a - 1
        result := r
      end
    ensure
      known_status: result.status = "OK" or result.status = "FAI
    end
end
```

38.4.3. 3.3 Knowledge Reliability: Safe Query Fallback

Continuation from Studio 4 query/validation services.

```
class Knowledge_Port
feature
  query_available: Boolean

  fetch_top(query: String): String
    require
      query_present: query /= ""
    do
      if query_available then
        result := "DOC:K-101"
      else
        result := "UNAVAILABLE"
      end
    ensure
      non_empty: result /= ""
    end
end

class Knowledge_Reliability_Service
feature
  port: Knowledge_Port
  log: Reliability_Log

  safe_query(query: String): Op_Result
    require
      query_present: query /= ""
    do
      let r: Op_Result := create Op_Result
      let raw: String := port.fetch_top(query)

      if raw = "UNAVAILABLE" then
        log.record("knowledge_fallback_used")
        r.status := "DEGRADED"
        r.value := "DOC:CACHED-001"
        r.attempts := 1
      else
        r.status := "OK"
        r.value := raw
        r.attempts := 1
```

```
      end

      result := r
    ensure
      known_status: result.status = "OK" or result.status = "DEG
    end
end
```

38.4.4. 3.4 World Reliability: Guarded Update + Recovery Status

Continuation from Studio 4 world update service.

```
class World_Port
feature
  object_id: String
  x: Integer
  max_x: Integer

  move(delta: Integer): String
    do
      let next: Integer := x + delta
      if next < 0 then
        x := 0
      elseif next > max_x then
        x := max_x
      else
        x := next
      end
      result := "UPDATED"
    ensure
      bounded: x >= 0 and x <= max_x
    end
end

class World_Reliability_Service
feature
  port: World_Port
  log: Reliability_Log

  safe_step(delta: Integer): Op_Result
    do
```

```
      let r: Op_Result := create Op_Result
      let status: String := port.move(delta)
      log.record("world_step_delta=" + delta)

      r.status := "OK"
      r.value := status
      r.attempts := 1
      result := r
    ensure
      known_status: result.status = "OK"
    end
end
```

38.4.5. 3.5 Reliability Driver + Regression Checks

```
class App
feature
  run() do
    let l: Reliability_Log := create Reliability_Log
    l.last_event := "init"

    -- Delivery reliability run
    let dp: Delivery_Port := create Delivery_Port
    dp.fail_until_attempt := 2

    let dr: Delivery_Reliability_Service := create Delivery_
    dr.port := dp
    dr.log := l
    dr.last_dispatched_task := "NONE"

    let d1: Op_Result := dr.dispatch_with_retry("T-7", 3)
    print("Delivery result: " + d1.status + ", attempts=" +

    let d2: Op_Result := dr.dispatch_with_retry("T-7", 3)
    print("Delivery idempotent result: " + d2.status + ", va

    -- Knowledge reliability run
    let kp: Knowledge_Port := create Knowledge_Port
    kp.query_available := false

    let kr: Knowledge_Reliability_Service := create Knowledg
```

```
    kr.port := kp
    kr.log := l

    let k1: Op_Result := kr.safe_query("contracts")
    print("Knowledge result: " + k1.status + ", value=" + k1.val

    -- World reliability run
    let wp: World_Port := create World_Port
    wp.object_id := "E-1"
    wp.x := 9
    wp.max_x := 10

    let wr: World_Reliability_Service := create World_Reliabilit
    wr.port := wp
    wr.log := l

    let w1: Op_Result := wr.safe_step(5)
    print("World result: " + w1.status + ", value=" + w1.value +

    -- Simple regression assertions
    if d1.status = "OK" and d1.attempts = 3 and d2.status = "IDE
       k1.status = "DEGRADED" and wp.x = 10 then
      print("RELIABILITY_CHECK:PASS")
    else
      print("RELIABILITY_CHECK:FAIL")
    end
  end
end
```

Expected outcomes:

- delivery handles transient failures with bounded retry
- repeated dispatch request is idempotent
- knowledge service degrades gracefully with fallback
- world updates preserve invariants under extreme deltas

38.5. 4. Studio Challenges

38.5.1. Level 1 — Single Reliability Mechanism

- choose one Studio 4 service

- add either retry, idempotency, or fallback logic
- add one regression check proving behavior under failure

38.5.2. Level 2 — Three-Domain Reliability Pass

- implement one reliability mechanism per domain:
  - delivery: retry/idempotency
  - knowledge: degraded fallback
  - world: invariant-preserving recovery
- show failure-path run logs

38.5.3. Level 3 — Incident Drill

- define one simulated production incident per domain
- capture reproduction steps
- patch through service boundary
- add regression checks to prevent recurrence

38.6. 5. Postmortem

Discuss with evidence:

- Which reliability mechanism produced the largest stability gain?
- Which failures should not be retried?
- Which fallback behavior is acceptable vs dangerous?
- Which diagnostics made root-cause fastest?

38.7. Deliverables

- reliability policy sheet (retry, idempotency, fallback, non-retryable failures)
- refactored Nex reliability services and run outputs
- incident playbook with reproduction + fix + regression tests
- release gate checklist for reliability sign-off

38.8. Exit Criteria

You are ready for Part VIII if:

- key operations have explicit failure semantics
- retry logic is bounded and test-backed
- idempotent behaviors are enforced where side effects exist
- degraded-mode behavior is intentional and contract-defined
- regression checks cover critical incident classes

Part VIII.

Part VIII — Systems That Grow

39. Part VIII — Systems That Grow — Managing Complexity

39.1. 31. Managing Complexity

By Studio 5, the systems are more reliable.

Now the next challenge is growth: keeping the code understandable and changeable as teams, features, and integrations expand.

Complexity is not just size. It is the cost of reasoning about behavior.

39.2. Complexity As A Budget

Every system has a complexity budget.

When that budget is exceeded, symptoms appear:

- fragile changes
- long onboarding time
- unclear dependency impact
- rising incident rate after routine edits

Managing complexity means making structure intentional.

39.3. Layered Reasoning

A practical layering model:

- domain layer: business rules and invariants
- application layer: orchestration/use cases
- interface/infrastructure layer: adapters, IO, transport

Key rule:

- dependencies should flow inward toward domain intent

This supports local reasoning and safer change.

39.4. Worked Design Path

Requirement:

> "Add priority reroute for premium tasks without breaking normal dispatch."

Without layering, reroute logic leaks into adapters and transport code.

With layering:

1. domain service decides reroute eligibility
2. app service orchestrates reroute flow
3. adapter layer executes storage and notification

Result: policy change stays mostly in domain/application layers.

39.5. Nex Implementation Sketch

```
class Task
feature
  task_id: String
  tier: String
  status: String
invariant
  id_present: task_id /= ""
  valid_tier: tier = "STANDARD" or tier = "PREMIUM"
end

class Reroute_Policy
feature
  should_reroute(t: Task): Boolean
```

```
    require
      task_present: t.task_id /= ""
    do
      result := t.tier = "PREMIUM" and t.status = "IN_TRANSI
    ensure
      bool_result: result = true or result = false
    end
end

class Dispatch_App_Service
feature
  policy: Reroute_Policy

  process_reroute(t: Task): String
    require
      task_present: t.task_id /= ""
    do
      if policy.should_reroute(t) then
        result := "REROUTE_TRIGGERED"
      else
        result := "NO_REROUTE"
      end
    ensure
      known_result: result = "REROUTE_TRIGGERED" or result =
    end
end
```

This design keeps rule decisions separate from infrastructure actions.

39.6. Complexity Signals Across The Three Systems

39.6.1. Delivery

- dispatch logic split across too many modules

39.6.2. Knowledge

- ranking, retrieval, and caching tangled

39.6.3. Virtual World

- update, collision, and render logic interwoven

Complexity management starts with explicit boundaries and dependency direction.

39.7. Common Mistakes

39.7.1. Mistake 1: Layer bypasses

Symptom:

- UI or adapter code modifies domain state directly

Recovery:

- route state changes through domain/application services

39.7.2. Mistake 2: Shared utility dumping ground

Symptom:

- one module accumulates unrelated logic

Recovery:

- split by domain responsibility, not convenience

39.7.3. Mistake 3: Cyclic dependencies

Symptom:

- modules require each other to compile/change

Recovery:

- extract ports/interfaces and invert dependency

Exercise Apply the section task and record your results before reading the solution notes.

39.8. Quick Exercise (12 Minutes)

For one subsystem:

1. map current modules
2. mark domain/application/interface boundaries
3. identify one dependency pointing the wrong direction
4. define one seam to fix it
5. estimate complexity impact

39.9. Connection to Nex

Nex contracts at layer boundaries make complexity-reduction refactors safer by preserving behavior intent while structure changes.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

39.10. Chapter Takeaways

- Complexity must be managed as an explicit engineering budget.
- Layered boundaries improve local reasoning.
- Dependency direction determines long-term maintainability.
- Reliability degrades when complexity structure is ignored.

In Chapter 32, we focus on designing those boundaries to absorb future change safely.

40. Part VIII — Systems That Grow — Designing for Change

40.1. 32. Designing for Change

Managing complexity keeps today stable.

Designing for change keeps tomorrow affordable.

The goal is not predicting every future requirement. The goal is building seams where likely changes can land without system-wide rewrite.

40.2. Change Seams

A change seam is a boundary where behavior can vary while contracts remain stable.

Common seam locations:

- policy/strategy interfaces
- adapter boundaries
- versioned data contracts
- feature toggle points

Good seams isolate volatility.

40.3. Versioning Strategy

Change-safe systems define explicit evolution policy:

- additive fields over breaking replacements
- deprecation windows

- compatibility adapters for old clients

Without versioning discipline, small improvements become migration crises.

40.4. Worked Design Path

Requirement:

> "Add new ranking strategy without breaking existing query clients."

Design path:

1. keep stable query interface (`rank(query): result`)
2. extract ranking strategy port
3. implement V1 and V2 strategies behind same contract
4. choose strategy via configuration/feature flag

Clients remain unchanged while behavior evolves.

40.5. Nex Implementation Sketch

```
class Rank_Strategy
feature
  rank(query: String): String
    require
      query_present: query /= ""
    do
      result := "NOT_IMPLEMENTED"
    ensure
      non_empty: result /= ""
    end
end

class Rank_V1
feature
  rank(query: String): String
```

```
    require
      query_present: query /= ""
    do
      result := "DOC:LEGACY-1"
    ensure
      non_empty: result /= ""
    end
end

class Rank_V2
feature
  rank(query: String): String
    require
      query_present: query /= ""
    do
      result := "DOC:MODERN-1"
    ensure
      non_empty: result /= ""
    end
end

class Rank_Service
feature
  strategy: Rank_V1

  run(query: String): String
    require
      query_present: query /= ""
    do
      result := strategy.rank(query)
    ensure
      non_empty: result /= ""
    end
end
```

The seam is clear: change ranking behavior without breaking service contract.

40.6. Designing For Change Across The Three Systems

40.6.1. Delivery

- swap route policies without changing dispatch API

40.6.2. Knowledge

- evolve ranking/scoring strategies behind stable query contract

40.6.3. Virtual World

- add new simulation rules behind existing step interface

Safe change depends on stable contracts and isolated variation points.

40.7. Common Mistakes

40.7.1. Mistake 1: Premature abstraction

Symptom:

- many empty seams with no real variation

Recovery:

- add seams where volatility is likely and evidenced

40.7.2. Mistake 2: No compatibility story

Symptom:

- new version breaks existing consumers

Recovery:

- plan versioning and transitional adapters

40.7.3. Mistake 3: Hidden feature flags

Symptom:

- behavior varies silently across environments

Recovery:

- make strategy selection explicit and observable

Exercise Apply the section task and record your results before reading the solution notes.

40.8. Quick Exercise (12 Minutes)

Pick one likely future change and define:

1. stable contract to preserve
2. seam to introduce
3. old/new implementation variants
4. rollout plan
5. rollback condition

Then identify one compatibility test you must automate.

40.9. Connection to Nex

Nex contract checks keep seams honest: old and new variants can be validated against the same behavior guarantees.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

40.10. Chapter Takeaways

- Design for change means designing for controlled variation.
- Stable contracts are the backbone of safe evolution.

- Versioning and compatibility are architecture concerns, not release chores.
- Seams should be intentional, not speculative.

In Chapter 33, we close Part VIII with refactoring practices that preserve behavior while restructuring systems.

41. Part VIII — Systems That Grow — Refactoring Without Fear

41.1. 33. Refactoring Without Fear

Growth requires change.

Refactoring is how we improve structure without changing observable behavior.

Fear appears when teams lack safety nets and clear process.

This chapter makes refactoring an engineering routine, not a risky event.

41.2. Behavior Preservation First

Refactoring is valid only when behavior remains equivalent for intended contracts.

Core safety tools:

- contract checks
- regression tests
- incremental scope
- before/after comparators

Without these, "refactor" becomes disguised rewrite.

41.3. Incremental Refactor Pattern

A reliable sequence:

1. capture current behavior with tests/contracts

2. extract seam (interface/adapter/function boundary)
3. move one responsibility at a time
4. run parity checks after each step
5. remove dead path only after confidence

This pattern minimizes risk and improves reversibility.

41.4. Worked Design Path

Requirement:

> "Split one large knowledge service into retrieval and ranking services."

Plan:

1. add tests for existing outputs
2. extract `retrieve_candidates` method
3. extract `rank_candidates` method
4. wire through coordinator
5. run before/after result comparison

Outcome: improved structure with preserved external behavior.

41.5. Nex Implementation Sketch

```
class Legacy_Knowledge_Service
feature
  query(q: String): String
    require
      query_present: q /= ""
    do
      if q = "graphs" then
        result := "DOC:G-1"
      else
        result := "DOC:GENERIC"
      end
    ensure
```

```
      non_empty: result /= ""
    end
end

class Retrieval_Service
feature
  retrieve(q: String): String
    require
      query_present: q /= ""
    do
      if q = "graphs" then
        result := "CAND:G-1"
      else
        result := "CAND:GENERIC"
      end
    ensure
      non_empty: result /= ""
    end
end

class Ranking_Service
feature
  rank(candidate: String): String
    require
      candidate_present: candidate /= ""
    do
      if candidate = "CAND:G-1" then
        result := "DOC:G-1"
      else
        result := "DOC:GENERIC"
      end
    ensure
      non_empty: result /= ""
    end
end

class Refactored_Knowledge_Service
feature
  retrieval: Retrieval_Service
  ranking: Ranking_Service

  query(q: String): String
```

```
    require
      query_present: q /= ""
    do
      let c: String := retrieval.retrieve(q)
      result := ranking.rank(c)
    ensure
      non_empty: result /= ""
    end
end
```

Both services expose the same outward query contract while internal structure improves.

41.6. Refactoring Risks Across The Three Systems

41.6.1. Delivery

- transition logic moved incorrectly across components

41.6.2. Knowledge

- retrieval/ranking split changes subtle ordering behavior

41.6.3. Virtual World

- update-loop extraction breaks deterministic order

Refactoring safety depends on explicit behavior contracts and parity checks.

41.7. Common Mistakes

41.7.1. Mistake 1: Big-bang refactor

Symptom:

- many failures with unclear root cause

Recovery:

- refactor in small verified slices

41.7.2. Mistake 2: Missing parity tests

Symptom:

- "cleaner" design changes business behavior unintentionally

Recovery:

- compare before/after outputs on representative cases

41.7.3. Mistake 3: Early deletion of old path

Symptom:

- no rollback option when regressions appear

Recovery:

- keep transitional path until confidence threshold

Exercise Apply the section task and record your results before reading the solution notes.

41.8. Quick Exercise (12 Minutes)

Choose one module and plan a refactor:

1. current behavior contract
2. seam to extract
3. first small structural change
4. parity check set
5. rollback criterion

Then execute only step 1 and step 2 before making deeper edits.

41.9. Connection to Nex

Nex contracts and invariants provide a built-in behavior guardrail for incremental refactoring, reducing fear and guesswork.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

41.10. Chapter Takeaways

- Refactoring is behavior-preserving structural improvement.
- Incremental, contract-backed steps reduce risk.
- Parity testing is the key confidence mechanism.
- Fear decreases when rollback and verification are explicit.

Part VIII established growth discipline: complexity control, change-oriented design, and safe refactoring.

Next is **System Milestone 6 — Evolution**, where these practices are applied to long-term system adaptation in Studio 6.

42. Studio 6 — Evolution

42.1. Subtitle

Evolving the reliable Studio 5 system into a long-lived platform.

Studio Focus This studio chapter is hands-on: implement, verify behavior, and document tradeoffs as you iterate.

42.2. 1. The Situation

Studio 5 made the architecture reliable: retries, idempotency, fallback behavior, and reproducible regression checks.

Now the challenge is not reliability alone. It is sustained evolution under new requirements, new clients, and changing policies.

Typical pressure points:

- new business rules that must coexist with legacy behavior
- API evolution without breaking existing consumers
- data migrations that must be reversible
- feature rollout risk in production environments

Studio 6 is the logical next step: evolve safely without losing reliability or architectural clarity.

42.3. 2. Engineering Brief

Deliver new capabilities while preserving Studio 5 trust guarantees.

Required outcomes:

- introduce versioned behavior behind stable contracts
- support compatibility mode for old clients
- add explicit migration path and rollback trigger
- preserve reliability checks from Studio 5
- define evolution governance (deprecation, sunset, ownership)

Implementation guidance:

- use additive changes before breaking replacements
- isolate v1/v2 differences behind strategy or adapter seam
- treat migration as code, not informal operations notes
- gate rollout with measurable signals and exit criteria

42.4. 3. Implementation In Nex

Suggested files:

- `evolution_types.nex`
- `delivery_evolution.nex`
- `knowledge_evolution.nex`
- `world_evolution.nex`
- `migration_runner.nex`
- `studio_6_main.nex`

If using the web IDE, place all classes in one file and run `App.run`.

42.4.1. 3.1 Shared Evolution Types

```
class Evolution_Result
feature
  status: String
  value: String
  mode: String
invariant
  status_present: status /= ""
  value_present: value /= ""
  mode_present: mode /= ""
end

class Rollout_Config
feature
  use_v2: Boolean
  compatibility_mode: Boolean
end
```

42.4.2. 3.2 Delivery Evolution (Policy V1 -> V2)

Continuation from Studio 5 delivery reliability flow.

```
class Delivery_Policy_V1
feature
  decide_priority(tier: String): String
    require
      tier_present: tier /= ""
    do
      if tier = "PREMIUM" then
        result := "FAST_TRACK"
      else
        result := "STANDARD_TRACK"
      end
    ensure
      known_result: result = "FAST_TRACK" or result = "STAND
    end
end

class Delivery_Policy_V2
feature
  decide_priority(tier: String; risk_score: Integer): String
    require
      tier_present: tier /= ""
      risk_non_negative: risk_score >= 0
    do
      if tier = "PREMIUM" and risk_score < 50 then
        result := "FAST_TRACK"
      elseif tier = "PREMIUM" then
        result := "SAFE_TRACK"
      else
        result := "STANDARD_TRACK"
      end
    ensure
      known_result:
        result = "FAST_TRACK" or
        result = "SAFE_TRACK" or
        result = "STANDARD_TRACK"
    end
end

class Delivery_Evolution_Service
feature
  v1: Delivery_Policy_V1
  v2: Delivery_Policy_V2
```

```
  config: Rollout_Config

  route_mode(tier: String; risk_score: Integer): Evolution_Resul
    require
      tier_present: tier /= ""
      risk_non_negative: risk_score >= 0
    do
      let r: Evolution_Result := create Evolution_Result

      if config.use_v2 then
        r.status := "OK"
        r.value := v2.decide_priority(tier, risk_score)
        r.mode := "V2"
      else
        r.status := "OK"
        r.value := v1.decide_priority(tier)
        r.mode := "V1"
      end

      result := r
    ensure
      ok_status: result.status = "OK"
    end
end
```

42.4.3. 3.3 Knowledge Evolution (Compatibility Adapter)

Continuation from Studio 5 safe query + fallback logic.

```
class Knowledge_Query_V1
feature
  run(query: String): String
    require
      query_present: query /= ""
    do
      result := "DOC:K-LEGACY"
    ensure
      non_empty: result /= ""
    end
end

class Knowledge_Query_V2
```

```
feature
  run(query: String; intent: String): String
    require
      query_present: query /= ""
      intent_present: intent /= ""
    do
      if intent = "deep" then
        result := "DOC:K-DEEP"
      else
        result := "DOC:K-FAST"
      end
    ensure
      non_empty: result /= ""
    end
end

class Knowledge_Compatibility_Adapter
feature
  v2: Knowledge_Query_V2

  run_v1_shape(query: String): String
    require
      query_present: query /= ""
    do
      result := v2.run(query, "fast")
    ensure
      non_empty: result /= ""
    end
end

class Knowledge_Evolution_Service
feature
  v1: Knowledge_Query_V1
  v1_on_v2: Knowledge_Compatibility_Adapter
  config: Rollout_Config

  query(query: String): Evolution_Result
    require
      query_present: query /= ""
    do
      let r: Evolution_Result := create Evolution_Result
```

```
      if config.use_v2 and not config.compatibility_mode then
        r.status := "OK"
        r.value := "DOC:K-DEEP"
        r.mode := "V2_NATIVE"
      elseif config.use_v2 then
        r.status := "OK"
        r.value := v1_on_v2.run_v1_shape(query)
        r.mode := "V2_COMPAT"
      else
        r.status := "OK"
        r.value := v1.run(query)
        r.mode := "V1"
      end

      result := r
    ensure
      ok_status: result.status = "OK"
    end
end
```

42.4.4. 3.4 World Evolution (Feature Flagged Rule Extension)

Continuation from Studio 5 safe world-step behavior.

```
class World_Rules_V1
feature
  apply(x: Integer; delta: Integer; max_x: Integer): Integer
    require
      max_valid: max_x >= 0
    do
      let next: Integer := x + delta
      if next < 0 then
        result := 0
      elseif next > max_x then
        result := max_x
      else
        result := next
      end
    ensure
      bounded: result >= 0 and result <= max_x
    end
```

```
end

class World_Rules_V2
feature
  apply(x: Integer; delta: Integer; max_x: Integer): Integer
    require
      max_valid: max_x >= 0
    do
      -- New damping rule for safer high-velocity transition
      let effective_delta: Integer := delta
      if delta > 3 then
        effective_delta := 3
      elseif delta < -3 then
        effective_delta := -3
      end

      let next: Integer := x + effective_delta
      if next < 0 then
        result := 0
      elseif next > max_x then
        result := max_x
      else
        result := next
      end
    ensure
      bounded: result >= 0 and result <= max_x
    end
end

class World_Evolution_Service
feature
  v1: World_Rules_V1
  v2: World_Rules_V2
  config: Rollout_Config

  step(x: Integer; delta: Integer; max_x: Integer): Evolutio
    require
      max_valid: max_x >= 0
    do
      let r: Evolution_Result := create Evolution_Result
      let out_x: Integer := 0
```

```
      if config.use_v2 then
        out_x := v2.apply(x, delta, max_x)
        r.mode := "V2"
      else
        out_x := v1.apply(x, delta, max_x)
        r.mode := "V1"
      end

      r.status := "OK"
      r.value := "X=" + out_x
      result := r
    ensure
      ok_status: result.status = "OK"
    end
end
```

42.4.5. 3.5 Migration Runner + Rollback Trigger

```
class Migration_Runner
feature
  migrated_count: Integer

  run(canary_passed: Boolean): String
    do
      if canary_passed then
        migrated_count := migrated_count + 100
        result := "MIGRATION_CONTINUE"
      else
        result := "ROLLBACK"
      end
    ensure
      known_result: result = "MIGRATION_CONTINUE" or result = "R
    end
end

class App
feature
  run() do
    let cfg: Rollout_Config := create Rollout_Config
    cfg.use_v2 := true
    cfg.compatibility_mode := true
```

```
-- Delivery evolution
let d1: Delivery_Policy_V1 := create Delivery_Policy_V1
let d2: Delivery_Policy_V2 := create Delivery_Policy_V2
let de: Delivery_Evolution_Service := create Delivery_Ev
de.v1 := d1
de.v2 := d2
de.config := cfg

let dr: Evolution_Result := de.route_mode("PREMIUM", 70)
print("Delivery: mode=" + dr.mode + " value=" + dr.value

-- Knowledge evolution
let k1: Knowledge_Query_V1 := create Knowledge_Query_V1
let kv2: Knowledge_Query_V2 := create Knowledge_Query_V2
let ka: Knowledge_Compatibility_Adapter := create Knowle
ka.v2 := kv2

let ke: Knowledge_Evolution_Service := create Knowledge_
ke.v1 := k1
ke.v1_on_v2 := ka
ke.config := cfg

let kr: Evolution_Result := ke.query("contracts")
print("Knowledge: mode=" + kr.mode + " value=" + kr.valu

-- World evolution
let w1: World_Rules_V1 := create World_Rules_V1
let w2: World_Rules_V2 := create World_Rules_V2
let we: World_Evolution_Service := create World_Evolutio
we.v1 := w1
we.v2 := w2
we.config := cfg

let wr: Evolution_Result := we.step(8, 6, 10)
print("World: mode=" + wr.mode + " value=" + wr.value)

-- Migration + rollback gate
let m: Migration_Runner := create Migration_Runner
m.migrated_count := 0
print("Migration gate: " + m.run(true))
print("Migration count: " + m.migrated_count)
print("Rollback gate: " + m.run(false))
```

```
  end
end
```

Expected outcomes:

- v2 behavior can be rolled out without breaking v1 clients
- compatibility mode supports gradual client migration
- migration continues only when canary checks pass
- rollback condition is explicit and executable

42.5. 4. Studio Challenges

42.5.1. Level 1 — One Evolution Path

- evolve one Studio 5 service from v1 to v2 behind a stable contract
- add compatibility mode
- prove old client behavior still works

42.5.2. Level 2 — Coordinated Multi-Domain Evolution

- add v2 behavior for delivery, knowledge, and world
- define one shared rollout policy
- show canary/rollback behavior with run logs

42.5.3. Level 3 — Future-Proofing Drill

- simulate two additional future changes
- evaluate if current seams absorb them cleanly
- define deprecation timeline and ownership

42.6. 5. Postmortem

Discuss with evidence:

- Which seam was most valuable for safe evolution?
- Which compatibility costs were worth paying?
- What should be deprecated next, and when?
- Which metrics should gate future rollouts?

42.7. Deliverables

- versioning plan (v1/v2 behavior map)
- compatibility matrix (client version vs service mode)
- migration + rollback playbook with gate conditions
- updated Nex code and evolution run logs
- deprecation roadmap with dates and owners

42.8. Exit Criteria

You are ready for Part IX if:

- critical services evolve through versioned seams, not rewrites
- compatibility behavior is explicit and tested
- migration and rollback are operationally defined
- reliability guarantees from Studio 5 remain intact during evolution

Part IX.

Part IX — Programming in the Age of AI

43. Part IX — Programming in the Age of AI — Working With AI Coding Assistants

43.1. 34. Working With AI Coding Assistants

AI coding assistants can accelerate implementation, but speed without structure creates fragile systems.

The key lesson is not "use AI" or "avoid AI." It is: use AI within engineering constraints.

43.2. AI As A Collaborator, Not An Authority

An assistant is strongest when given:

- clear problem framing
- explicit contracts and invariants
- architecture boundaries
- acceptance criteria

If prompts are vague, outputs are usually plausible but misaligned.

43.3. Constraint-Driven Prompting

Good prompts define:

- required behavior
- non-negotiable constraints
- integration boundaries
- validation steps

Example prompt pattern:

1. task goal
2. existing interfaces/contracts
3. forbidden changes
4. required tests/checks

This keeps AI output aligned with system intent.

43.4. Worked Design Path

Requirement:

> "Add fallback behavior to knowledge query service without changing current interface."

AI workflow:

1. provide current interface contract
2. specify allowed internal changes only
3. request fallback logic with explicit statuses
4. require regression checks for existing behavior
5. review output for contract adherence

This produces usable drafts and reduces integration rework.

43.5. Nex Implementation Sketch

```
class Knowledge_Query_Service
feature
  query(q: String): String
    require
      query_present: q /= ""
    do
      if q = "contracts" then
        result := "DOC:K-101"
      else
        result := "UNAVAILABLE"
```

```
      end
    ensure
      non_empty: result /= ""
    end
end

class Knowledge_Fallback_Wrapper
feature
  core: Knowledge_Query_Service

  query_safe(q: String): String
    require
      query_present: q /= ""
    do
      let raw: String := core.query(q)
      if raw = "UNAVAILABLE" then
        result := "DOC:CACHED-001"
      else
        result := raw
      end
    ensure
      non_empty: result /= ""
    end
end
```

This pattern is ideal for AI assistance: bounded scope, stable interface, explicit behavior.

43.6. AI Workflow Across The Three Systems

43.6.1. Delivery

- propose route policy variants under fixed dispatch contract

43.6.2. Knowledge

- propose ranking/fallback variants under stable query API

43.6.3. Virtual World

- propose update-rule variants under bounded state invariants

In each case, constraints define safe collaboration.

43.7. Common Mistakes

43.7.1. Mistake 1: Prompting without system context

Symptom:

- output conflicts with architecture or contracts

Recovery:

- include interfaces, invariants, and exclusions in prompt

43.7.2. Mistake 2: Accepting first draft blindly

Symptom:

- subtle regressions discovered later

Recovery:

- require parity checks and targeted tests

43.7.3. Mistake 3: Asking AI to redesign everything at once

Symptom:

- high-risk broad changes with weak validation

Recovery:

- scope to one bounded change per iteration

Exercise Apply the section task and record your results before reading the solution notes.

43.8. Quick Exercise (10 Minutes)

Choose one small change and write an AI prompt containing:

1. objective
2. current contract
3. constraints/forbidden changes
4. expected tests
5. acceptance criteria

Then compare AI output quality with and without constraints.

43.9. Connection to Nex

Nex contracts and invariants are excellent prompt anchors because they encode behavior intent in machine-checkable form.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

43.10. Chapter Takeaways

- AI assistants are most effective under explicit engineering constraints.
- Prompt quality directly affects code quality.
- Contracts and interfaces are the control surface for safe AI collaboration.
- Human review remains mandatory.

In Chapter 35, we focus on systematic review of AI-generated code.

44. Part IX — Programming in the Age of AI — Reviewing AI-Generated Code

44.1. 35. Reviewing AI-Generated Code

AI can generate code quickly.

Review determines whether that code is safe, correct, and maintainable.

In AI workflows, review quality is the difference between acceleration and debt.

44.2. What To Review First

Prioritize review by risk, not by line count.

High-priority review targets:

- contract correctness
- invariant preservation
- behavior regressions
- security/reliability implications
- integration boundary violations

Cosmetic style issues come after behavioral safety.

44.3. Review Checklist For AI Code

A practical checklist:

1. Does it satisfy stated requirement?
2. Does it preserve existing contracts?

3. Does it handle failure states explicitly?
4. Does it introduce hidden coupling?
5. Are tests added for changed behavior?

This checklist catches most high-impact problems early.

44.4. Worked Design Path

Requirement:

> "AI added retry logic to delivery dispatch."

Review steps:

1. verify retry bound exists
2. verify non-retryable errors do not loop
3. verify idempotency guard remains correct
4. verify existing success path unchanged
5. add regression tests for retry and duplicate requests

44.5. Nex Implementation Sketch

```
class Dispatch_Port
feature
  fail_code: String

  send(task_id: String): String
    require
      id_present: task_id /= ""
    do
      if fail_code = "NONE" then
        result := "SENT"
      elseif fail_code = "TRANSIENT" then
        result := "TRANSIENT_FAILURE"
      else
        result := "PERMANENT_FAILURE"
      end
    ensure
```

```
      known_result:
        result = "SENT" or
        result = "TRANSIENT_FAILURE" or
        result = "PERMANENT_FAILURE"
    end
end

class Dispatch_With_Retry
feature
  port: Dispatch_Port

  send(task_id: String; max_attempts: Integer): String
    require
      id_present: task_id /= ""
      attempts_valid: max_attempts >= 1
    do
      let attempt: Integer := 1
      let status: String := "TRANSIENT_FAILURE"

      from
      until attempt > max_attempts or status = "SENT" or sta
      loop
        status := port.send(task_id)
        attempt := attempt + 1
      end

      result := status
    ensure
      known_result:
        result = "SENT" or
        result = "TRANSIENT_FAILURE" or
        result = "PERMANENT_FAILURE"
    end
end
```

Review focus here is correctness under all failure paths, not syntax.

44.6. AI Review Across The Three Systems

44.6.1. Delivery

- transition legality and retry safety

44.6.2. Knowledge

- ranking correctness and fallback semantics

44.6.3. Virtual World

- deterministic update guarantees and bound safety

Review must be domain-aware, not generic.

44.7. Common Mistakes

44.7.1. Mistake 1: Style-first review

Symptom:

- severe behavioral bugs survive

Recovery:

- start with contracts, invariants, failure paths

44.7.2. Mistake 2: No regression focus

Symptom:

- existing behavior broken by new code

Recovery:

- compare before/after observable behavior explicitly

44.7.3. Mistake 3: Missing adversarial cases

Symptom:

- retry loops, edge crashes, or invalid outputs in production

Recovery:

- add failure-oriented tests before approval

Exercise Apply the section task and record your results before reading the solution notes.

44.8. Quick Exercise (12 Minutes)

Take one AI-generated patch and review it with this order:

1. contract alignment
2. invariant impact
3. failure path handling
4. regression test coverage
5. maintainability concerns

Document one blocker-level issue and one minor issue.

44.9. Connection to Nex

Nex contracts make semantic review more objective by converting behavior expectations into explicit checks.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

44.10. Chapter Takeaways

- AI-generated code requires risk-first review.
- Contract and invariant validation should lead review order.
- Regression safety is non-negotiable.
- Review quality defines AI workflow reliability.

In Chapter 36, we close with the role of human engineering judgment in AI-assisted software development.

45. Part IX — Programming in the Age of AI — Human Judgment in an AI World

45.1. 36. Human Judgment in an AI World

AI can assist with implementation, refactoring, testing, and documentation.

It cannot own accountability.

Final responsibility for correctness, safety, ethics, and long-term system quality remains human.

45.2. What Human Judgment Owns

Engineers must own decisions about:

- problem framing
- architecture direction
- contract definitions
- risk acceptance
- rollout and rollback policy

AI can propose options. Humans choose and justify.

45.3. Decision Quality In AI Workflows

A strong AI-era engineering loop:

1. define intent and constraints
2. generate alternatives
3. evaluate tradeoffs with evidence

4. select and validate
5. monitor and adapt in production

This is still engineering judgment, now with faster iteration support.

45.4. Worked Design Path

Requirement:

> "Enable v2 policy rollout for premium delivery routing."

Human decisions required:

- what counts as success metric
- rollout canary threshold
- fallback/rollback trigger
- deprecation timeline for v1

AI can draft code and tests, but these policy decisions are human governance work.

45.5. Nex Implementation Sketch

```
class Rollout_Governance
feature
  canary_success_rate: Integer
  min_required_rate: Integer
  rollback_triggered: Boolean

  decide(): String
    require
      rates_valid: canary_success_rate >= 0 and min_required_rat
    do
      if canary_success_rate >= min_required_rate then
        rollback_triggered := false
        result := "PROCEED"
      else
```

```
        rollback_triggered := true
        result := "ROLLBACK"
      end
    ensure
      known_result: result = "PROCEED" or result = "ROLLBACK
    end
invariant
  rate_bounds: canary_success_rate >= 0 and min_required_rat
end

class Engineering_Decision_Log
feature
  decision: String
  rationale: String

  record(d, r: String): String
    require
      inputs_present: d /= "" and r /= ""
    do
      decision := d
      rationale := r
      result := "RECORDED"
    ensure
      persisted: decision = d and rationale = r
    end
end
```

The point is explicit governance and traceable decision ownership.

45.6. Human Judgment Across The Three Systems

45.6.1. Delivery

- safety and service-level tradeoffs during policy evolution

45.6.2. Knowledge

- ranking fairness, explainability, and fallback acceptability

45.6.3. Virtual World

- determinism vs performance tradeoffs in simulation rules

These are not purely technical optimizations. They are product and ethics decisions too.

45.7. Common Mistakes

45.7.1. Mistake 1: Delegating decision ownership to AI

Symptom:

- unclear accountability after incidents

Recovery:

- document human owner for each critical decision

45.7.2. Mistake 2: Treating AI output as objective truth

Symptom:

- plausible but wrong design choices adopted quickly

Recovery:

- require evidence, tests, and tradeoff analysis

45.7.3. Mistake 3: Missing governance trail

Symptom:

- team cannot explain why risky rollout happened

Recovery:

- maintain lightweight decision logs and gate criteria

Exercise Apply the section task and record your results before reading the solution notes.

45.8. Quick Exercise (12 Minutes)

Pick one recent AI-assisted change and document:

1. decision owner
2. alternatives considered
3. chosen option and rationale
4. validation evidence
5. rollback condition

Then identify one governance improvement for future changes.

45.9. Connection to Nex

Nex encourages explicitness through contracts and invariants, which supports accountable, auditable human decisions in AI-assisted workflows.

Takeaways Capture the key principles from this chapter and one action you will apply immediately.

45.10. Chapter Takeaways

- AI accelerates implementation, not accountability transfer.
- Human judgment owns intent, risk, and governance.
- Reliable AI workflows require explicit constraints and evidence-based decisions.
- Engineering maturity in the AI era is measured by decision quality.

This book's central thread remains unchanged across every part:

- model clearly
- specify behavior explicitly
- measure and verify continuously
- evolve safely with accountable judgment

That is the foundation of real-world software engineering, with or without AI.

46. Epilogue — The Systems Behind Everything

The robot in the intersection moves again.

The researcher finds the right paper in seconds.

The virtual economy recovers and stays stable under pressure.

Not because someone wrote more code quickly.

Because the teams learned to see the system clearly.

46.1. What Changed

At the start, each team had working features and failing systems.

By the end, they had something different:

- explicit models of entities and relationships
- contracts and invariants that prevented silent corruption
- algorithm and data-structure choices tied to workload reality
- component boundaries that supported change
- reliability and evolution practices that scaled over time

This is the shift from coding to engineering.

46.2. One Pattern, Many Domains

The delivery network, knowledge engine, and virtual world were never truly separate stories.

They were variations of the same structure:

- entities
- relationships
- events
- decisions under constraints

Once you can model and reason about that structure, you can work across domains with confidence.

That is why real software engineering skill transfers.

46.3. The Role of Nex

In the prologue, we said Nex was chosen on purpose.

That remains true at the end.

Nex gave us a practical environment to study real engineering behavior:

- functional and object-oriented thinking in one language
- contracts and invariants near the code they govern
- exploratory workflows with optional dynamic typing
- accessible web-based development with minimal setup friction

Most importantly, Nex was used here as a vehicle for habits that matter in any production language.

The goal was never language lock-in.

The goal was durable engineering judgment.

46.4. Programming in the AI Era

AI changed the speed of implementation.

It did not change ownership.

You still own:

- problem framing
- architecture choices
- risk acceptance
- rollout and rollback decisions
- accountability for outcomes

AI can accelerate your work.

It cannot replace your judgment.

If you keep contracts explicit, review risk-first, and evolve systems intentionally, AI becomes a force multiplier instead of a source of hidden debt.

46.5. What To Do Next

Carry this practice into your next real system:

1. Write the problem before writing the code.
2. Name invariants before scaling features.
3. Choose data structures and algorithms by access patterns, not habit.
4. Refactor with parity checks, not hope.
5. Treat reliability and evolution as first-class design concerns.

If you do that consistently, your systems will survive growth, change, and pressure.

46.6. Last Question

The prologue began with a question behind every failure:

What problem am I actually solving?

The epilogue adds the companion question:

How will this system stay correct, useful, and evolvable when everything around it changes?

Keep both questions close.

They are the difference between software that demos well and software that endures.

47. Glossary

This glossary defines core terms used throughout the book.

47.1. Algorithm

A finite, explicit procedure that transforms inputs into outputs while meeting stated guarantees.

47.2. Asymptotic Complexity

A description of how runtime or memory grows as input size grows (for example: `O(1),O(log n),O(n),O(n log n),O(n^2)`).

47.3. Class Invariant

A condition that must hold for every valid object state of a class.

47.4. Contract

A precise behavioral agreement in code, usually expressed with preconditions (`require`), postconditions (`ensure`), and invariants.

47.5. Coupling

The degree to which one module depends on details of another.

47.6. Data Model

A structured representation of entities, relationships, and constraints in a problem domain.

47.7. Deduplication by Ancestor Class

When combining inherited items (such as invariants), each ancestor class contributes once even in multiple-inheritance graphs.

47.8. Encapsulation

A design principle that hides internal representation and exposes stable behavior through clear interfaces.

47.9. Feature (Method)

A callable behavior defined on a class.

47.10. Inheritance

A mechanism where a class reuses and extends behavior or structure from one or more parent classes.

47.11. Invariant

A property that must remain true across allowed state transitions.

47.12. Multiple Inheritance

A form of inheritance where a class has more than one immediate parent class.

47.13. Postcondition (`ensure`)

A condition that must be true after successful feature execution.

47.14. Precondition (`require`)

A condition that must be true before feature execution is valid.

47.15. Recursive Inheritance Merge

A rule where inherited contracts are collected transitively across parent classes.

47.16. Refactor

A structural code change that preserves externally visible behavior while improving design qualities.

47.17. Reliability

The ability of software to behave correctly under expected load and adverse conditions over time.

47.18. Scaling

The behavior of a system as data volume, traffic, or operational complexity grows.

47.19. Search Space

The set of candidate states or paths an algorithm may need to inspect.

47.20. Time Complexity

How execution steps grow as input size grows.

47.21. Transition Rule

A rule that defines which state changes are legal for an entity.

48. Index of Terms

This is a practical index of major topics and where to review them.

48.1. A

- **Algorithms**: Part III — The Shape of Algorithms, Algorithm Lab — The First Experiments, Algorithm Lab — When Algorithms Compete
- **Architecture boundaries**: Part VI — Building Real Software, Studio 4 — The Architecture Refactor

48.2. C

- **Complexity (Big-O intuition)**: Part III — Measuring Algorithm Behavior, Part V — Algorithms That Power Systems
- **Contracts (`require` / `ensure`)**: Part I — From Stories to Specifications, Part VII — Making Software Trustworthy, see `docs/CONTRACTS.md`

48.3. D

- **Data modeling**: Part II — Modeling the World, Studio 2 — The Model Redesign
- **Data structures**: Part IV — Organizing Data

48.4. I

- **Inheritance contracts**: Part VII — Making Software Trustworthy, see `docs/CONTRACTS.md`
- **Invariants**: Part II — Modeling Change, Part VII — Making Software Trustworthy, see `docs/CONTRACTS.md`

48.5. R

- **Reliability and recovery**: Studio 5 — Reliability, Part VII
- **Refactoring for scale**: Studio 3 — The Scaling Crisis, Part IV

48.6. S

- **Scaling behavior**: Studio 3 — The Scaling Crisis, Part V
- **System evolution**: Part VIII — Systems That Grow, Studio 6 — Evolution

48.7. T

- **Testing and edge cases**: Part I — Edge Cases: Where Systems Break, Part VII

48.8. V

- **Verification mindset**: Part VII — Making Software Trustworthy, Part IX — Programming in the Age of AI

49. References

This chapter is a placeholder for citations and bibliography.

Current status:

- No formal citations have been added yet.
- Chapter references are currently embedded as narrative links.

Planned bibliography policy:

- Add primary sources for algorithms and data structures.
- Add references for software design by contract and reliability engineering.
- Add references for programming language design and pedagogy where relevant to Nex.

Citation workflow (planned):

1. Add BibTeX entries to `docs/book/references.bib`.
2. Cite in chapters using `[@citation_key]`.
3. Rebuild with `make book` or `make print`.